Lecture 4: Probability and Statistics

Math for Machine Learning

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Introduction

Machine learning is about learning from data, and data is noisy, incomplete and random: measurements contain errors, labels can be wrong, and the training set is only a sample of all the data we might see. Probability theory is the mathematical language for uncertainty, and statistics is the art of drawing conclusions from random data. Both are everywhere in machine learning:

- classifiers such as logistic regression and softmax regression (Lecture 3) output probabilities;

- the loss functions of Lecture 3 are negative log-likelihoods: least squares corresponds to Gaussian noise, and cross-entropy to categorical labels;
- regularization corresponds to prior beliefs about the parameters;
- stochastic gradient descent uses random mini-batches, and works because averages of many random terms concentrate around their mean;
- generative models, such as normalizing flows (Lecture 3), learn probability distributions.

In this lecture, we introduce probability spaces, conditional probability and Bayes' theorem, random variables, expectation and variance, the Poisson process, the variance of a neural network layer (which explains the Xavier and He initializations), conditional independence, graphical models and naive Bayes, moment generating functions, and random vectors with their covariance matrices, where the linear algebra of Lectures 1 and 2 enters. We then study the Gaussian distribution in detail, the law of large numbers and the central limit theorem, Monte Carlo sampling and the convergence of stochastic gradient descent, and Markov chains with Markov chain Monte Carlo (Metropolis and Gibbs sampling), the Bellman equations of reinforcement learning, and stationary time series with autoregressive models. The second half of the lecture is about learning from data: estimation (the method of moments, maximum likelihood, the Cramér–Rao bound, sufficiency and the Rao–Blackwell theorem, exponential families and generalized linear models, the empirical CDF and kernel density estimates, and survival analysis with censored data), hypothesis testing (p-values, power, the Neyman–Pearson lemma, t-tests and χ^2-tests, standard errors, t-tests and heteroscedasticity in linear regression, multiple testing, the bootstrap and permutation tests), the evaluation of models and experiments (cross-validation, classification metrics and ROC curves, generalization bounds, and A/B testing with Simpson's paradox), Bayesian inference (including Gaussian processes), information theory (entropy, cross-entropy and the Kullback–Leibler divergence), and latent variable models (Gaussian mixtures, the EM algorithm, k-means, probabilistic PCA and variational autoencoders). Finally, we add Gaussian noise to the ordinary differential equations of Lecture 3: Brownian motion, stochastic differential equations and the Ornstein–Uhlenbeck process, and Langevin dynamics, which samples from a distribution by noisy gradient descent and is behind diffusion models. The appendix to this lecture builds probability theory on measure theory, which treats discrete and continuous distributions, and everything in between, in a single framework.

Notation. As in the previous lectures, vectors of $\mathbb{R}^n$ are bold. Random variables are denoted by capital letters such as X and Y, their values by small letters such as x and y, and random vectors by bold capital letters such as $\mathbf{X}$. We write $P(A)$ for the probability of an event A, $\mathbf{1}_A$ for its indicator function, which is 1 on A and 0 outside A, $\mathbb{E}[X]$ for the expectation of X, and log for the natural logarithm.

1 Probability Spaces

A random experiment has a set of possible outcomes, and we want to assign probabilities to sets of outcomes.

Definition 1.1 (Probability space)**.** A probability space consists of a sample space Ω, the set of all possible outcomes, and a probability P, which assigns a number $P(A)$ to every event $A \subseteq \Omega$, such that:

1. $P(A) \geq 0$ for every event A;

2. $P(\Omega) = 1$;

3. if $A_1, A_2, \ldots$ are pairwise disjoint events, then $P(A_1 \cup A_2 \cup \cdots) = P(A_1) + P(A_2) + \cdots$.

Remark 1.2 (Measure theory). If Ω is uncountable, such as $\Omega = \mathbb{R}$, it is impossible to assign a probability to every subset of Ω in a consistent way. A rigorous treatment specifies a collection of events (a σ-algebra) and uses measure theory. All sets that we meet in practice, such as intervals, balls and their unions and complements, are events, and we ignore these technicalities in this lecture; the appendix to this lecture treats them.

Proposition 1.3. *For all events A, B and $A_1, A_2, \ldots$:*

1. *$P(A^c) = 1 - P(A)$, where $A^c = \Omega \setminus A$, and $P(\emptyset) = 0$;*

2. *if $A \subseteq B$, then $P(A) \leq P(B)$;*

3. *$P(A \cup B) = P(A) + P(B) - P(A \cap B)$;*

4. *(union bound) $P(A_1 \cup A_2 \cup \cdots) \leq P(A_1) + P(A_2) + \cdots$.*

Proof. (1) A and A^c are disjoint with union Ω, so $P(A)+P(A^c) = 1$; for $A = \Omega$, this gives $P(\emptyset) = 0$. (2) B is the disjoint union of A and $B \setminus A$, so $P(B) = P(A) + P(B \setminus A) \geq P(A)$. (3) $A \cup B$ is the disjoint union of A and $B \setminus A$, and B is the disjoint union of $B \setminus A$ and $A \cap B$; so $P(A \cup B) = P(A) + P(B \setminus A) = P(A) + P(B) - P(A \cap B)$. (4) Let $B_1 = A_1$ and $B_k = A_k \setminus (A_1 \cup \cdots \cup A_{k-1})$. The B_k are disjoint, $B_k \subseteq A_k$, and they have the same union as the A_k; now use the third axiom and (2). □

Example 1.4 (Equally likely outcomes)**.** If Ω is finite and all outcomes are equally likely, then $P(A) = \frac{|A|}{|\Omega|}$. For two fair dice, $\Omega = \{1, \ldots, 6\}^2$ has 36 outcomes. The sum is 7 for the 6 outcomes $(1,6), (2,5), \ldots, (6,1)$, so $P(\text{sum} = 7) = \frac{6}{36} = \frac{1}{6}$. The event "at least one six" is the complement of "no six", which has $5 \cdot 5 = 25$ outcomes, so its probability is $1 - \frac{25}{36} = \frac{11}{36}$.

1.1 Conditional probability and Bayes' theorem

If we learn that an event B has happened, the probabilities of other events change: only the outcomes in B are still possible.

Definition 1.5 (Conditional probability)**.** If $P(B) > 0$, the conditional probability of A given B is

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)}.$$

For fixed B, the function $A \mapsto P(A \mid B)$ is again a probability. Rearranging the definition gives the product rule $P(A \cap B) = P(A \mid B)\, P(B)$.

Theorem 1.6 (Total probability and Bayes' theorem)**.** *Let $B_1, \ldots, B_n$ be disjoint events with union Ω and $P(B_i) > 0$. Then for every event A,*

$$P(A) = \sum_{i=1}^{n} P(A \mid B_i)\, P(B_i),$$

and if $P(A) > 0$, then

$$P(B_j \mid A) = \frac{P(A \mid B_j)\, P(B_j)}{P(A)} = \frac{P(A \mid B_j)\, P(B_j)}{\sum_{i=1}^{n} P(A \mid B_i)\, P(B_i)}.$$

Proof. A is the disjoint union of the events $A \cap B_i$, so $P(A) = \sum_i P(A \cap B_i) = \sum_i P(A \mid B_i)P(B_i)$ by the product rule. For the second formula, use the product rule twice: $P(B_j \mid A)P(A) = P(A \cap B_j) = P(A \mid B_j)P(B_j)$. □

Bayes' theorem reverses conditional probabilities: from the probability of the evidence A given each hypothesis B_j, it computes the probability of each hypothesis given the evidence. The probabilities $P(B_j)$ are called prior probabilities, and the $P(B_j \mid A)$ posterior probabilities.

Example 1.7 (A medical test)**.** A disease affects 1% of a population. A test detects it in 99% of the sick people, but also gives a positive result for 5% of the healthy people. What is the probability that a person with a positive test is sick? With D = "sick" and $+$ = "positive test",

$$P(D \mid +) = \frac{P(+ \mid D)P(D)}{P(+ \mid D)P(D) + P(+ \mid D^c)P(D^c)} = \frac{0.99 \cdot 0.01}{0.99 \cdot 0.01 + 0.05 \cdot 0.99} = \frac{1}{6}.$$

Although the test is quite accurate, only one in six people with a positive test is sick, because the disease is rare: among 10 000 people, about 99 sick people and 495 healthy people test positive. Ignoring the prior probability is a common mistake, called the base rate fallacy. In machine learning, the same effect makes classifiers for rare classes (such as fraud detection) produce many false alarms.

1.2 Independence

Definition 1.8 (Independence)**.** Two events A and B are independent if $P(A \cap B) = P(A)P(B)$. Events $A_1, \dots, A_n$ are (mutually) independent if $P(A_{i_1} \cap \cdots \cap A_{i_k}) = P(A_{i_1}) \cdots P(A_{i_k})$ for every choice of indices $i_1 < \cdots < i_k$.

If $P(B) > 0$, then A and B are independent if and only if $P(A \mid B) = P(A)$: learning that B happened does not change the probability of A. For example, for two fair dice, the events A = "the first die is even" and B = "the sum is 7" are independent: $A \cap B$ consists of the 3 outcomes $(2,5), (4,3), (6,1)$, so $P(A \cap B) = \frac{3}{36} = \frac{1}{2} \cdot \frac{1}{6} = P(A)P(B)$. Independence is not the same as disjointness: disjoint events with positive probabilities are never independent, since $P(A \cap B) = 0$.

2 Random Variables and Their Distributions

Definition 2.1 (Random variable)**.** A random variable is a function $X : \Omega \to \mathbb{R}$ on a probability space. Its distribution describes the probabilities $P(X \in I) = P(\{\omega : X(\omega) \in I\})$ for intervals I. Its cumulative distribution function (CDF) is $F(x) = P(X \le x)$.

The CDF is nondecreasing and right-continuous, tends to 0 as $x \to -\infty$ and to 1 as $x \to \infty$, and $P(a < X \le b) = F(b) - F(a)$. In practice, we rarely look at the sample space: a random variable is described by its distribution. Most distributions in machine learning are of one of two kinds.

Definition 2.2 (Discrete and continuous random variables)**.**

1. X is discrete if it takes only finitely or countably many values $x_1, x_2, \dots$. Its probability mass function (pmf) is $p(x) = P(X = x)$, and $\sum_i p(x_i) = 1$.

2. X is continuous if there is a function $p \ge 0$, its probability density function (pdf), with

$$P(a \le X \le b) = \int_a^b p(x)\,dx \qquad \text{for all } a \le b, \qquad \text{and} \qquad \int_{-\infty}^{\infty} p(x)\,dx = 1.$$

For a continuous random variable, $P(X = x) = 0$ for every single value x, and the density is the derivative of the CDF, $p = F'$ (where p is continuous). A density is not a probability: it can be larger than 1. For a small interval, $P(x \le X \le x + h) \approx p(x)h$.

Example 2.3 (Discrete distributions)**.** 1. Bernoulli(p): $X \in \{0, 1\}$ with $P(X = 1) = p$; for example, a coin flip, or a binary label.

2. Binomial(n, p): the number of successes in n independent Bernoulli(p) trials,

$$P(X = k) = \binom{n}{k} p^k (1-p)^{n-k}, \qquad k = 0, 1, \dots, n,$$

since each of the $\binom{n}{k}$ sequences with k successes has probability $p^k(1-p)^{n-k}$. For example, for $n = 10$ and $p = 0.3$, $P(X = 3) = 120 \cdot 0.3^3 \cdot 0.7^7 \approx 0.267$ (Figure 1).

3. Geometric(p): the number of independent Bernoulli(p) trials until the first success, $P(X = k) = (1-p)^{k-1}p$ for $k = 1, 2, \dots$.

4. Poisson(λ): $P(X = k) = e^{-\lambda}\frac{\lambda^k}{k!}$ for $k = 0, 1, 2, \dots$; it describes counts of rare events, such as the number of clicks on an ad in an hour. It is the limit of Binomial($n, \frac{\lambda}{n}$) as $n \to \infty$.

5. Categorical($\mathbf{p}$): $X \in \{1, \dots, K\}$ with $P(X = k) = p_k$, where $p_k \ge 0$ and $\sum_k p_k = 1$; for example, the class predicted by a softmax classifier.

Example 2.4 (Continuous distributions)**.** 1. Uniform(a, b): $p(x) = \frac{1}{b-a}$ for $a \le x \le b$, and 0 otherwise.

2. Exponential(λ): $p(x) = \lambda e^{-\lambda x}$ for $x \ge 0$; then $P(X > t) = e^{-\lambda t}$. It describes waiting times, and is memoryless: $P(X > s + t \mid X > s) = \frac{e^{-\lambda(s+t)}}{e^{-\lambda s}} = P(X > t)$.

3. Normal (Gaussian) $\mathcal{N}(\mu, \sigma^2)$, with mean μ and variance $\sigma^2 > 0$:

$$p(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\Big(-\frac{(x-\mu)^2}{2\sigma^2}\Big).$$

It integrates to 1 by the Gaussian integral of Lecture 3. The standard normal distribution $\mathcal{N}(0, 1)$ has the density $\varphi(z) = \frac{1}{\sqrt{2\pi}} e^{-z^2/2}$ and the CDF $\Phi(z) = \int_{-\infty}^{z} \varphi(t)\, dt$, which has no formula in terms of elementary functions.

2.1 Transformations of random variables

If X is a random variable and $g : \mathbb{R} \to \mathbb{R}$ is a function, then $Y = g(X)$ is again a random variable. For continuous random variables and invertible g, the density changes by the derivative of g, just as areas change by the Jacobian determinant in Lecture 3.

Proposition 2.5 (Change of variables)**.** *Let X have the density p_X, and let g be differentiable and strictly increasing (or strictly decreasing). Then $Y = g(X)$ has the density*

$$p_Y(y) = p_X(x)\, \frac{1}{|g'(x)|}, \qquad \textit{where } x = g^{-1}(y).$$

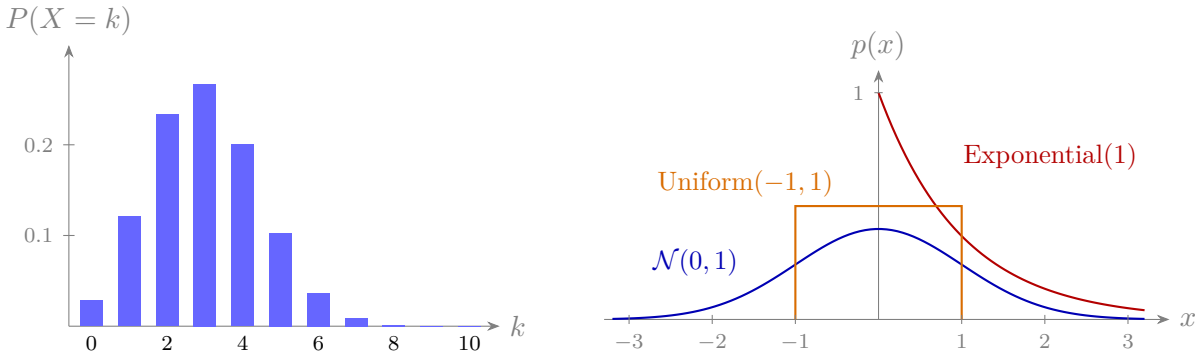


Figure 1: Left: the pmf of the Binomial$(10, 0.3)$ distribution. Right: the densities of the standard normal distribution, the Exponential(1) distribution and the Uniform$(-1, 1)$ distribution.

Proof. For increasing g, $F_Y(y) = P(g(X) \le y) = P(X \le g^{-1}(y)) = F_X(g^{-1}(y))$. Differentiating with the chain rule gives $p_Y(y) = p_X(g^{-1}(y))\,(g^{-1})'(y) = \frac{p_X(x)}{g'(x)}$. For decreasing g, $F_Y(y) = 1 - F_X(g^{-1}(y))$, which gives the same formula with $|g'(x)|$. □

Example 2.6. 1. (Standardization.) If $X \sim \mathcal{N}(\mu, \sigma^2)$, then $Z = \frac{X-\mu}{\sigma}$ has the density $\sigma\, p_X(\mu + \sigma z) = \varphi(z)$, so $Z \sim \mathcal{N}(0, 1)$. Conversely, $X = \mu + \sigma Z$. Hence $P(|X - \mu| \le k\sigma) = P(|Z| \le k)$, which is 0.683, 0.954 and 0.997 for $k = 1, 2, 3$, and 0.95 for $k = 1.96$.

2. (Log-normal distribution.) If $X \sim \mathcal{N}(0, 1)$, then $Y = e^X$ has the density $\frac{\varphi(\log y)}{y}$ for $y > 0$.

Theorem 2.7 (Inverse transform sampling)**.** *Let F be nondecreasing and right-continuous with the limits 0 at $-\infty$ and 1 at ∞, let*

$$F^{-1}(u) = \min\{x \in \mathbb{R} : F(x) \ge u\} \qquad \text{for } 0 < u < 1$$

be its quantile function, and let $U \sim$ Uniform$(0, 1)$. Then $X = F^{-1}(U)$ has the CDF F.

Proof. For $0 < u < 1$, the set $\{x : F(x) \ge u\}$ is nonempty and bounded below, since F tends to 1 and to 0. Since F is nondecreasing, it is an interval that extends to ∞, and since F is right-continuous, it contains its left endpoint. So the minimum exists, and $F^{-1}(u) \le x$ if and only if $F(x) \ge u$. Hence $P(X \le x) = P(U \le F(x)) = F(x)$. □

If F is continuous and strictly increasing, F^{-1} is the inverse function of F. So a computer that can generate uniform random numbers can generate random numbers with any distribution on $\mathbb{R}$. For example, the Exponential(λ) distribution has $F(x) = 1 - e^{-\lambda x}$, so $X = -\frac{1}{\lambda}\log(1 - U)$ is exponentially distributed. For a Bernoulli(p) variable, F jumps from 0 to $1 - p$ at 0 and from $1 - p$ to 1 at 1, so $F^{-1}(u) = 0$ for $u \le 1 - p$ and $F^{-1}(u) = 1$ for $u > 1 - p$: we return 1 if $U > 1 - p$, which has the probability p. Normalizing flows (Lecture 3) generalize this idea to many dimensions: they transform simple random vectors by learned invertible maps.

3 Expectation and Variance

Definition 3.1 (Expectation)**.** The expectation (expected value, or mean) of a random variable X is

$$\mathbb{E}[X] = \sum_i x_i\, p(x_i) \qquad \text{or} \qquad \mathbb{E}[X] = \int_{-\infty}^{\infty} x\, p(x)\, dx,$$

for discrete and continuous X, respectively, provided that the sum or integral converges absolutely.

The expectation is the center of mass of the distribution: if we put the mass $p(x_i)$ at each point x_i, the balance point is $\mathbb{E}[X]$. To compute the expectation of a function of X, we do not need the distribution of $g(X)$; one can show that

$$\mathbb{E}[g(X)] = \sum_i g(x_i)\, p(x_i) \qquad \text{or} \qquad \mathbb{E}[g(X)] = \int_{-\infty}^{\infty} g(x)\, p(x)\, dx.$$

The expectation is linear: $\mathbb{E}[aX + bY + c] = a\mathbb{E}[X] + b\mathbb{E}[Y] + c$ for all random variables X, Y with finite expectations, and all numbers a, b, c (we prove this in Section 4 for discrete random variables).

Example 3.2 (Tail sums)**.** Let X take values in $\{0, 1, 2, \ldots\}$. Then

$$\mathbb{E}[X] = \sum_{k=0}^{\infty} P(X > k).$$

Indeed, $P(X > k) = \sum_{j>k} P(X = j)$, and in the double series $\sum_{k\geq 0} \sum_{j>k} P(X = j)$, whose terms are nonnegative and can therefore be rearranged, each $P(X = j)$ appears j times, for $k = 0, \ldots, j-1$; so the sum is $\sum_j j\, P(X = j) = \mathbb{E}[X]$. In other words, $X = \sum_{k\geq 0} \mathbf{1}_{\{X>k\}}$, since exactly the first X terms are 1, and we may take expectations term by term. For a Geometric(p) variable, $P(X > k) = (1-p)^k$, the probability that the first k trials fail, so $\mathbb{E}[X] = \sum_{k\geq 0}(1-p)^k = \frac{1}{p}$. Example 4.7 gives the analogous formula for continuous random variables.

Definition 3.3 (Variance)**.** The variance of X with mean $\mu = \mathbb{E}[X]$ is $\mathrm{Var}(X) = \mathbb{E}[(X-\mu)^2]$, and its standard deviation is $\sigma = \sqrt{\mathrm{Var}(X)}$.

The variance measures how far X typically is from its mean. Expanding the square and using linearity,

$$\mathrm{Var}(X) = \mathbb{E}[X^2 - 2\mu X + \mu^2] = \mathbb{E}[X^2] - 2\mu^2 + \mu^2 = \mathbb{E}[X^2] - \mathbb{E}[X]^2,$$

and $\mathrm{Var}(aX + b) = \mathbb{E}[(aX - a\mu)^2] = a^2 \mathrm{Var}(X)$: shifting does not change the spread, and scaling by a scales the standard deviation by $|a|$.

Proposition 3.4. *The distributions of Section 2 have the following means and variances:*

distribution	$\mathbb{E}[X]$	$\mathrm{Var}(X)$
Bernoulli(p)	p	$p(1-p)$
Binomial(n, p)	np	$np(1-p)$
Geometric(p)	$\frac{1}{p}$	$\frac{1-p}{p^2}$
Poisson(λ)	λ	λ
Uniform(a, b)	$\frac{a+b}{2}$	$\frac{(b-a)^2}{12}$
Exponential(λ)	$\frac{1}{\lambda}$	$\frac{1}{\lambda^2}$
$\mathcal{N}(\mu, \sigma^2)$	μ	σ^2

Proof of some cases. Bernoulli: $\mathbb{E}[X] = 0 \cdot (1-p) + 1 \cdot p = p$, and $X^2 = X$, so $\mathrm{Var}(X) = p - p^2$. Uniform: $\mathbb{E}[X] = \frac{1}{b-a}\int_a^b x\, dx = \frac{b^2-a^2}{2(b-a)} = \frac{a+b}{2}$, and similarly $\mathbb{E}[X^2] = \frac{a^2+ab+b^2}{3}$, so $\mathrm{Var}(X) = \frac{(b-a)^2}{12}$. Exponential: integrating by parts, $\mathbb{E}[X] = \int_0^\infty x\lambda e^{-\lambda x}\, dx = \left[-xe^{-\lambda x}\right]_0^\infty + \int_0^\infty e^{-\lambda x}\, dx = \frac{1}{\lambda}$. Poisson:

$\mathbb{E}[X] = \sum_{k\ge1} k e^{-\lambda}\frac{\lambda^k}{k!} = \lambda\sum_{k\ge1} e^{-\lambda}\frac{\lambda^{k-1}}{(k-1)!} = \lambda$. Normal: by Example 2.6, $X = \mu + \sigma Z$ with $Z \sim \mathcal{N}(0,1)$, so it suffices to show $\mathbb{E}[Z] = 0$ and $\mathrm{Var}(Z) = 1$. The first holds since $z\varphi(z)$ is an odd function, and for the second, integration by parts with $\varphi'(z) = -z\varphi(z)$ gives

$$\mathbb{E}[Z^2] = \int_{-\infty}^{\infty} z \cdot z\varphi(z)\,dz = \big[-z\varphi(z)\big]_{-\infty}^{\infty} + \int_{-\infty}^{\infty} \varphi(z)\,dz = 1.$$

The binomial case follows from the Bernoulli case and Proposition 4.9 below. □

Example 3.5. For a fair die, $\mathbb{E}[X] = \frac{1+2+\cdots+6}{6} = 3.5$, which is not a possible value, and $\mathbb{E}[X^2] = \frac{91}{6}$, so $\mathrm{Var}(X) = \frac{91}{6} - \frac{49}{4} = \frac{35}{12} \approx 2.92$.

3.1 Inequalities

The variance controls how likely large deviations from the mean are.

Proposition 3.6 (Markov and Chebyshev)**.** *1. (Markov's inequality.) If $X \ge 0$ and $a > 0$, then $P(X \ge a) \le \frac{\mathbb{E}[X]}{a}$.*

2. (Chebyshev's inequality.) If X has mean μ and variance σ^2, then $P(|X-\mu| \ge k) \le \frac{\sigma^2}{k^2}$ for every $k > 0$.

Proof. (1) Since $X \ge 0$, we have $X \ge a\mathbf{1}_{\{X\ge a\}}$. Taking expectations, $\mathbb{E}[X] \ge aP(X \ge a)$. (2) Apply (1) to $(X-\mu)^2$ and $a = k^2$: $P(|X-\mu| \ge k) = P((X-\mu)^2 \ge k^2) \le \frac{\sigma^2}{k^2}$. □

For example, every random variable is at least 2 standard deviations from its mean with probability at most $\frac{1}{4}$. This bound holds for all distributions, so it is weak for specific ones: for a normal distribution, the probability is only 0.046.

Theorem 3.7 (Jensen's inequality)**.** *If f is convex, then $f(\mathbb{E}[X]) \le \mathbb{E}[f(X)]$. If f is concave, then $f(\mathbb{E}[X]) \ge \mathbb{E}[f(X)]$.*

Proof. We prove this for differentiable f. Let $\mu = \mathbb{E}[X]$. By the first-order characterization of convexity in Lecture 3, $f(x) \ge f(\mu) + f'(\mu)(x-\mu)$ for all x. So $f(X) \ge f(\mu) + f'(\mu)(X-\mu)$, and taking expectations gives $\mathbb{E}[f(X)] \ge f(\mu) + f'(\mu)\cdot 0$. For concave f, apply this to $-f$. □

For example, $f(x) = x^2$ gives $\mathbb{E}[X]^2 \le \mathbb{E}[X^2]$, i.e., $\mathrm{Var}(X) \ge 0$, and the concave function log gives $\mathbb{E}[\log X] \le \log \mathbb{E}[X]$ for $X > 0$. We will use the latter to show that the Kullback–Leibler divergence is nonnegative (Section 13).

Example 3.8 (The Chernoff bound)**.** Markov's inequality becomes much stronger when it is applied to an exponential: for every $t > 0$, $P(X \ge a) = P(e^{tX} \ge e^{ta}) \le e^{-ta}\mathbb{E}[e^{tX}]$, and we may choose the best t. For $S \sim \mathrm{Binomial}(n, \frac{1}{2})$, the sum of n independent fair coin flips, $\mathbb{E}[e^{tS}] = \big(\frac{1+e^t}{2}\big)^n$ (by Proposition 4.4 below). For $n = 100$ and $a = 60$, the best choice is $e^t = \frac{3}{2}$, which gives $P(S \ge 60) \le \big(\frac{5}{4}\big)^{100}\big(\frac{2}{3}\big)^{60} \approx 0.134$, while Chebyshev's inequality only gives $P(S \ge 60) \le P(|S-50| \ge 10) \le \frac{25}{100} = 0.25$; the exact value is 0.028. As n grows, Chernoff bounds decrease exponentially, while Chebyshev's bound only decreases like $\frac{1}{n}$. Hoeffding's inequality (Theorem 7.5) is proved in this way.

4 Joint Distributions and Independence

To describe how several random variables depend on each other, we need their joint distribution.

Definition 4.1 (Joint, marginal and conditional distributions)**.** The joint pmf of two discrete random variables X and Y is $p(x,y) = P(X = x, Y = y)$. Two continuous random variables have a joint density $p(x,y) \geq 0$ if $P((X,Y) \in A) = \iint_A p(x,y)\,dx\,dy$ for all (reasonable) sets $A \subseteq \mathbb{R}^2$. The marginal distributions of X and Y are

$$p_X(x) = \sum_y p(x,y) \qquad \text{or} \qquad p_X(x) = \int_{-\infty}^{\infty} p(x,y)\,dy,$$

and similarly for p_Y. The conditional distribution of Y given $X = x$ (for $p_X(x) > 0$) is

$$p(y \mid x) = \frac{p(x,y)}{p_X(x)}.$$

The formulas $p_X(x) = \sum_y p(x,y)$ (the sum rule) and $p(x,y) = p(y \mid x)\, p_X(x)$ (the product rule) are the two basic rules of probabilistic machine learning, and Bayes' theorem $p(x \mid y) = \frac{p(y|x)p_X(x)}{p_Y(y)}$ follows from them. The same definitions apply to more than two random variables.

Definition 4.2 (Independent random variables)**.** X and Y are independent if $p(x,y) = p_X(x)\, p_Y(y)$ for all x and y; equivalently, $p(y \mid x) = p_Y(y)$ whenever $p_X(x) > 0$. Random variables $X_1, \ldots, X_n$ are independent if their joint pmf or density is the product of the marginal ones. They are independent and identically distributed (i.i.d.) if moreover they all have the same distribution.

The standard assumption in machine learning is that the training examples are i.i.d. samples from an unknown distribution: each example is drawn from the same distribution, independently of the others.

Example 4.3. Let $X, Y \in \{0,1\}$ have the joint pmf

	$Y=0$	$Y=1$	p_X
$X=0$	0.3	0.2	0.5
$X=1$	0.1	0.4	0.5
p_Y	0.4	0.6	

The marginals are the row and column sums. The conditional probabilities are $P(Y = 1 \mid X = 1) = \frac{0.4}{0.5} = 0.8$ and $P(Y = 1 \mid X = 0) = \frac{0.2}{0.5} = 0.4$. Since $p(1,1) = 0.4 \neq 0.5 \cdot 0.6 = p_X(1)p_Y(1)$, X and Y are not independent: learning X changes the probabilities of Y.

For a function $g(X,Y)$, the expectation is $\mathbb{E}[g(X,Y)] = \sum_{x,y} g(x,y)\, p(x,y)$, or the corresponding integral. In particular, linearity of expectation follows: for discrete X and Y,

$$\mathbb{E}[X+Y] = \sum_{x,y} (x+y)\, p(x,y) = \sum_x x \sum_y p(x,y) + \sum_y y \sum_x p(x,y) = \mathbb{E}[X] + \mathbb{E}[Y].$$

Proposition 4.4. *If X and Y are independent, then $\mathbb{E}[XY] = \mathbb{E}[X]\,\mathbb{E}[Y]$, and more generally $\mathbb{E}[g(X)h(Y)] = \mathbb{E}[g(X)]\,\mathbb{E}[h(Y)]$.*

Proof. For discrete random variables,

$$\mathbb{E}[g(X)h(Y)] = \sum_{x,y} g(x)h(y)p_X(x)p_Y(y) = \Big(\sum_x g(x)p_X(x)\Big)\Big(\sum_y h(y)p_Y(y)\Big);$$

for densities, the same computation works with integrals. □

Example 4.5 (Sums of independent random variables)**.** If X and Y are independent with the densities p_X and p_Y, then $X+Y$ has the density

$$p_{X+Y}(z) = \int_{-\infty}^{\infty} p_X(x)\, p_Y(z-x)\, dx,$$

the convolution of p_X and p_Y. Indeed, substituting $y = z - x$ in the inner integral and exchanging the order of integration, which is allowed for nonnegative functions (Fubini's theorem, Lecture 3),

$$P(X+Y \le t) = \iint \mathbf{1}_{\{x+y\le t\}}\, p_X(x)\, p_Y(y)\, dy\, dx = \int_{-\infty}^{t}\int_{-\infty}^{\infty} p_X(x)\, p_Y(z-x)\, dx\, dz.$$

For two independent Uniform$(0,1)$ variables, the integrand is 1 for $\max(0, z-1) \le x \le \min(1, z)$ and 0 otherwise, so $p_{X+Y}(z) = z$ for $0 \le z \le 1$ and $p_{X+Y}(z) = 2 - z$ for $1 \le z \le 2$: a triangular density, and $P(X+Y \le \frac{1}{2}) = \frac{1}{8}$. For independent $X \sim \text{Poisson}(\lambda)$ and $Y \sim \text{Poisson}(\mu)$, the same computation with sums and the binomial theorem gives

$$P(X+Y=n) = \sum_{k=0}^{n} e^{-\lambda}\frac{\lambda^k}{k!}\, e^{-\mu}\frac{\mu^{n-k}}{(n-k)!} = e^{-(\lambda+\mu)}\frac{(\lambda+\mu)^n}{n!},$$

so $X+Y \sim \text{Poisson}(\lambda+\mu)$. For example, if a website gets independent Poisson numbers of clicks from search engines and from ads, with the means 3 and 2 per minute, then the total number of clicks per minute is Poisson(5).

Example 4.6 (The Poisson process)**.** Customers arrive at a shop, or requests at a server, with independent Exponential(λ) waiting times $T_1, T_2, \ldots$ between consecutive arrivals. Let $S_n = T_1 + \cdots + T_n$ be the time of the n-th arrival, and $N(t)$ the number of arrivals up to time t. By the convolution formula and induction, S_n has the density

$$p_{S_n}(t) = \frac{\lambda^n t^{n-1}}{(n-1)!}e^{-\lambda t} \qquad (t \ge 0),$$

since $\int_0^t \frac{\lambda^n s^{n-1}}{(n-1)!}e^{-\lambda s}\,\lambda e^{-\lambda(t-s)}\,ds = \lambda^{n+1}e^{-\lambda t}\int_0^t \frac{s^{n-1}}{(n-1)!}\,ds = \frac{\lambda^{n+1}t^n}{n!}e^{-\lambda t}$. Integration by parts gives

$$P(S_{n+1} \le t) = \int_0^t \frac{\lambda^{n+1}s^n}{n!}e^{-\lambda s}\,ds = -\frac{(\lambda t)^n}{n!}e^{-\lambda t} + \int_0^t \frac{\lambda^n s^{n-1}}{(n-1)!}e^{-\lambda s}\,ds = P(S_n \le t) - \frac{(\lambda t)^n}{n!}e^{-\lambda t}.$$

Since $N(t) \ge n$ exactly when $S_n \le t$, we get $P(N(t)=n) = P(S_n \le t) - P(S_{n+1} \le t) = e^{-\lambda t}\frac{(\lambda t)^n}{n!}$ for $n \ge 1$, and $P(N(t)=0) = P(T_1 > t) = e^{-\lambda t}$: the number of arrivals in the time t is Poisson(λt). This is the Poisson process with rate λ. One can show more: by the memorylessness of the exponential distribution, the numbers of arrivals in disjoint time intervals are independent and Poisson distributed, with means λ times the lengths of the intervals. This explains why counts of many independent rare events, such as clicks, emails or radioactive decays, are approximately Poisson distributed.

Example 4.7 (Tail integrals)**.** For a random variable $X \geq 0$ with a density p, exchanging the order of integration (Fubini's theorem for nonnegative functions) gives

$$\int_0^\infty P(X > t)\,dt = \int_0^\infty \int_t^\infty p(x)\,dx\,dt = \int_0^\infty \Big(\int_0^x dt\Big) p(x)\,dx = \int_0^\infty x\,p(x)\,dx = \mathbb{E}[X],$$

the continuous version of Example 3.2; the appendix to this lecture shows that $\mathbb{E}[X] = \int_0^\infty P(X > t)\,dt$ for every random variable $X \geq 0$. For example, for $X \sim \text{Exponential}(\lambda)$, $\mathbb{E}[X] = \int_0^\infty e^{-\lambda t}\,dt = \frac{1}{\lambda}$, without integration by parts. Similarly, $\mathbb{E}[X^p] = \int_0^\infty p t^{p-1} P(X > t)\,dt$ for $p > 0$, which computes moments from tail bounds.

4.1 Covariance and correlation

Definition 4.8 (Covariance and correlation)**.** The covariance of X and Y, with means μ_X and μ_Y, is

$$\text{Cov}(X, Y) = \mathbb{E}[(X - \mu_X)(Y - \mu_Y)] = \mathbb{E}[XY] - \mathbb{E}[X]\,\mathbb{E}[Y],$$

and if both have positive variances, their correlation is $\text{Corr}(X, Y) = \frac{\text{Cov}(X,Y)}{\sigma_X \sigma_Y}$. If $\text{Cov}(X, Y) = 0$, they are uncorrelated.

The covariance is positive if X and Y tend to be larger than their means at the same time, and negative if one tends to be large when the other is small. Note that $\text{Cov}(X, X) = \text{Var}(X)$, and the covariance is bilinear and symmetric. By Proposition 4.4, independent random variables are uncorrelated. The converse is false: if X is uniform on $\{-1, 0, 1\}$ and $Y = X^2$, then $\text{Cov}(X, Y) = \mathbb{E}[X^3] - \mathbb{E}[X]\mathbb{E}[X^2] = 0$, but Y is a function of X. Correlation only measures linear dependence.

Proposition 4.9. $\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y) + 2\,\text{Cov}(X, Y)$. *In particular, if $X_1, \ldots, X_n$ are pairwise uncorrelated (for example, independent), then* $\text{Var}(X_1 + \cdots + X_n) = \text{Var}(X_1) + \cdots + \text{Var}(X_n)$.

Proof. With $\tilde{X} = X - \mu_X$ and $\tilde{Y} = Y - \mu_Y$: $\text{Var}(X+Y) = \mathbb{E}[(\tilde{X}+\tilde{Y})^2] = \mathbb{E}[\tilde{X}^2] + \mathbb{E}[\tilde{Y}^2] + 2\mathbb{E}[\tilde{X}\tilde{Y}]$. The second statement follows by induction. □

For example, a Binomial(n, p) random variable is the sum of n independent Bernoulli(p) random variables, so its mean is np and its variance is $np(1 - p)$, as claimed in Proposition 3.4.

The covariance behaves like an inner product. For random variables U and V with finite variances, $\langle U, V\rangle = \mathbb{E}[UV]$ is bilinear, symmetric, and $\langle U, U\rangle = \mathbb{E}[U^2] \geq 0$, with equality only if $U = 0$ with probability 1. So, if we do not distinguish random variables that are equal with probability 1, it is an inner product as in Lecture 2. For centered random variables, the inner product is the covariance, the norm is the standard deviation, and the correlation is the cosine of the angle between them.

Theorem 4.10. $|\,\text{Cov}(X, Y)| \leq \sigma_X \sigma_Y$, *i.e.,* $-1 \leq \text{Corr}(X, Y) \leq 1$. *Moreover,* $|\,\text{Corr}(X, Y)| = 1$ *if and only if $Y = aX + b$ with probability* 1*, for some $a \neq 0$ and b; the sign of the correlation is the sign of a.*

Proof. This is the Cauchy–Schwarz inequality for the centered random variables $\tilde{X}$ and $\tilde{Y}$. Directly: for every $t \in \mathbb{R}$,

$$0 \leq \mathbb{E}[(\tilde{Y} - t\tilde{X})^2] = \sigma_Y^2 - 2t\,\text{Cov}(X, Y) + t^2\sigma_X^2.$$

A quadratic polynomial in t that is never negative has a discriminant ≤ 0, so $4\,\text{Cov}(X, Y)^2 - 4\sigma_X^2\sigma_Y^2 \leq 0$. Equality holds if and only if the polynomial has a zero $t = a$, i.e., $\mathbb{E}[(\tilde{Y} - a\tilde{X})^2] = 0$, i.e., $\tilde{Y} = a\tilde{X}$ with probability 1; then $Y = aX + (\mu_Y - a\mu_X)$, and $\text{Cov}(X, Y) = a\sigma_X^2$. □

Example 4.11. In Example 4.3, $\mathbb{E}[XY] = P(X = 1, Y = 1) = 0.4$, so $\mathrm{Cov}(X, Y) = 0.4 - 0.5 \cdot 0.6 = 0.1$. With $\mathrm{Var}(X) = 0.25$ and $\mathrm{Var}(Y) = 0.24$, the correlation is $\frac{0.1}{\sqrt{0.06}} \approx 0.41$.

Initializing neural networks. Variances of sums explain how the weights of a neural network have to be scaled when training starts. A unit of a layer computes $z = W_1 x_1 + \cdots + W_n x_n$ from the n outputs x_j of the previous layer, and passes $a(z)$ on, for an activation function a such as $\mathrm{ReLU}(z) = \max(z, 0)$ (Lecture 3). At initialization, the weights are random, and the following proposition tracks the size of the signal.

Proposition 4.12 (Variance of a layer)**.** *Let $x_1, \ldots, x_n$ be random variables with $\mathbb{E}[x_j^2] = q$ for all j, and let $W_1, \ldots, W_n$ have mean 0 and variance v, such that $W_1, \ldots, W_n$ and the vector $(x_1, \ldots, x_n)$ are independent. Then $z = W_1 x_1 + \cdots + W_n x_n$ has*

$$\mathbb{E}[z] = 0 \qquad \textit{and} \qquad \mathrm{Var}(z) = nvq.$$

If moreover $-W_j$ has the same distribution as W_j for every j, then $-z$ has the same distribution as z, and $\mathbb{E}[\mathrm{ReLU}(z)^2] = \frac{1}{2}\mathrm{Var}(z)$.

Proof. By Proposition 4.4, $\mathbb{E}[W_j x_j] = \mathbb{E}[W_j]\,\mathbb{E}[x_j] = 0$, and for $j \neq k$, $\mathbb{E}[W_j x_j W_k x_k] = \mathbb{E}[W_j]\,\mathbb{E}[W_k x_j x_k] = 0$, since W_j is independent of (W_k, x_j, x_k). So the terms $W_j x_j$ are uncorrelated, with $\mathrm{Var}(W_j x_j) = \mathbb{E}[W_j^2]\,\mathbb{E}[x_j^2] = vq$, and $\mathrm{Var}(z) = nvq$ by Proposition 4.9. If every W_j is symmetric, then $(-W_1, \ldots, -W_n, x_1, \ldots, x_n)$ has the same joint distribution as $(W_1, \ldots, W_n, x_1, \ldots, x_n)$, by independence, so $-z$ has the same distribution as z. Then $\mathbb{E}[z^2 \mathbf{1}_{\{z>0\}}] = \mathbb{E}[z^2 \mathbf{1}_{\{z<0\}}]$, and the two add up to $\mathbb{E}[z^2]$, so $\mathbb{E}[\mathrm{ReLU}(z)^2] = \mathbb{E}[z^2 \mathbf{1}_{\{z>0\}}] = \frac{1}{2}\mathbb{E}[z^2] = \frac{1}{2}\mathrm{Var}(z)$. □

Note that $q = \mathbb{E}[x_j^2]$ is the second moment, not the variance: the outputs of a ReLU layer are nonnegative and do not have mean 0, but only their second moments matter.

Example 4.13 (Xavier and He initialization)**.** Consider a network of ReLU layers, where layer l has n_{l-1} inputs and weights with mean 0 and variance v_l, drawn independently and symmetrically. By Proposition 4.12, the second moments q_l of the outputs of layer l satisfy $q_l = \frac{1}{2} n_{l-1} v_l q_{l-1}$, so after L layers,

$$q_L = q_0 \prod_{l=1}^{L} \frac{n_{l-1} v_l}{2}.$$

With $v_l = \frac{2}{n_{l-1}}$ (He initialization), every factor is 1, and the signal keeps its size through any number of layers. With $v_l = \frac{1}{n_{l-1}}$, it is halved in every layer, and after 50 layers, it is multiplied by $2^{-50} \approx 9 \cdot 10^{-16}$: it vanishes. With standard normal weights ($v_l = 1$) and 512 units per layer, it is multiplied by 256 in every layer, and explodes. The same computation for backpropagation, which multiplies the gradients by $W_l^\top$ and sums over the n_l outputs of layer l (Lecture 3), asks for $n_l v_l \approx 2$; for activations such as tanh, which are linear near 0 (no factor $\frac{1}{2}$), the forward and backward conditions are $n_{l-1} v_l = 1$ and $n_l v_l = 1$, and Xavier (Glorot) initialization compromises with $v_l = \frac{2}{n_{l-1} + n_l}$. For weights that are uniform on $[-b, b]$, the variance is $\frac{b^2}{3}$, so the uniform He initialization uses $b = \sqrt{6/n_{l-1}}$. These are the default initializations of deep learning libraries, and normalization layers such as batch normalization keep the scale under control during training, by standardizing z.

4.2 Conditional expectation

Definition 4.14 (Conditional expectation)**.** The conditional expectation of Y given $X = x$ is the mean of the conditional distribution,

$$m(x) = \mathbb{E}[Y \mid X = x] = \sum_y y\, p(y \mid x) \qquad \text{or} \qquad \int_{-\infty}^{\infty} y\, p(y \mid x)\, dy.$$

The random variable $\mathbb{E}[Y \mid X] = m(X)$ is the conditional expectation of Y given X.

Theorem 4.15. *1. (Tower property.)* $\mathbb{E}\big[\mathbb{E}[Y \mid X]\big] = \mathbb{E}[Y]$.

2. (Best prediction.) For every function g, $\mathbb{E}[(Y - g(X))^2] \geq \mathbb{E}[(Y - m(X))^2]$, *with* $m(x) = \mathbb{E}[Y \mid X = x]$. *More precisely,*

$$\mathbb{E}[(Y - g(X))^2] = \mathbb{E}[(Y - m(X))^2] + \mathbb{E}[(m(X) - g(X))^2].$$

Proof. We give the proof for discrete random variables. (1) $\mathbb{E}[m(X)] = \sum_x p_X(x) \sum_y y\, p(y \mid x) = \sum_{x,y} y\, p(x, y) = \mathbb{E}[Y]$. (2) For every function h, the error $Y - m(X)$ is orthogonal to $h(X)$:

$$\mathbb{E}[(Y - m(X))h(X)] = \sum_x p_X(x)h(x) \sum_y (y - m(x))p(y \mid x) = \sum_x p_X(x)h(x)\big(m(x) - m(x)\big) = 0.$$

Now write $Y - g(X) = (Y - m(X)) + (m(X) - g(X))$, expand the square, and use this with $h = m - g$; the mixed term vanishes, as in the theorem of Pythagoras. □

So $\mathbb{E}[Y \mid X]$ is the best prediction of Y from X in the sense of the mean squared error: it is the orthogonal projection of Y onto the space of functions of X, with respect to the inner product $\mathbb{E}[UV]$ (compare with orthogonal projections in Lecture 2). A regression model trained with the squared loss on enough data therefore estimates the conditional mean $\mathbb{E}[Y \mid X = \mathbf{x}]$. In Example 4.3, $\mathbb{E}[Y \mid X = 0] = 0.4$ and $\mathbb{E}[Y \mid X = 1] = 0.8$, and indeed $0.5 \cdot 0.4 + 0.5 \cdot 0.8 = 0.6 = \mathbb{E}[Y]$.

Example 4.16. 1. (Symmetry.) Let $X_1, \ldots, X_n$ be i.i.d., and $S = X_1 + \cdots + X_n$. The pairs (X_i, S) all have the same joint distribution, so the $\mathbb{E}[X_i \mid S]$ are the same function of S, and they add up to $\mathbb{E}[S \mid S] = S$. Hence $\mathbb{E}[X_1 \mid S] = \frac{S}{n}$. For example, the expected value of the first of two dice, given that their sum is 8, is 4.

2. (Law of total variance.) By part (2) of Theorem 4.15, with the constant function $g = \mathbb{E}[Y]$,

$$\mathrm{Var}(Y) = \mathbb{E}\big[(Y - m(X))^2\big] + \mathbb{E}\big[(m(X) - \mathbb{E}[Y])^2\big].$$

By the tower property, $\mathbb{E}[m(X)] = \mathbb{E}[Y]$, so the second term is $\mathrm{Var}(\mathbb{E}[Y \mid X])$. Summing first over y, as in the proof of the tower property, shows that the first term is $\mathbb{E}[v(X)]$, where $v(x)$ is the variance of the conditional distribution of Y given $X = x$. With the conditional variance $\mathrm{Var}(Y \mid X) = v(X)$, this is the law of total variance

$$\mathrm{Var}(Y) = \mathbb{E}\big[\,\mathrm{Var}(Y \mid X)\big] + \mathrm{Var}\left(\mathbb{E}[Y \mid X]\right) :$$

the variance of Y is the mean of the variances within the groups $\{X = x\}$ plus the variance of the group means. For example, let the number of customers of a shop on a day be $N \sim \mathrm{Poisson}(\lambda)$, and let customer i spend Y_i, where the Y_i are i.i.d. with mean μ and variance σ^2, and independent of N. The revenue $T = Y_1 + \cdots + Y_N$ has $\mathbb{E}[T \mid N] = N\mu$ and $\mathrm{Var}(T \mid N) = N\sigma^2$, so $\mathbb{E}[T] = \lambda\mu$ and $\mathrm{Var}(T) = \mathbb{E}[N]\sigma^2 + \mathrm{Var}(N)\mu^2 = \lambda(\sigma^2 + \mu^2)$. For $\lambda = 50$, $\mu = 20$ and $\sigma^2 = 100$, the revenue has the mean 1000 and the standard deviation $\sqrt{25\,000} \approx 158$.

4.3 Conditional independence and graphical models

Many models in machine learning are built from the assumption that some variables are independent once others are known.

Definition 4.17 (Conditional independence)**.** Random variables X and Y are conditionally independent given Z, written $X \perp\!\!\!\perp Y \mid Z$, if

$$p(x, y \mid z) = p(x \mid z)\, p(y \mid z) \qquad \text{for all } x, y \text{ and all } z \text{ with } p_Z(z) > 0.$$

Equivalently, $p(x \mid y, z) = p(x \mid z)$ whenever $p(y, z) > 0$: once Z is known, Y carries no further information about X. Indeed, $p(x \mid y, z) = \frac{p(x,y,z)}{p(y,z)} = \frac{p(x,y|z)}{p(y|z)}$, which equals $p(x \mid z)$ if and only if $p(x, y \mid z) = p(x \mid z)p(y \mid z)$. Conditional independence and independence are different properties.

- (A common cause.) Choose one of two coins with probability $\frac{1}{2}$ each, a fair one ($Z = 0$) and one with heads probability 0.9 ($Z = 1$), and let X and Y indicate heads in two flips of the chosen coin. Then $X \perp\!\!\!\perp Y \mid Z$. But $P(X = 1) = \frac{1}{2}(0.5 + 0.9) = 0.7$ and $P(X = 1, Y = 1) = \frac{1}{2}(0.25 + 0.81) = 0.53 \neq 0.7^2$, so X and Y are dependent: heads in the first flip makes the biased coin more likely, and $P(Y = 1 \mid X = 1) = \frac{0.53}{0.7} \approx 0.76$. Bayesian inference has this structure, with the parameter θ in the role of Z: the data are i.i.d. given θ, and we learn about future data from past data.

- (Explaining away.) Let X and Y be independent fair bits, and $Z = X + Y$. Then $X \perp\!\!\!\perp Y$, but not given Z: $P(X = 1 \mid Z = 1) = \frac{1}{2}$, while $P(X = 1 \mid Y = 1, Z = 1) = 0$. If an alarm ($Z$) can be set off by a burglary (X) or an earthquake (Y), then learning about the earthquake explains the alarm away and makes the burglary less likely.

- The defining property of a Markov chain (Section 8) is that the future is independent of the past given the present.

Applying the product rule repeatedly gives the chain rule of probability,

$$p(x_1, \dots, x_n) = p(x_1)\, p(x_2 \mid x_1)\, p(x_3 \mid x_1, x_2) \cdots p(x_n \mid x_1, \dots, x_{n-1}).$$

A directed graphical model, or Bayesian network, assumes that each variable only depends on a few of the earlier ones, its parents $\mathrm{pa}(i) \subseteq \{1, \dots, i-1\}$:

$$p(x_1, \dots, x_n) = \prod_{i=1}^{n} p(x_i \mid x_{\mathrm{pa}(i)}).$$

It is drawn as a graph with an arrow from each parent to its child. The factorization saves parameters: a general distribution of n binary variables has $2^n - 1$ free parameters, but if every variable has at most k parents, the conditional tables have at most $n2^k$ entries. Such factorizations are equivalent to conditional independence statements; the three basic patterns are the following.

Proposition 4.18. *Let X, Y, Z have a joint pmf or density of one of the following forms.*

1. *(Chain $X \to Z \to Y$.) $p(x, y, z) = p(x)\, p(z \mid x)\, p(y \mid z)$. Then $X \perp\!\!\!\perp Y \mid Z$.*
2. *(Fork $X \leftarrow Z \to Y$.) $p(x, y, z) = p(z)\, p(x \mid z)\, p(y \mid z)$. Then $X \perp\!\!\!\perp Y \mid Z$.*
3. *(Collider $X \to Z \leftarrow Y$.) $p(x, y, z) = p(x)\, p(y)\, p(z \mid x, y)$. Then $X \perp\!\!\!\perp Y$, but in general not given Z.*

Proof. (1) By Bayes' theorem, $p(x)p(z \mid x) = p_Z(z)p(x \mid z)$, so $p(x, y \mid z) = \frac{p(x,y,z)}{p_Z(z)} = p(x \mid z)p(y \mid z)$. (2) Divide by $p(z)$. (3) Summing over z gives $p(x, y) = p(x)p(y) \sum_z p(z \mid x, y) = p(x)p(y)$; the explaining-away example above shows that X and Y can be dependent given Z. □

Many models of this lecture are graphical models: a mixture model $z \to \mathbf{x}$ (Section 14), Bayesian inference $\theta \to x_1, \dots, x_N$ (a fork with N branches), a hidden Markov model, in which hidden states form a Markov chain $z_1 \to z_2 \to \cdots$ and each z_t emits an observation x_t, and the following classifier.

Example 4.19 (Naive Bayes)**.** A naive Bayes classifier predicts a class $y \in \{1, \dots, K\}$ from features $x_1, \dots, x_d$ under the assumption that the features are conditionally independent given the class: $p(\mathbf{x}, y) = p(y) \prod_{j=1}^{d} p(x_j \mid y)$. By Bayes' theorem,

$$p(y \mid \mathbf{x}) \propto p(y) \prod_{j=1}^{d} p(x_j \mid y),$$

and maximum likelihood estimates each factor by counting. For a spam filter, let $P(\text{spam}) = 0.4$, let the word "free" appear in 30% of spam and 2% of other emails, and the word "meeting" in 1% of spam and 10% of other emails. An email that contains "free" but not "meeting" has the posterior odds

$$\frac{P(\text{spam} \mid \mathbf{x})}{P(\text{other} \mid \mathbf{x})} = \frac{0.4}{0.6} \cdot \frac{0.3}{0.02} \cdot \frac{0.99}{0.9} = 11,$$

so $P(\text{spam} \mid \mathbf{x}) = \frac{11}{12} \approx 0.92$. With binary features $x_j \in \{0, 1\}$ and two classes, the log-odds

$$\log \frac{p(y = 1 \mid \mathbf{x})}{p(y = 0 \mid \mathbf{x})} = \log \frac{p(y = 1)}{p(y = 0)} + \sum_{j=1}^{d} \Big(x_j \log \frac{p(x_j = 1 \mid y = 1)}{p(x_j = 1 \mid y = 0)} + (1 - x_j) \log \frac{p(x_j = 0 \mid y = 1)}{p(x_j = 0 \mid y = 0)} \Big)$$

are an affine function of $\mathbf{x}$, as for logistic regression (Lecture 3). Naive Bayes is a generative model, which fits $p(\mathbf{x}, y)$ by counting, while logistic regression is the corresponding discriminative model, which fits $p(y \mid \mathbf{x})$ directly; naive Bayes needs less data, but its probabilities are overconfident when the features are dependent. A word that never appears in the spam of the training data would get $p(x_j = 1 \mid \text{spam}) = 0$ and veto the class; adding a pseudo-count to all counts (Laplace smoothing, a MAP estimate with a Beta prior, Section 12) avoids this.

4.4 Moment generating functions

The Chernoff bound (Example 3.8) used the expectation of e^{tX}. As a function of t, it contains all the moments of X, and it turns sums of independent random variables into products.

Definition 4.20 (Moment generating function)**.** The moment generating function (MGF) of a random variable X is $M_X(t) = \mathbb{E}[e^{tX}]$, for all $t \in \mathbb{R}$ for which this expectation is finite. Its logarithm $K_X(t) = \log M_X(t)$ is the cumulant generating function.

Always $M_X(0) = 1$. For a Bernoulli(p) variable, $M_X(t) = 1 - p + pe^t$; for $X \sim \text{Poisson}(\lambda)$, $M_X(t) = \sum_k e^{tk} e^{-\lambda} \frac{\lambda^k}{k!} = e^{-\lambda} e^{\lambda e^t} = e^{\lambda(e^t - 1)}$; for $X \sim \text{Exponential}(\lambda)$, $M_X(t) = \int_0^\infty e^{tx} \lambda e^{-\lambda x}\, dx = \frac{\lambda}{\lambda - t}$ for $t < \lambda$ (and ∞ for $t \geq \lambda$); and for $X \sim \mathcal{N}(\mu, \sigma^2)$, completing the square, $tx - \frac{(x-\mu)^2}{2\sigma^2} = \mu t + \frac{\sigma^2 t^2}{2} - \frac{(x - \mu - \sigma^2 t)^2}{2\sigma^2}$, so

$$M_X(t) = e^{\mu t + \sigma^2 t^2/2} \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi\sigma^2}} e^{-(x - \mu - \sigma^2 t)^2/(2\sigma^2)}\, dx = e^{\mu t + \sigma^2 t^2/2}, \qquad K_X(t) = \mu t + \frac{\sigma^2 t^2}{2}.$$

Proposition 4.21. *1.* $M_{aX+b}(t) = e^{bt}M_X(at)$.

2. If X and Y are independent, then $M_{X+Y}(t) = M_X(t)M_Y(t)$ and $K_{X+Y} = K_X + K_Y$.

3. If $M_X(t)$ is finite for all t in an interval $(-t_0, t_0)$ with $t_0 > 0$, then M_X has derivatives of all orders on $(-t_0, t_0)$, with $M_X^{(k)}(t) = \mathbb{E}[X^k e^{tX}]$. In particular, $\mathbb{E}[X^k] = M_X^{(k)}(0)$, and $K_X'(0) = \mathbb{E}[X]$, $K_X''(0) = \mathrm{Var}(X)$.

Proof. (1) $\mathbb{E}[e^{t(aX+b)}] = e^{bt}\mathbb{E}[e^{(at)X}]$. (2) By Proposition 4.4, $\mathbb{E}[e^{tX}e^{tY}] = \mathbb{E}[e^{tX}]\mathbb{E}[e^{tY}]$.

(3) Let $|t| < t_0$, and choose a with $|t| < a < t_0$; let $s = \frac{1}{2}(a - |t|)$, and $0 < |h| \le s$. For every $k \ge 0$, the number $c_k = \max_{x\in\mathbb{R}} |x|^{k+2}e^{-s|x|}$ is finite. By Taylor's theorem (Lecture 3), $|e^u - 1 - u| \le \frac{1}{2}u^2 e^{|u|}$, so for all x,

$$\begin{aligned}\left|x^k\big(e^{(t+h)x} - e^{tx} - hxe^{tx}\big)\right| &= |x|^k e^{tx}\left|e^{hx} - 1 - hx\right| \le \frac{h^2}{2}|x|^{k+2}e^{(|t|+s)|x|} \\ &\le \frac{h^2 c_k}{2}\, e^{a|x|} \le \frac{h^2 c_k}{2}\big(e^{ax} + e^{-ax}\big).\end{aligned}$$

Similarly, $|x^k e^{tx}| \le c_k'(e^{ax} + e^{-ax})$ with $c_k' = \max_x |x|^k e^{-(a-|t|)|x|}$, so $m_k(t) = \mathbb{E}[X^k e^{tX}]$ is finite. Taking expectations, $|m_k(t+h) - m_k(t) - h\, m_{k+1}(t)| \le \frac{h^2 c_k}{2}\big(M_X(a) + M_X(-a)\big)$, so m_k is differentiable with $m_k' = m_{k+1}$. Since $m_0 = M_X$, induction gives $M_X^{(k)} = m_k$. Finally, $K_X' = \frac{M_X'}{M_X}$ and $K_X'' = \frac{M_X''}{M_X} - \big(\frac{M_X'}{M_X}\big)^2$, which at $t = 0$ give $\mathbb{E}[X]$ and $\mathbb{E}[X^2] - \mathbb{E}[X]^2$. □

For example, for $X \sim \mathrm{Poisson}(\lambda)$, $K_X(t) = \lambda(e^t - 1)$ has $K_X'(0) = K_X''(0) = \lambda$: the mean and the variance are both λ. For $S \sim \mathrm{Binomial}(n, p)$, a sum of n independent Bernoulli variables, $M_S(t) = (1 - p + pe^t)^n$, and $K_S(t) = n\log(1 - p + pe^t)$ gives $\mathbb{E}[S] = np$ and $\mathrm{Var}(S) = np(1-p)$. Cumulant generating functions of independent variables add, just like means and variances. This gives a heuristic explanation of the central limit theorem: for i.i.d. standardized X_i (with mean 0 and variance 1), $K_{(X_1+\cdots+X_N)/\sqrt{N}}(t) = NK_X(t/\sqrt{N}) \approx N\frac{t^2}{2N} = \frac{t^2}{2}$ by Taylor's theorem, since $K_X(0) = K_X'(0) = 0$ and $K_X''(0) = 1$, and $\frac{t^2}{2}$ is the cumulant generating function of $\mathcal{N}(0,1)$.

Chernoff bounds. By Markov's inequality, $P(X \ge a) \le e^{-ta}M_X(t) = e^{-(ta - K_X(t))}$ for every $t > 0$, so

$$P(X \ge a) \le \exp\Big(-\sup_{t>0}\big(ta - K_X(t)\big)\Big).$$

For $X \sim \mathcal{N}(0, \sigma^2)$ and $a > 0$, the supremum of $ta - \frac{\sigma^2 t^2}{2}$ is $\frac{a^2}{2\sigma^2}$, attained at $t = \frac{a}{\sigma^2}$, so $P(X \ge a) \le e^{-a^2/(2\sigma^2)}$. Random variables with $K_X(t) \le \frac{\sigma^2 t^2}{2}$ for all t, i.e., with tails at least as light as those of $\mathcal{N}(0, \sigma^2)$, are called sub-Gaussian. Hoeffding's lemma (Lemma 7.6) says that a bounded variable with mean 0 and values in $[a, b]$ is sub-Gaussian with $\sigma^2 = \frac{(b-a)^2}{4}$, and the proof of Hoeffding's inequality adds the cumulant generating functions of the summands.

5 Random Vectors and Covariance Matrices

A random vector $\mathbf{X} = (X_1, \ldots, X_d)^\top$ is a vector of random variables, such as the features of a randomly chosen data point. Its expectation is taken entrywise.

Definition 5.1 (Mean and covariance matrix)**.** The mean of a random vector $\mathbf{X}$ is $\boldsymbol{\mu} = \mathbb{E}[\mathbf{X}] = (\mathbb{E}[X_1], \dots, \mathbb{E}[X_d])^\top$, and its covariance matrix is the $d \times d$ matrix

$$\Sigma = \operatorname{Cov}(\mathbf{X}) = \mathbb{E}\big[(\mathbf{X} - \boldsymbol{\mu})(\mathbf{X} - \boldsymbol{\mu})^\top\big], \qquad \text{with entries } \Sigma_{ij} = \operatorname{Cov}(X_i, X_j).$$

The diagonal entries of Σ are the variances $\operatorname{Var}(X_i)$. Since $(\mathbf{X}-\boldsymbol{\mu})(\mathbf{X}-\boldsymbol{\mu})^\top$ is an outer product, linear algebra enters immediately.

Proposition 5.2. *Let $\mathbf{X}$ have mean $\boldsymbol{\mu}$ and covariance matrix Σ.*

1. $\Sigma = \mathbb{E}[\mathbf{X}\mathbf{X}^\top] - \boldsymbol{\mu}\boldsymbol{\mu}^\top$.
2. *For every matrix $A \in \mathbb{R}^{m\times d}$ and vector $\mathbf{b} \in \mathbb{R}^m$, $\mathbb{E}[A\mathbf{X} + \mathbf{b}] = A\boldsymbol{\mu} + \mathbf{b}$ and $\operatorname{Cov}(A\mathbf{X} + \mathbf{b}) = A\Sigma A^\top$.*
3. *For every $\mathbf{u} \in \mathbb{R}^d$, $\operatorname{Var}(\mathbf{u}^\top\mathbf{X}) = \mathbf{u}^\top\Sigma\mathbf{u}$. In particular, Σ is symmetric and positive semidefinite.*

Proof. (2) Each entry of $A\mathbf{X} + \mathbf{b}$ is a linear combination of the X_j plus a constant, so linearity of expectation gives $\mathbb{E}[A\mathbf{X} + \mathbf{b}] = A\boldsymbol{\mu} + \mathbf{b}$, and in the same way $\mathbb{E}[AMB] = A\,\mathbb{E}[M]\,B$ for a random matrix M and fixed matrices A and B. Hence

$$\operatorname{Cov}(A\mathbf{X} + \mathbf{b}) = \mathbb{E}\big[A(\mathbf{X} - \boldsymbol{\mu})(\mathbf{X} - \boldsymbol{\mu})^\top A^\top\big] = A\Sigma A^\top.$$

(1) is proved like the formula $\operatorname{Var}(X) = \mathbb{E}[X^2] - \mathbb{E}[X]^2$. (3) is (2) with $A = \mathbf{u}^\top$, and $\operatorname{Var}(\mathbf{u}^\top\mathbf{X}) \geq 0$; Σ is symmetric since $\operatorname{Cov}(X_i, X_j) = \operatorname{Cov}(X_j, X_i)$. □

By (3), Σ fails to be positive definite exactly if $\operatorname{Var}(\mathbf{u}^\top\mathbf{X}) = 0$ for some $\mathbf{u} \neq \mathbf{0}$, i.e., if $\mathbf{u}^\top\mathbf{X}$ is constant: then $\mathbf{X}$ lies in an affine hyperplane (with probability 1). Part (3) also describes the variance in every direction: for a unit vector $\mathbf{u}$, the variance of the projection $\mathbf{u}^\top\mathbf{X}$ is $\mathbf{u}^\top\Sigma\mathbf{u}$. By the Rayleigh quotient (Lecture 2), it is largest for an eigenvector of Σ with the largest eigenvalue. This is the population version of principal component analysis, and the total variance $\sum_i \operatorname{Var}(X_i) = \operatorname{tr}\Sigma$ is the sum of the eigenvalues of Σ.

Example 5.3. Let $\Sigma = \begin{pmatrix} 4 & 2 \\ 2 & 3 \end{pmatrix}$. Then $\operatorname{Var}(X_1 + X_2) = (1,1)\Sigma(1,1)^\top = 4 + 3 + 2\cdot 2 = 11$, $\operatorname{Var}(X_1 - X_2) = 4 + 3 - 4 = 3$, and $\operatorname{Corr}(X_1, X_2) = \frac{2}{\sqrt{12}} \approx 0.58$. The eigenvalues of Σ are $\frac{7 \pm \sqrt{17}}{2} \approx 5.56$ and 1.44: the variance in the direction of the first principal component is 5.56, out of the total variance $\operatorname{tr}\Sigma = 7$.

Remark 5.4 (Moments as tensors). In the language of the section on tensors in Lecture 1, $(\mathbf{X} - \boldsymbol{\mu})(\mathbf{X}-\boldsymbol{\mu})^\top = (\mathbf{X}-\boldsymbol{\mu}) \otimes (\mathbf{X}-\boldsymbol{\mu})$, so the covariance matrix is the expectation of a random tensor in $\mathbb{R}^d \otimes \mathbb{R}^d$. Part 2 of Proposition 5.2 says that it transforms like such a tensor: in Einstein notation, $\operatorname{Cov}(A\mathbf{X})_{ab} = A_{ai}A_{bj}\Sigma_{ij}$, which is $A\Sigma A^\top$. Higher moments are tensors of higher order. The third central moment $T = \mathbb{E}\big[(\mathbf{X} - \boldsymbol{\mu}) \otimes (\mathbf{X} - \boldsymbol{\mu}) \otimes (\mathbf{X} - \boldsymbol{\mu})\big]$ has the entries $T_{ijk} = \mathbb{E}[(X_i - \mu_i)(X_j - \mu_j)(X_k - \mu_k)]$, it changes to $A_{ai}A_{bj}A_{ck}T_{ijk}$ under $\mathbf{X} \mapsto A\mathbf{X}$, and it is 0 for every distribution that is symmetric about its mean, such as a Gaussian (since $\mathbf{X} - \boldsymbol{\mu}$ and $\boldsymbol{\mu} - \mathbf{X}$ have the same distribution, $T = -T$). Moment tensors are computed with `np.einsum`, and their decompositions into sums of pure tensors (the tensor rank of Lecture 1) are used to estimate the parameters of mixture models.

Definition 5.5 (Whitening and Mahalanobis distance)**.** Let Σ be symmetric positive definite, with spectral decomposition $\Sigma = Q\Lambda Q^\top$ (Lecture 2), and let $\Sigma^{-1/2} = Q\Lambda^{-1/2}Q^\top$, where $\Lambda^{-1/2} = \operatorname{diag}(\lambda_1^{-1/2}, \dots, \lambda_d^{-1/2})$. The whitened random vector is $\mathbf{Z} = \Sigma^{-1/2}(\mathbf{X} - \boldsymbol{\mu})$, and the Mahalanobis distance of $\mathbf{x}$ from $\boldsymbol{\mu}$ is

$$d_\Sigma(\mathbf{x}, \boldsymbol{\mu}) = \sqrt{(\mathbf{x} - \boldsymbol{\mu})^\top \Sigma^{-1} (\mathbf{x} - \boldsymbol{\mu})} = \|\Sigma^{-1/2}(\mathbf{x} - \boldsymbol{\mu})\|.$$

The matrix $\Sigma^{-1/2}$ is symmetric with $(\Sigma^{-1/2})^2 = \Sigma^{-1}$, so by Proposition 5.2, $\operatorname{Cov}(\mathbf{Z}) = \Sigma^{-1/2}\Sigma\Sigma^{-1/2} = I$: whitening removes all correlations and scales all directions to unit variance. The Mahalanobis distance is the norm of the inner product $\langle \mathbf{x}, \mathbf{y} \rangle_{\Sigma^{-1}} = \mathbf{x}^\top \Sigma^{-1} \mathbf{y}$ (Lecture 2): it measures distances in units of standard deviations along each principal axis. Standardizing or whitening the features of a data set is a common preprocessing step; by Lecture 3, it improves the condition number and speeds up gradient descent.

Sample statistics. Given data points $\mathbf{x}_1, \dots, \mathbf{x}_N \in \mathbb{R}^d$, the sample mean and the sample covariance matrix are

$$\bar{\mathbf{x}} = \frac{1}{N} \sum_{i=1}^{N} \mathbf{x}_i \qquad \text{and} \qquad S = \frac{1}{N} \sum_{i=1}^{N} (\mathbf{x}_i - \bar{\mathbf{x}})(\mathbf{x}_i - \bar{\mathbf{x}})^\top = \frac{1}{N} X_c^\top X_c,$$

where X_c is the centered data matrix with rows $(\mathbf{x}_i - \bar{\mathbf{x}})^\top$. These are the mean and covariance matrix of the empirical distribution, which gives probability $\frac{1}{N}$ to each data point; principal component analysis in Lecture 2 used exactly this matrix. In Section 9, we will see why one often divides by $N - 1$ instead of N.

6 The Gaussian Distribution

The Gaussian (normal) distribution is the most important distribution in statistics and machine learning. It appears as the limit of sums of many small random effects (Section 7), it is completely described by its mean and covariance matrix, and its marginals and conditionals can be computed with linear algebra.

Definition 6.1 (Multivariate Gaussian distribution)**.** Let $\boldsymbol{\mu} \in \mathbb{R}^d$, and let Σ be a symmetric positive definite $d \times d$ matrix. A random vector $\mathbf{X}$ has the Gaussian distribution $\mathcal{N}(\boldsymbol{\mu}, \Sigma)$ if it has the density

$$p(\mathbf{x}) = \frac{1}{(2\pi)^{d/2} \det(\Sigma)^{1/2}} \exp\Big(-\frac{1}{2} (\mathbf{x} - \boldsymbol{\mu})^\top \Sigma^{-1} (\mathbf{x} - \boldsymbol{\mu}) \Big).$$

For $d = 1$ and $\Sigma = \sigma^2$, this is the density of Section 2. For $\boldsymbol{\mu} = \mathbf{0}$ and $\Sigma = I$, the density is $(2\pi)^{-d/2} e^{-\|\mathbf{z}\|^2/2} = \prod_i \varphi(z_i)$: a standard Gaussian vector $\mathbf{Z} \sim \mathcal{N}(\mathbf{0}, I)$ has i.i.d. standard normal entries. All other Gaussians are obtained from it by affine maps.

Theorem 6.2. *Let $\mathbf{Z} \sim \mathcal{N}(\mathbf{0}, I)$, let A be an invertible $d \times d$ matrix, and let $\boldsymbol{\mu} \in \mathbb{R}^d$. Then $\mathbf{X} = A\mathbf{Z} + \boldsymbol{\mu} \sim \mathcal{N}(\boldsymbol{\mu}, AA^\top)$. Consequently, the density of Definition 6.1 integrates to 1, and $\mathcal{N}(\boldsymbol{\mu}, \Sigma)$ has mean $\boldsymbol{\mu}$ and covariance matrix Σ.*

Proof. By the change of variables formula (Lecture 3), with $\mathbf{z} = A^{-1}(\mathbf{x} - \boldsymbol{\mu})$, whose Jacobian has the determinant $\det(A)^{-1}$,

$$p_{\mathbf{X}}(\mathbf{x}) = p_{\mathbf{Z}}\big(A^{-1}(\mathbf{x} - \boldsymbol{\mu})\big) \, \frac{1}{|\det A|} = \frac{1}{(2\pi)^{d/2} |\det A|} \exp\Big(-\frac{1}{2} \|A^{-1}(\mathbf{x} - \boldsymbol{\mu})\|^2 \Big).$$

With $\Sigma = AA^\top$, we have $\|A^{-1}\mathbf{y}\|^2 = \mathbf{y}^\top A^{-\top}A^{-1}\mathbf{y} = \mathbf{y}^\top\Sigma^{-1}\mathbf{y}$ and $\det\Sigma = \det(A)^2$, which gives the density of $\mathcal{N}(\boldsymbol{\mu},\Sigma)$. Every symmetric positive definite Σ can be written as $AA^\top$, for example with $A = \Sigma^{1/2} = Q\Lambda^{1/2}Q^\top$. So every Gaussian density is the density of some random vector, and integrates to 1. By Proposition 5.2, $\mathbb{E}[\mathbf{X}] = A\mathbb{E}[\mathbf{Z}] + \boldsymbol{\mu} = \boldsymbol{\mu}$ and $\mathrm{Cov}(\mathbf{X}) = A\,I\,A^\top = \Sigma$. □

This is also how Gaussian vectors are sampled: compute a factorization $\Sigma = AA^\top$ (in practice, the Cholesky factorization $\Sigma = LL^\top$ with a lower triangular L, as in `np.linalg.cholesky`), draw $\mathbf{z}$ with i.i.d. standard normal entries, and return $\boldsymbol{\mu} + L\mathbf{z}$. If $\mathbf{X} \sim \mathcal{N}(\boldsymbol{\mu},\Sigma)$ and B is invertible, then $B\mathbf{X} + \mathbf{b} = (BA)\mathbf{Z} + (B\boldsymbol{\mu} + \mathbf{b})$, so $B\mathbf{X} + \mathbf{b} \sim \mathcal{N}(B\boldsymbol{\mu} + \mathbf{b}, B\Sigma B^\top)$.

Remark 6.3 (The reparameterization trick). Writing a Gaussian random vector as $\boldsymbol{\mu} + \boldsymbol{\sigma} \odot \mathbf{Z}$ with a fixed source of randomness $\mathbf{Z}$ makes it a differentiable function of the parameters $\boldsymbol{\mu}$ and $\boldsymbol{\sigma}$. Variational autoencoders use this to backpropagate (Lecture 3) through random sampling steps.

The density of $\mathcal{N}(\boldsymbol{\mu},\Sigma)$ is constant on the ellipsoids $(\mathbf{x}-\boldsymbol{\mu})^\top\Sigma^{-1}(\mathbf{x}-\boldsymbol{\mu}) = r^2$, i.e., on the sets of points with Mahalanobis distance r from $\boldsymbol{\mu}$. By the spectral theorem (Lecture 2), their axes point in the directions of the eigenvectors $\mathbf{q}_i$ of Σ, with half-lengths $r\sqrt{\lambda_i}$, i.e., r standard deviations in the direction $\mathbf{q}_i$ (Figure 2).

Theorem 6.4 (Independence of blocks)**.** *Let* $\mathbf{X} = (\mathbf{X}_a, \mathbf{X}_b) \sim \mathcal{N}(\boldsymbol{\mu},\Sigma)$*, and let* $\Sigma = \left(\begin{smallmatrix}\Sigma_{aa} & 0\\ 0 & \Sigma_{bb}\end{smallmatrix}\right)$ *be block diagonal. Then* $\mathbf{X}_a$ *and* $\mathbf{X}_b$ *are independent, with* $\mathbf{X}_a \sim \mathcal{N}(\boldsymbol{\mu}_a,\Sigma_{aa})$ *and* $\mathbf{X}_b \sim \mathcal{N}(\boldsymbol{\mu}_b,\Sigma_{bb})$*.*

Proof. Σ^{-1} is block diagonal with blocks Σ_{aa}^{-1} and Σ_{bb}^{-1}, and $\det\Sigma = \det\Sigma_{aa}\det\Sigma_{bb}$. Hence the quadratic form in the exponent splits into a sum of two terms, and the density is the product of the densities of $\mathcal{N}(\boldsymbol{\mu}_a,\Sigma_{aa})$ and $\mathcal{N}(\boldsymbol{\mu}_b,\Sigma_{bb})$. Integrating over $\mathbf{x}_b$ shows that the first factor is the marginal density of $\mathbf{X}_a$, and similarly for $\mathbf{X}_b$. □

So for jointly Gaussian random vectors, uncorrelated parts are independent. This is special to Gaussians, and it is the key to the following theorem, which is used constantly in Gaussian processes, Kalman filters and Bayesian linear regression.

Theorem 6.5 (Marginals and conditionals)**.** *Let* $\mathbf{X} = (\mathbf{X}_a, \mathbf{X}_b) \sim \mathcal{N}(\boldsymbol{\mu},\Sigma)$ *with*

$$\boldsymbol{\mu} = \begin{pmatrix}\boldsymbol{\mu}_a\\ \boldsymbol{\mu}_b\end{pmatrix}, \qquad \Sigma = \begin{pmatrix}\Sigma_{aa} & \Sigma_{ab}\\ \Sigma_{ba} & \Sigma_{bb}\end{pmatrix}.$$

Then $\mathbf{X}_a \sim \mathcal{N}(\boldsymbol{\mu}_a,\Sigma_{aa})$*,* $\mathbf{X}_b \sim \mathcal{N}(\boldsymbol{\mu}_b,\Sigma_{bb})$*, and the conditional distribution of* $\mathbf{X}_a$ *given* $\mathbf{X}_b = \mathbf{x}_b$ *is*

$$\mathcal{N}(\boldsymbol{\mu}_a + \Sigma_{ab}\Sigma_{bb}^{-1}(\mathbf{x}_b - \boldsymbol{\mu}_b),\ \Sigma_{aa} - \Sigma_{ab}\Sigma_{bb}^{-1}\Sigma_{ba}).$$

Proof. The block Σ_{bb} is positive definite (since $\mathbf{u}^\top\Sigma_{bb}\mathbf{u} = (\mathbf{0},\mathbf{u})^\top\Sigma(\mathbf{0},\mathbf{u}) > 0$ for $\mathbf{u} \neq \mathbf{0}$), so $K = \Sigma_{ab}\Sigma_{bb}^{-1}$ is defined. Let $\mathbf{W} = \mathbf{X}_a - K\mathbf{X}_b$, i.e.,

$$\begin{pmatrix}\mathbf{W}\\ \mathbf{X}_b\end{pmatrix} = B\mathbf{X}, \qquad B = \begin{pmatrix}I & -K\\ 0 & I\end{pmatrix}.$$

B is invertible, so $(\mathbf{W},\mathbf{X}_b)$ is Gaussian with covariance matrix $B\Sigma B^\top$. Its off-diagonal block is $\mathrm{Cov}(\mathbf{W},\mathbf{X}_b) = \Sigma_{ab} - K\Sigma_{bb} = 0$, and its upper left block is

$$\mathrm{Cov}(\mathbf{W}) = \Sigma_{aa} - K\Sigma_{ba} - \Sigma_{ab}K^\top + K\Sigma_{bb}K^\top = \Sigma_{aa} - \Sigma_{ab}\Sigma_{bb}^{-1}\Sigma_{ba}.$$

By Theorem 6.4, $\mathbf{X}_b \sim \mathcal{N}(\boldsymbol{\mu}_b,\Sigma_{bb})$, and $\mathbf{W}$ is independent of $\mathbf{X}_b$, with $\mathbf{W} \sim \mathcal{N}(\boldsymbol{\mu}_a - K\boldsymbol{\mu}_b, \Sigma_{aa} - \Sigma_{ab}\Sigma_{bb}^{-1}\Sigma_{ba})$. Since $\det B = 1$, the joint density of $\mathbf{X}$ is $p(\mathbf{x}_a,\mathbf{x}_b) = p_{\mathbf{W}}(\mathbf{x}_a - K\mathbf{x}_b)\,p_{\mathbf{X}_b}(\mathbf{x}_b)$, so the conditional density is $p(\mathbf{x}_a \mid \mathbf{x}_b) = p_{\mathbf{W}}(\mathbf{x}_a - K\mathbf{x}_b)$: given $\mathbf{X}_b = \mathbf{x}_b$, $\mathbf{X}_a = \mathbf{W} + K\mathbf{x}_b$, which has the claimed distribution. The marginal of $\mathbf{X}_a$ follows by exchanging the roles of a and b. □

The conditional mean is an affine function of $\mathbf{x}_b$, and the conditional covariance $\Sigma_{aa} - \Sigma_{ab}\Sigma_{bb}^{-1}\Sigma_{ba}$ (the Schur complement of Σ_{bb}, as in the subsection on block matrices in Lecture 1, with the roles of the two blocks exchanged) does not depend on $\mathbf{x}_b$. It is never larger than Σ_{aa}, since $\Sigma_{ab}\Sigma_{bb}^{-1}\Sigma_{ba}$ is positive semidefinite: observing $\mathbf{X}_b$ reduces the uncertainty about $\mathbf{X}_a$.

Example 6.6 (Regression to the mean)**.** Let (X, Y) be Gaussian with means 0, variances 1, and correlation ρ. By Theorem 6.5, given $X = x$, $Y \sim \mathcal{N}(\rho x, 1 - \rho^2)$. For example, for $\rho = 0.5$ and $x = 2$, $Y \sim \mathcal{N}(1, 0.75)$. If X and Y are the heights of parents and their children (measured in standard deviations from the mean), then children of very tall parents are on average tall, but less extreme than their parents. Francis Galton observed this "regression to the mean" in 1886, and it gave linear regression its name. Note that the regression line $y = \rho x$ is not the major axis of the ellipses of constant density, which is the line $y = x$ (Figure 2).

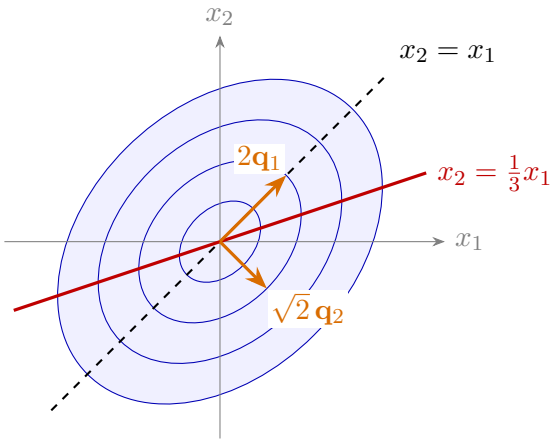


Figure 2: Level curves of the density of $\mathcal{N}(\mathbf{0}, \Sigma)$ with $\Sigma = \left(\begin{smallmatrix} 3 & 1 \\ 1 & 3 \end{smallmatrix}\right)$, for Mahalanobis distances $r = 0.5, 1, 1.5, 2$. The eigenvectors $\mathbf{q}_1 = \frac{1}{\sqrt{2}}(1, 1)$ and $\mathbf{q}_2 = \frac{1}{\sqrt{2}}(1, -1)$ of Σ have the eigenvalues 4 and 2, so the axes have half-lengths $2r$ and $\sqrt{2}\,r$ (orange: one standard deviation in each direction). The conditional mean $\mathbb{E}[X_2 \mid X_1 = x_1] = \frac{\Sigma_{21}}{\Sigma_{11}}x_1 = \frac{1}{3}x_1$ (red) is flatter than the major axis $x_2 = x_1$ (dashed).

Corollary 6.7. *Let $\mathbf{X} \sim \mathcal{N}(\boldsymbol{\mu}, \Sigma)$, and let $A \in \mathbb{R}^{m \times d}$ have rank m. Then $A\mathbf{X} + \mathbf{b} \sim \mathcal{N}(A\boldsymbol{\mu} + \mathbf{b}, A\Sigma A^\top)$. In particular, if $X \sim \mathcal{N}(\mu_1, \sigma_1^2)$ and $Y \sim \mathcal{N}(\mu_2, \sigma_2^2)$ are independent, then $X + Y \sim \mathcal{N}(\mu_1 + \mu_2, \sigma_1^2 + \sigma_2^2)$.*

Proof. The rows of A are linearly independent, so they can be extended to a basis of $\mathbb{R}^d$ (Lecture 1): there is a matrix C such that $B = \left(\begin{smallmatrix} A \\ C \end{smallmatrix}\right)$ is invertible. Then $B\mathbf{X} + (\mathbf{b}, \mathbf{0})$ is Gaussian, and $A\mathbf{X} + \mathbf{b}$ is its first block, which is Gaussian by Theorem 6.5, with the mean and covariance of Proposition 5.2. For the second statement, (X, Y) is Gaussian with the covariance matrix $\operatorname{diag}(\sigma_1^2, \sigma_2^2)$ (its density is the product of the two densities), and $X + Y = (1, 1)(X, Y)^\top$. □

7 Limit Theorems and Monte Carlo Methods

Machine learning relies on averages: the training loss is an average over the data, a mini-batch gradient is an average over a few examples, and the test accuracy is an average over the test set. The limit theorems explain why averages of many random quantities are reliable.

Proposition 7.1. *Let $X_1, \dots, X_N$ be i.i.d. with mean μ and variance σ^2, and let $\bar{X}_N = \frac{1}{N}(X_1 + \cdots + X_N)$ be their sample mean. Then*

$$\mathbb{E}[\bar{X}_N] = \mu \qquad \textit{and} \qquad \mathrm{Var}(\bar{X}_N) = \frac{\sigma^2}{N}.$$

Proof. By linearity, $\mathbb{E}[\bar{X}_N] = \frac{1}{N} \cdot N\mu$, and by Proposition 4.9, $\mathrm{Var}(\bar{X}_N) = \frac{1}{N^2} \cdot N\sigma^2$. □

So the standard deviation of the sample mean, its standard error, is $\frac{\sigma}{\sqrt{N}}$: to halve the error, we need four times as much data.

Theorem 7.2 (Weak law of large numbers)**.** *In the situation of Proposition 7.1, for every $\varepsilon > 0$,*

$$P(|\bar{X}_N - \mu| \geq \varepsilon) \leq \frac{\sigma^2}{N\varepsilon^2} \to 0 \qquad \textit{as } N \to \infty.$$

Proof. Chebyshev's inequality (Proposition 3.6) for $\bar{X}_N$, which has mean μ and variance $\frac{\sigma^2}{N}$. □

The law of large numbers says that the sample mean is very likely close to μ for large N. (The strong law of large numbers, which we discuss in the appendix to this lecture, even says that $\bar{X}_N \to \mu$ with probability 1.) The central limit theorem describes the size and shape of the remaining fluctuations: they are approximately Gaussian, whatever the distribution of the X_i.

Theorem 7.3 (Central limit theorem)**.** *In the situation of Proposition 7.1, with $\sigma > 0$, for every $z \in \mathbb{R}$,*

$$P\Big(\frac{\sqrt{N}(\bar{X}_N - \mu)}{\sigma} \leq z\Big) \to \Phi(z) \qquad \textit{as } N \to \infty.$$

We prove the central limit theorem in Section 7.1, after introducing convergence in distribution. Informally, $\bar{X}_N$ is approximately $\mathcal{N}(\mu, \frac{\sigma^2}{N})$, and the sum $X_1 + \cdots + X_N$ is approximately $\mathcal{N}(N\mu, N\sigma^2)$. This explains why Gaussian distributions are so common: a measurement error that is the sum of many small independent effects is approximately Gaussian.

Example 7.4. We flip a fair coin 100 times. The number of heads S is Binomial$(100, \frac{1}{2})$, with mean 50 and standard deviation $\sqrt{100 \cdot \frac{1}{4}} = 5$. By the central limit theorem, $P(S \geq 60) = P(S \geq 59.5) \approx 1 - \Phi(\frac{59.5-50}{5}) = 1 - \Phi(1.9) \approx 0.0287$, while the exact value is 0.0284. (Replacing 60 by 59.5, the "continuity correction", improves the approximation of a discrete distribution by a continuous one.) Figure 3 compares a binomial distribution with its Gaussian approximation.

Monte Carlo estimation. To compute an expectation $\mathbb{E}[g(X)]$ that we cannot compute exactly, we can draw i.i.d. samples $x_1, \dots, x_N$ of X and use the average $\frac{1}{N}\sum_i g(x_i)$. By the central limit theorem, its error is approximately Gaussian with standard deviation $\frac{\sigma_g}{\sqrt{N}}$, where σ_g is the standard deviation of $g(X)$, no matter how high-dimensional X is. For example, the fraction of N uniformly random points of the square $[-1, 1]^2$ that land in the unit disk estimates $\frac{\pi}{4}$, so 4 times this fraction estimates π, with standard error $4\sqrt{\frac{\pi}{4}(1 - \frac{\pi}{4})/N}$, which is 0.016 for $N = 10\,000$. Each additional correct digit costs 100 times more samples.

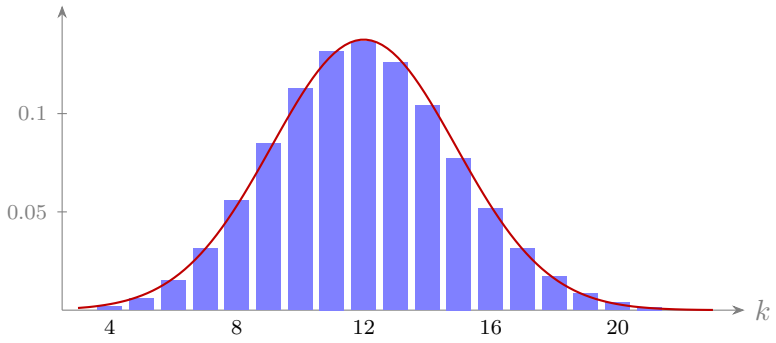


Figure 3: The pmf of the Binomial$(40, 0.3)$ distribution, the sum of 40 independent Bernoulli(0.3) variables, and the density of its Gaussian approximation $\mathcal{N}(12, 8.4)$ with the same mean and variance (red).

Concentration inequalities. The central limit theorem is only an approximation for finite N. For bounded random variables, there is an exact bound that is much stronger than Chebyshev's inequality.

Theorem 7.5 (Hoeffding's inequality)**.** *Let $X_1, \dots, X_N$ be i.i.d. with mean μ and values in $[a, b]$. Then for every $\varepsilon > 0$,*

$$P(|\bar{X}_N - \mu| \geq \varepsilon) \leq 2 \exp\Big(- \frac{2N\varepsilon^2}{(b-a)^2} \Big).$$

The proof uses the Chernoff bound of Example 3.8, together with a bound for the expectation of an exponential.

Lemma 7.6 (Hoeffding's lemma)**.** *Let X have values in $[a, b]$ and $\mathbb{E}[X] = 0$. Then $\mathbb{E}[e^{tX}] \leq e^{t^2(b-a)^2/8}$ for every $t \in \mathbb{R}$.*

Proof. If $a = b$, then $X = 0$. Otherwise, $a \leq 0 \leq b$, since $\mathbb{E}[X] = 0$. The function $x \mapsto e^{tx}$ is convex, so on $[a, b]$ it lies below the chord between (a, e^{ta}) and (b, e^{tb}):

$$e^{tx} \leq \frac{b-x}{b-a}\, e^{ta} + \frac{x-a}{b-a}\, e^{tb} \qquad \text{for } a \leq x \leq b.$$

Taking expectations and using $\mathbb{E}[X] = 0$ gives, with $p = \frac{-a}{b-a} \in [0, 1]$ and $u = t(b-a)$, so that $ta = -pu$,

$$\mathbb{E}[e^{tX}] \leq \frac{b\, e^{ta} - a\, e^{tb}}{b-a} = e^{ta}(1 - p + pe^u) = e^{L(u)}, \qquad L(u) = -pu + \log\,(1 - p + pe^u).$$

We have $L(0) = 0$ and $L'(u) = -p + q(u)$ with $q(u) = \frac{pe^u}{1-p+pe^u} \in [0, 1]$, so $L'(0) = 0$, and $L''(u) = q(u)(1 - q(u)) \leq \frac{1}{4}$. By Taylor's theorem (Lecture 3), $L(u) = \frac{1}{2}L''(\xi)u^2 \leq \frac{u^2}{8} = \frac{t^2(b-a)^2}{8}$ for some ξ. □

Proof of Theorem 7.5. Let $t > 0$ and $S = \sum_i (X_i - \mu)$. Each $X_i - \mu$ has mean 0 and values in the interval $[a - \mu, b - \mu]$ of length $b - a$. By Markov's inequality for e^{tS} (as in Example 3.8), independence and Lemma 7.6,

$$P(\bar{X}_N - \mu \geq \varepsilon) = P(S \geq N\varepsilon) \leq e^{-tN\varepsilon}\, \mathbb{E}[e^{tS}] = e^{-tN\varepsilon} \prod_{i=1}^{N} \mathbb{E}[e^{t(X_i - \mu)}] \leq \exp\Big(-tN\varepsilon + \frac{Nt^2(b-a)^2}{8} \Big).$$

The choice $t = \frac{4\varepsilon}{(b-a)^2}$ gives the bound $\exp\big(-\frac{2N\varepsilon^2}{(b-a)^2}\big)$. The variables $-X_i$, with values in $[-b,-a]$, give the same bound for $P(\bar{X}_N - \mu \le -\varepsilon)$, and the union bound gives the factor 2. □

For example, let a classifier have the (unknown) error rate μ, and let $\bar{X}_N$ be its error rate on a test set of $N = 1000$ examples drawn independently of the training data, where $X_i \in \{0,1\}$ indicates an error. Then $P(|\bar{X}_N - \mu| \ge 0.05) \le 2e^{-5} \approx 0.013$, while Chebyshev's inequality with $\sigma^2 \le \frac{1}{4}$ only gives 0.1.

Remark 7.7 (Limit theorems in machine learning).

1. The average loss on the training data, $\frac{1}{N}\sum_i \ell(\boldsymbol{\theta}; \mathbf{x}_i, y_i)$, is a sample mean. It estimates the expected loss $\mathbb{E}[\ell(\boldsymbol{\theta}; \mathbf{X}, Y)]$ on new data, which is the quantity we actually care about. For a fixed $\boldsymbol{\theta}$, the limit theorems apply. But the trained parameters depend on the training data, so the training loss of the trained model is a biased, usually too optimistic, estimate (overfitting). This is why we evaluate on separate test data, and why learning theory combines Hoeffding's inequality with the union bound (Proposition 1.3) to control all candidate models simultaneously (Section 11.3).

2. In stochastic gradient descent (Lecture 3), the mini-batch gradient of B randomly chosen examples is an unbiased estimate of the full gradient, and its covariance matrix decreases like $\frac{1}{B}$ (for sampling with replacement), by Proposition 7.1 applied to each coordinate. Larger batches give less noisy gradients, but each step costs more.

7.1 Slutsky's theorem and the delta method

The law of large numbers and the central limit theorem are statements about limits of probabilities. We give names to these two kinds of convergence.

Definition 7.8 (Convergence in probability and in distribution)**.** Let $Y, Y_1, Y_2, \ldots$ be random variables. We say that $Y_n \to Y$

1. in probability if $P(|Y_n - Y| > \varepsilon) \to 0$ for every $\varepsilon > 0$;
2. in distribution if $P(Y_n \le y) \to P(Y \le y)$ at every point y at which the CDF of Y is continuous.

Convergence in distribution only depends on the distributions, and we also write $Y_n \to \mathcal{N}(0,1)$, for example. The law of large numbers (Theorem 7.2) says that $\bar{X}_N \to \mu$ in probability, and the central limit theorem (Theorem 7.3) says that $\frac{\sqrt{N}(\bar{X}_N - \mu)}{\sigma} \to \mathcal{N}(0,1)$ in distribution; the appendix to this lecture discusses further modes of convergence. Convergence in distribution can also be tested with expectations of functions.

Lemma 7.9 (Test functions)**.** *Let $Y, Y_1, Y_2, \ldots$ be random variables.*

1. *If $\mathbb{E}[h(Y_n)] \to \mathbb{E}[h(Y)]$ for every three times continuously differentiable function $h : \mathbb{R} \to \mathbb{R}$ for which h, h', h'' and h''' are bounded, then $Y_n \to Y$ in distribution.*
2. *Conversely, if $Y_n \to Y$ in distribution, then $\mathbb{E}[h(Y_n)] \to \mathbb{E}[h(Y)]$ for every bounded continuous function h.*

Proof. Let F and F_n be the CDFs of Y and Y_n. A CDF has at most countably many points of discontinuity, since it has at most k jumps of size larger than $\frac{1}{k}$ (the jumps add up to at most 1). Hence every interval contains continuity points of F.

(1) Let $\psi(s) = 1$ for $s \le 0$, $\psi(s) = 0$ for $s \ge 1$, and $\psi(s) = 1 - 35s^4 + 84s^5 - 70s^6 + 20s^7$ for $0 < s < 1$. Then $\psi'(s) = -140s^3(1-s)^3 \le 0$ on $(0,1)$, so the derivatives of orders 1, 2 and 3 of the polynomial vanish at 0 and 1, and ψ is a nonincreasing, three times continuously differentiable function with values in $[0,1]$ and bounded derivatives. Let y be a continuity point of F, and let $\delta > 0$. The function $h(x) = \psi(\frac{x-y}{\delta})$ satisfies $\mathbf{1}_{\{x \le y\}} \le h(x) \le \mathbf{1}_{\{x \le y+\delta\}}$, so

$$P(Y_n \le y) \le \mathbb{E}[h(Y_n)] \to \mathbb{E}[h(Y)] \le P(Y \le y + \delta) = F(y+\delta).$$

Similarly, $\tilde{h}(x) = \psi(\frac{x-y+\delta}{\delta})$ satisfies $\mathbf{1}_{\{x \le y-\delta\}} \le \tilde{h}(x) \le \mathbf{1}_{\{x \le y\}}$, which gives $P(Y_n \le y) \ge \mathbb{E}[\tilde{h}(Y_n)] \to \mathbb{E}[\tilde{h}(Y)] \ge F(y-\delta)$. So all limit points of the sequence $P(Y_n \le y)$ lie between $F(y-\delta)$ and $F(y+\delta)$, and these tend to $F(y)$ as $\delta \to 0$, since F is continuous at y.

(2) Let $|h| \le C$, and let $\varepsilon > 0$. Choose continuity points $a < b$ of F with $F(a) < \varepsilon$ and $1 - F(b) < \varepsilon$. The function h is uniformly continuous on $[a,b]$ (by the proposition on uniform continuity in the appendix to Lecture 3), so there are continuity points $a = t_0 < t_1 < \cdots < t_K = b$ of F such that $|h(x) - h(t_k)| \le \varepsilon$ for $t_{k-1} \le x \le t_k$. Let s be the step function with $s(x) = h(t_k)$ for $t_{k-1} < x \le t_k$, and $s(x) = 0$ for $x \le a$ and for $x > b$. Then $|h - s| \le \varepsilon$ on $(a,b]$ and $|h - s| \le C$ elsewhere, so for every random variable V,

$$\big|\mathbb{E}[h(V)] - \mathbb{E}[s(V)]\big| \le \varepsilon + C\,P(V \le a) + C\,P(V > b).$$

Now $\mathbb{E}[s(Y_n)] = \sum_k h(t_k)(F_n(t_k) - F_n(t_{k-1})) \to \sum_k h(t_k)(F(t_k) - F(t_{k-1})) = \mathbb{E}[s(Y)]$, and $P(Y_n \le a) + P(Y_n > b) = F_n(a) + 1 - F_n(b) \to F(a) + 1 - F(b) < 2\varepsilon$. Hence all limit points of $\big|\mathbb{E}[h(Y_n)] - \mathbb{E}[h(Y)]\big|$ are at most $2(\varepsilon + 2C\varepsilon)$. Since ε was arbitrary, $\mathbb{E}[h(Y_n)] \to \mathbb{E}[h(Y)]$. □

Proof of the central limit theorem (Theorem 7.3). Let $Y_i = \frac{X_i - \mu}{\sigma}$. These are i.i.d. with mean 0 and variance 1, and $S_N = \frac{\sqrt{N}(\bar{X}_N - \mu)}{\sigma} = \frac{1}{\sqrt{N}}(Y_1 + \cdots + Y_N)$. Let $Z \sim \mathcal{N}(0,1)$. Since Φ is continuous, by Lemma 7.9 it suffices to show that $\mathbb{E}[h(S_N)] \to \mathbb{E}[h(Z)]$ for every h as in part 1 of the lemma; let C_2 and C_3 be bounds for $|h''|$ and $|h'''|$.

A Taylor bound. Let $r(w,x) = h(w+x) - h(w) - h'(w)x - \frac{1}{2}h''(w)x^2$. Taylor's theorem (Lecture 3) with $k = 2$ gives $r(w,x) = \frac{1}{6}h'''(\xi)x^3$, and with $k = 1$ it gives $r(w,x) = \frac{1}{2}(h''(\xi') - h''(w))x^2$, for some ξ and ξ' between w and $w + x$. Hence

$$|r(w,x)| \le m(x) = \min\Big(\frac{C_3}{6}|x|^3,\, C_2 x^2\Big).$$

Replacing one variable at a time. Let $Z_1, \ldots, Z_N$ be i.i.d. $\mathcal{N}(0,1)$ variables, independent of the Y_i. By Corollary 6.7, $T_N = \frac{1}{\sqrt{N}}(Z_1 + \cdots + Z_N) \sim \mathcal{N}(0,1)$, so $\mathbb{E}[h(Z)] = \mathbb{E}[h(T_N)]$. For $k = 0, \ldots, N$, let $V_k = \frac{1}{\sqrt{N}}(Y_1 + \cdots + Y_k + Z_{k+1} + \cdots + Z_N)$, so that $V_N = S_N$ and $V_0 = T_N$, and let $W_k = V_k - \frac{Y_k}{\sqrt{N}} = V_{k-1} - \frac{Z_k}{\sqrt{N}}$ for $k \ge 1$. Then

$$\mathbb{E}[h(S_N)] - \mathbb{E}[h(T_N)] = \sum_{k=1}^{N} \big(\mathbb{E}[h(V_k)] - \mathbb{E}[h(V_{k-1})]\big) = \sum_{k=1}^{N} \Big(\mathbb{E}\Big[h\Big(W_k + \frac{Y_k}{\sqrt{N}}\Big)\Big] - \mathbb{E}\Big[h\Big(W_k + \frac{Z_k}{\sqrt{N}}\Big)\Big]\Big).$$

Expand $h(W_k + x) = h(W_k) + h'(W_k)x + \frac{1}{2}h''(W_k)x^2 + r(W_k, x)$ for $x = \frac{Y_k}{\sqrt{N}}$ and for $x = \frac{Z_k}{\sqrt{N}}$. The variable W_k is independent of Y_k and of Z_k, and Y_k and Z_k both have mean 0 and variance 1, so $\mathbb{E}[h'(W_k)Y_k] = \mathbb{E}[h'(W_k)]\mathbb{E}[Y_k] = 0 = \mathbb{E}[h'(W_k)Z_k]$ and $\mathbb{E}[h''(W_k)Y_k^2] = \mathbb{E}[h''(W_k)] = \mathbb{E}[h''(W_k)Z_k^2]$. So only the remainders remain:

$$\big|\mathbb{E}[h(S_N)] - \mathbb{E}[h(Z)]\big| \le \sum_{k=1}^{N} \Big(\mathbb{E}\Big[m\Big(\frac{Y_k}{\sqrt{N}}\Big)\Big] + \mathbb{E}\Big[m\Big(\frac{Z_k}{\sqrt{N}}\Big)\Big]\Big) = N\,\mathbb{E}\Big[m\Big(\frac{Y_1}{\sqrt{N}}\Big)\Big] + N\,\mathbb{E}\Big[m\Big(\frac{Z_1}{\sqrt{N}}\Big)\Big].$$

Bounding the remainders. Using the cubic bound, $N\,\mathbb{E}[m(Z_1/\sqrt{N})] \le \frac{C_3\mathbb{E}[|Z_1|^3]}{6\sqrt{N}} \to 0$. For the Y_1 term, let $\varepsilon > 0$, and use the cubic bound if $|Y_1| \le \varepsilon\sqrt{N}$ (then $|Y_1|^3 \le \varepsilon\sqrt{N}\,Y_1^2$) and the quadratic bound otherwise:

$$N\,\mathbb{E}\Big[m\Big(\frac{Y_1}{\sqrt{N}}\Big)\Big] \le \frac{C_3}{6}\,\varepsilon\,\mathbb{E}[Y_1^2] + C_2\,\mathbb{E}\big[Y_1^2\,\mathbf{1}_{\{|Y_1|>\varepsilon\sqrt{N}\}}\big].$$

Here $\mathbb{E}[Y_1^2] = 1$, and the last expectation tends to 0 as $N \to \infty$, since it is the tail of the convergent sum or integral $\mathbb{E}[Y_1^2]$. Hence all limit points of $|\mathbb{E}[h(S_N)] - \mathbb{E}[h(Z)]|$ are at most $\frac{C_3\varepsilon}{6}$, for every $\varepsilon > 0$, and $\mathbb{E}[h(S_N)] \to \mathbb{E}[h(Z)]$. □

The proof replaces the variables Y_k by Gaussian ones, one at a time; each replacement changes the expectation only by the remainder of order $N^{-3/2}$, because the first two moments match. This is Lindeberg's proof of the central limit theorem. Two rules allow us to combine limits in distribution.

Theorem 7.10 (Continuous mapping and Slutsky)**.** *Let $X_n \to X$ in distribution.*

1. *If g is continuous, then $g(X_n) \to g(X)$ in distribution.*
2. *(Slutsky) If $Y_n \to c$ in probability for a constant c, then $X_n + Y_n \to X + c$ and $X_nY_n \to cX$ in distribution.*
3. *If $Y_n \to c$ and $Z_n \to d$ in probability, and g is continuous at (c, d), then $g(Y_n, Z_n) \to g(c, d)$ in probability. The same holds for functions of one variable.*
4. *$Y_n \to c$ in distribution if and only if $Y_n \to c$ in probability.*

Proof. By Lemma 7.9, it suffices for parts 1 and 2 to show convergence of $\mathbb{E}[h(\cdot)]$ for every h as in part 1 of the lemma; let $|h| \le C_0$ and $|h'| \le C_1$.

(1) The function $h \circ g$ is bounded and continuous, so $\mathbb{E}[h(g(X_n))] \to \mathbb{E}[h(g(X))]$ by Lemma 7.9.

(2) Let $\varepsilon > 0$. By the mean value theorem, $|h(x+y) - h(x+c)| \le C_1|y-c|$, and the difference is at most $2C_0$ in any case. So

$$\big|\mathbb{E}[h(X_n + Y_n)] - \mathbb{E}[h(X_n + c)]\big| \le C_1\varepsilon + 2C_0\,P(|Y_n - c| > \varepsilon),$$

where the probability tends to 0. Since $x \mapsto h(x + c)$ is bounded and continuous, $\mathbb{E}[h(X_n + c)] \to \mathbb{E}[h(X + c)]$ by Lemma 7.9. So all limit points of $|\mathbb{E}[h(X_n + Y_n)] - \mathbb{E}[h(X + c)]|$ are at most $C_1\varepsilon$, for every $\varepsilon > 0$. For the product, let $\eta > 0$, choose $M > 0$ such that $\pm M$ are continuity points of the CDF F of X with $F(-M) + 1 - F(M) < \eta$ (see the proof of Lemma 7.9), and then $\varepsilon > 0$ with $C_1M\varepsilon < \eta$. If $|X_n| \le M$ and $|Y_n - c| \le \varepsilon$, then $|h(X_nY_n) - h(cX_n)| \le C_1|X_n||Y_n - c| \le \eta$, so

$$\big|\mathbb{E}[h(X_nY_n)] - \mathbb{E}[h(cX_n)]\big| \le \eta + 2C_0\big(P(|X_n| > M) + P(|Y_n - c| > \varepsilon)\big).$$

As $n \to \infty$, $P(|X_n| > M) \le P(X_n \le -M) + 1 - P(X_n \le M) \to F(-M) + 1 - F(M) < \eta$, and $\mathbb{E}[h(cX_n)] \to \mathbb{E}[h(cX)]$ by Lemma 7.9. So all limit points of $|\mathbb{E}[h(X_nY_n)] - \mathbb{E}[h(cX)]|$ are at most $\eta + 2C_0\eta$, for every $\eta > 0$.

(3) Let $\varepsilon > 0$. By continuity, there is a $\delta > 0$ with $|g(y, z) - g(c, d)| \le \varepsilon$ whenever $|y - c| \le \delta$ and $|z - d| \le \delta$. Hence $P(|g(Y_n, Z_n) - g(c, d)| > \varepsilon) \le P(|Y_n - c| > \delta) + P(|Z_n - d| > \delta) \to 0$.

(4) The CDF of the constant c is continuous everywhere except at c. If $Y_n \to c$ in distribution, then $P(|Y_n - c| > \varepsilon) \le P(Y_n \le c - \varepsilon) + 1 - P(Y_n \le c + \varepsilon) \to 0 + 1 - 1 = 0$. The converse is part 2, applied to the constant sequence 0 and to Y_n. □

Slutsky's theorem justifies the confidence intervals of Section 9, in which we replace σ by the sample standard deviation s_N. By the law of large numbers, applied to the X_i and the X_i^2 (if $\mathbb{E}[X_i^4] < \infty$, so that Theorem 7.2 applies to the X_i^2), and by part 3, $s_N \to \sigma$ and $\frac{\sigma}{s_N} \to 1$ in probability, so by part 2,

$$\frac{\sqrt{N}(\bar{X}_N - \mu)}{s_N} = \frac{\sigma}{s_N} \cdot \frac{\sqrt{N}(\bar{X}_N - \mu)}{\sigma} \to \mathcal{N}(0,1) \qquad \text{in distribution,}$$

and $\bar{x} \pm 1.96\frac{s}{\sqrt{N}}$ is an asymptotic 95% confidence interval.

Proposition 7.11 (Delta method)**.** *Let $\sqrt{N}(T_N - \theta) \to \mathcal{N}(0, \tau^2)$ in distribution, and let g be continuously differentiable. Then $\sqrt{N}(g(T_N) - g(\theta)) \to \mathcal{N}(0, g'(\theta)^2\tau^2)$ in distribution.*

Proof. By the mean value theorem, $g(T_N) - g(\theta) = g'(\xi_N)(T_N - \theta)$ for some ξ_N between θ and T_N. By part 2 of Theorem 7.10, $T_N - \theta = \frac{1}{\sqrt{N}} \cdot \sqrt{N}(T_N - \theta) \to 0 \cdot Y = 0$ in distribution, where $Y \sim \mathcal{N}(0, \tau^2)$, hence in probability by part 4. Since $|\xi_N - \theta| \le |T_N - \theta|$, also $\xi_N \to \theta$ in probability, and $g'(\xi_N) \to g'(\theta)$ in probability by part 3, since g' is continuous. By part 2, $\sqrt{N}(g(T_N) - g(\theta)) = g'(\xi_N) \cdot \sqrt{N}(T_N - \theta) \to g'(\theta)Y$ in distribution, and $g'(\theta)Y \sim \mathcal{N}(0, g'(\theta)^2\tau^2)$. □

Example 7.12 (Confidence intervals on the log-odds scale)**.** For a proportion $\hat{p} = \bar{X}_N$ of N Bernoulli(p) observations, $\sqrt{N}(\hat{p} - p) \to \mathcal{N}(0, p(1-p))$ by the central limit theorem. The log-odds $g(p) = \log \frac{p}{1-p}$ have the derivative $g'(p) = \frac{1}{p(1-p)}$, so the estimated log-odds $g(\hat{p})$ have approximately the standard error $\frac{1}{\sqrt{Np(1-p)}}$. For 30 successes in 100 trials, the log-odds are $\log \frac{0.3}{0.7} \approx -0.847$, with the estimated standard error $\frac{1}{\sqrt{21}} \approx 0.218$. This gives the approximate 95% confidence interval $-0.847 \pm 1.96 \cdot 0.218 \approx [-1.275, -0.420]$ for the log-odds, and, transforming back with the sigmoid function, $[0.218, 0.397]$ for p. Unlike the interval $\hat{p} \pm 1.96\sqrt{\hat{p}(1-\hat{p})/N} = [0.210, 0.390]$, it is not symmetric around $\hat{p}$, and it never leaves $[0, 1]$. Logistic regression (Lecture 3) models the log-odds, and its standard errors are computed on this scale.

7.2 Monte Carlo sampling

Monte Carlo estimation needs samples from the distribution of interest. We have already seen inverse transform sampling (Theorem 2.7) and sampling Gaussian vectors by affine maps (Section 6), and we will see Markov chain Monte Carlo in Section 8. Two further general methods are rejection sampling, which produces exact samples, and importance sampling, which reweights samples from a different distribution.

Theorem 7.13 (Rejection sampling)**.** *Let p and q be densities, and let $M \ge 1$ be such that $p(x) \le Mq(x)$ for all x. Repeat the following trial until a value is accepted: draw $X \sim q$ and, independently, $U \sim$ Uniform$(0,1)$, and accept X if $U \le \frac{p(X)}{Mq(X)}$. Then the accepted value has the density p, and each trial is accepted with probability $\frac{1}{M}$.*

Proof. For every set A, $P(X \in A \text{ and accepted}) = \int_A q(x)\,\frac{p(x)}{Mq(x)}\,dx = \frac{1}{M}\int_A p(x)\,dx$. For $A = \mathbb{R}$, this gives $P(\text{accepted}) = \frac{1}{M}$, so $P(X \in A \mid \text{accepted}) = \int_A p(x)\,dx$. Since the trials are independent, the accepted value has this conditional distribution. □

Geometrically, the points $(X, UMq(X))$ are uniformly distributed under the graph of Mq, and we keep those that lie under the graph of p; their first coordinates have the density p. The method also works if p is only known up to a constant. For example, the Beta$(2,2)$ density $6x(1-x)$ is at

most 1.5 on $[0,1]$, so with the uniform proposal and $M = 1.5$, two out of three trials are accepted. In high dimensions, however, M is usually enormous: for the target $\mathcal{N}(\mathbf{0}, I)$ in $\mathbb{R}^d$ and the proposal $\mathcal{N}(\mathbf{0}, 2I)$, the ratio $\frac{p}{q} = 2^{d/2} e^{-\|\mathbf{x}\|^2/4}$ forces $M \geq 2^{d/2}$, so only one in 1024 trials is accepted for $d = 20$. This is one reason why high-dimensional sampling relies on MCMC.

Proposition 7.14 (Importance sampling)**.** *Let p and q be densities such that $q(x) > 0$ for all x with $f(x)p(x) \neq 0$, and let $w = \frac{p}{q}$ (the importance weights). Then $\mathbb{E}_p[f(X)] = \mathbb{E}_q[f(Y)w(Y)]$. Hence, for i.i.d. samples $y_1, \ldots, y_N$ from q, the estimator*

$$\hat{I} = \frac{1}{N} \sum_{i=1}^{N} f(y_i) w(y_i)$$

is unbiased, with variance $\frac{1}{N} \operatorname{Var}_q(f(Y)w(Y))$.

Proof. $\int f(x)p(x)\,dx = \int f(y)\frac{p(y)}{q(y)}q(y)\,dy$; the variance follows from Proposition 7.1. □

The variance is small if q puts its mass where $|f|p$ is large; for $f \geq 0$, the ideal (but impractical) choice $q \propto fp$ even gives variance 0.

Example 7.15 (A rare event)**.** The tail probability $P(Z > 4) = 3.17 \cdot 10^{-5}$ of $Z \sim \mathcal{N}(0,1)$ is hard to estimate by plain Monte Carlo: among $N = 10\,000$ samples, we expect only 0.32 samples above 4, and the relative standard error of the estimate is about 178%. With the proposal $q = \mathcal{N}(4,1)$, about half of the samples land above 4, and they are reweighted by $w(y) = \frac{\varphi(y)}{\varphi(y-4)} = e^{8-4y}$. With the same N, the relative standard error drops to about 2%. Probabilities of rare failures are estimated in this way.

If p is only known up to a constant, one uses the self-normalized estimator $\sum_i f(y_i)w_i / \sum_i w_i$, which is slightly biased but consistent. If a few weights dominate, the estimate is unreliable; the effective sample size $\frac{(\sum_i w_i)^2}{\sum_i w_i^2}$ measures how many samples effectively contribute. In machine learning, importance weights correct for differences between distributions: in off-policy evaluation in reinforcement learning (data collected with one policy, used to evaluate another), and under covariate shift, where training examples are weighted by the ratio of the test and training densities.

7.3 Stochastic gradient descent

Stochastic gradient descent (Remark 10.10 of Lecture 3) replaces the gradient of the training loss by the gradient on a random mini-batch. Its iterates are random vectors, and we can now analyze it. We write SGD as

$$\mathbf{x}_{k+1} = \mathbf{x}_k - \eta_k \mathbf{g}_k,$$

where the stochastic gradient $\mathbf{g}_k$ is computed from a new random mini-batch in each step. We assume that it is unbiased and has bounded second moments: $\mathbb{E}[\mathbf{g}_k \mid \mathbf{x}_k] = \nabla f(\mathbf{x}_k)$ and $\mathbb{E}[\|\mathbf{g}_k\|^2 \mid \mathbf{x}_k] \leq G^2$. (For uniformly random mini-batches, the first condition holds as in Remark 10.10 of Lecture 3, and $\mathbf{x}_k$ takes finitely many values, so these conditional expectations are those of Section 4.2, applied entrywise.)

Theorem 7.16 (Convergence of SGD)**.** *Let f be differentiable and μ-strongly convex (Definition 11.10 of Lecture 3), with the minimum $\mathbf{x}^*$, and let $a_k = \mathbb{E}\big[\|\mathbf{x}_k - \mathbf{x}^*\|^2\big]$.*

1. *With the step sizes $\eta_k = \frac{1}{\mu(k+1)}$, $a_k \leq \dfrac{G^2}{\mu^2 k}$ for all $k \geq 1$.*

2. With a constant step size $\eta \le \frac{1}{2\mu}$, $a_k \le (1-2\mu\eta)^k a_0 + \dfrac{\eta G^2}{2\mu}$.

Proof. Expanding the square,

$$\|\mathbf{x}_{k+1} - \mathbf{x}^*\|^2 = \|\mathbf{x}_k - \mathbf{x}^*\|^2 - 2\eta_k \mathbf{g}_k^\top(\mathbf{x}_k - \mathbf{x}^*) + \eta_k^2 \|\mathbf{g}_k\|^2.$$

Summing over the values $\mathbf{x}$ of $\mathbf{x}_k$, as in the proof of the tower property (Theorem 4.15),

$$\mathbb{E}[\mathbf{g}_k^\top(\mathbf{x}_k - \mathbf{x}^*)] = \sum_{\mathbf{x}} P(\mathbf{x}_k = \mathbf{x})\, \mathbb{E}[\mathbf{g}_k \mid \mathbf{x}_k = \mathbf{x}]^\top (\mathbf{x} - \mathbf{x}^*) = \mathbb{E}[\nabla f(\mathbf{x}_k)^\top (\mathbf{x}_k - \mathbf{x}^*)],$$

which is at least μa_k, by Lemma 11.11 of Lecture 3. Similarly, $\mathbb{E}[\|\mathbf{g}_k\|^2] \le G^2$. Taking expectations,

$$a_{k+1} \le (1 - 2\mu\eta_k)\, a_k + \eta_k^2 G^2.$$

(1) With $\eta_k = \frac{1}{\mu(k+1)}$, the step $k = 0$ gives $a_1 \le -a_0 + \frac{G^2}{\mu^2} \le \frac{G^2}{\mu^2}$. If $a_k \le \frac{G^2}{\mu^2 k}$ for some $k \ge 1$, then

$$a_{k+1} \le \Big(1 - \frac{2}{k+1}\Big)\frac{G^2}{\mu^2 k} + \frac{G^2}{\mu^2 (k+1)^2} = \frac{G^2}{\mu^2(k+1)}\Big(\frac{k-1}{k} + \frac{1}{k+1}\Big) \le \frac{G^2}{\mu^2(k+1)},$$

since $\frac{1}{k+1} \le \frac{1}{k}$. (2) With $0 \le 1 - 2\mu\eta < 1$, applying the recursion k times gives $a_k \le (1-2\mu\eta)^k a_0 + \eta^2 G^2 \sum_{j=0}^{k-1} (1-2\mu\eta)^j \le (1-2\mu\eta)^k a_0 + \frac{\eta^2 G^2}{2\mu\eta}$. □

So SGD with decreasing step sizes converges, but only at the rate $\frac{1}{k}$, much slower than the geometric rate of gradient descent (Theorem 11.12 of Lecture 3). With a constant step size, the error first decreases geometrically, but then fluctuates at a level proportional to η: the noise of the stochastic gradients keeps the iterates in a ball around $\mathbf{x}^*$. This is why step sizes are decreased during training. Each step of SGD costs only $|B|$ gradient evaluations instead of N; for large data sets and moderate accuracy, this more than compensates for the slower rate. Larger mini-batches reduce the noise: $\mathbb{E}[\|\mathbf{g}\|^2] = \|\nabla f\|^2 + \operatorname{tr}\operatorname{Cov}(\mathbf{g})$, and the covariance matrix of a mini-batch gradient decreases like $\frac{1}{|B|}$ (Proposition 7.1).

Example 7.17 (SGD computes running means)**.** Let $f(\theta) = \frac{1}{2N}\sum_{i=1}^N (\theta - y_i)^2$, which is 1-strongly convex with the minimum $\bar{y}$, the mean of the data. SGD with mini-batches of size 1 uses $g_k = \theta_k - y_{I_k}$ with a uniformly random index I_k. With $\eta_k = \frac{1}{k+1}$,

$$\theta_{k+1} = \theta_k - \frac{\theta_k - y_{I_k}}{k+1} = \frac{k\,\theta_k + y_{I_k}}{k+1},$$

so $\theta_1 = y_{I_0}$, and by induction $\theta_k = \frac{1}{k}(y_{I_0} + \cdots + y_{I_{k-1}})$ is the mean of k data points drawn with replacement. By Proposition 7.1, $\mathbb{E}[(\theta_k - \bar{y})^2] = \frac{s^2}{k}$, where $s^2 = \frac{1}{N}\sum_i (y_i - \bar{y})^2$. So the rate $\frac{1}{k}$ of Theorem 7.16 cannot be improved in general.

8 Markov Chains

So far, we mostly considered independent random variables. Many random sequences have memory: tomorrow's weather depends on today's, the next word of a sentence depends on the previous ones, and a user browsing the web chooses the next page depending on the current one. The simplest model with memory is a Markov chain, in which the future depends on the past only through the present.

Definition 8.1 (Markov chain)**.** A sequence of random variables $X_0, X_1, X_2, \ldots$ with values in a finite set of states $\{1, \ldots, n\}$ is a Markov chain with transition matrix $P \in \mathbb{R}^{n \times n}$ if

$$P(X_{t+1} = j \mid X_t = i, X_{t-1} = i_{t-1}, \ldots, X_0 = i_0) = P_{ij}$$

for all times t and all states $i, j, i_0, \ldots, i_{t-1}$ (whenever the condition has positive probability).

The entry P_{ij} is the probability of moving from state i to state j in one step. So P has nonnegative entries, and each row sums to 1, i.e., $P\mathbf{1} = \mathbf{1}$; such matrices are called stochastic matrices.

Example 8.2 (Weather)**.** Let state 1 be "sunny" and state 2 "rainy". After a sunny day, the next day is sunny with probability 0.9; after a rainy day, it is sunny with probability 0.5:

$$P = \begin{pmatrix} 0.9 & 0.1 \\ 0.5 & 0.5 \end{pmatrix}.$$

The chain is drawn as a graph with an arrow $i \to j$ labeled P_{ij} for each possible transition (Figure 4).

We write the distribution of X_t as a row vector $\boldsymbol{\pi}_t = \big(P(X_t = 1), \ldots, P(X_t = n)\big)$. (The same weather model appears in the section on eigenvalues in Lecture 1, with the distributions written as columns and multiplied by the transposed matrix $P^\top$.)

Proposition 8.3. *$\boldsymbol{\pi}_{t+1} = \boldsymbol{\pi}_t P$, hence $\boldsymbol{\pi}_t = \boldsymbol{\pi}_0 P^t$. More generally, $P(X_{t+k} = j \mid X_t = i) = (P^k)_{ij}$.*

Proof. By the law of total probability (Theorem 1.6), $P(X_{t+1} = j) = \sum_i P(X_t = i)\, P(X_{t+1} = j \mid X_t = i) = \sum_i (\boldsymbol{\pi}_t)_i P_{ij}$. The second statement follows by induction on k, in the same way. □

So the distribution evolves by repeated multiplication with P, and the long-term behavior of the chain is a question about the powers of a matrix, i.e., about its eigenvalues (Lecture 1). For the weather chain, starting on a sunny day, $\boldsymbol{\pi}_0 = (1, 0)$, $\boldsymbol{\pi}_1 = (0.9, 0.1)$, $\boldsymbol{\pi}_2 = (0.86, 0.14)$, $\boldsymbol{\pi}_3 = (0.844, 0.156)$, and the distribution seems to approach a limit.

Definition 8.4 (Stationary distribution)**.** A probability distribution $\boldsymbol{\pi}$ (a row vector with $\pi_i \geq 0$ and $\sum_i \pi_i = 1$) is stationary for P if $\boldsymbol{\pi} P = \boldsymbol{\pi}$.

If X_0 has a stationary distribution, then so does every X_t. Transposing, $P^\top \boldsymbol{\pi}^\top = \boldsymbol{\pi}^\top$: a stationary distribution is an eigenvector of $P^\top$ with the eigenvalue 1 (a left eigenvector of P), scaled to have entries that sum to 1.

Proposition 8.5. *Let P be a stochastic matrix. Then 1 is an eigenvalue of P and of $P^\top$, and every (complex) eigenvalue λ of P satisfies $|\lambda| \leq 1$.*

Proof. $P\mathbf{1} = \mathbf{1}$, and $P^\top$ has the same characteristic polynomial as P, since $\det(A^\top) = \det(A)$ (Lecture 1). If $P\mathbf{v} = \lambda\mathbf{v}$ with $\mathbf{v} \neq \mathbf{0}$, choose i with $|v_i| = \max_j |v_j|$. Then $|\lambda||v_i| = |\sum_j P_{ij} v_j| \leq \sum_j P_{ij}|v_j| \leq \sum_j P_{ij}|v_i| = |v_i|$. □

Whether the distribution converges to a stationary distribution depends on two properties. A chain is irreducible if every state can be reached from every other state, i.e., for all i and j there is a t with $(P^t)_{ij} > 0$. It is aperiodic if for every state i, the numbers $t \geq 1$ with $(P^t)_{ii} > 0$ have greatest common divisor 1; for an irreducible chain, it suffices that some $P_{ii} > 0$.

Theorem 8.6 (Convergence of Markov chains)**.** *An irreducible Markov chain has a unique stationary distribution $\boldsymbol{\pi}$, and all $\pi_i > 0$. If the chain is also aperiodic, then $\boldsymbol{\pi}_0 P^t \to \boldsymbol{\pi}$ as $t \to \infty$, for every initial distribution $\boldsymbol{\pi}_0$.*

The key step of the proof is that some power of P has only positive entries.

Lemma 8.7. *If a Markov chain is irreducible and aperiodic, then there is an $m \geq 1$ such that all entries of P^m are positive.*

Proof. Fix a state i, and let $S_i = \{t \geq 1 : (P^t)_{ii} > 0\}$. Since $(P^{s+t})_{ii} = \sum_k (P^s)_{ik}(P^t)_{ki} \geq (P^s)_{ii}(P^t)_{ii}$, the set S_i is closed under addition. We show that it contains all large integers. Enumerate $S_i = \{s_1, s_2, \ldots\}$. The greatest common divisors of $s_1, \ldots, s_l$ form a nonincreasing sequence of positive integers, which is eventually constant; its final value divides every element of S_i, so it is 1, and $s_1, \ldots, s_l$ have the greatest common divisor 1 for some l. By Bézout's identity from elementary number theory, $1 = \sum_k a_k s_k$ for some integers a_k. Let u be the sum of the terms with $a_k > 0$, and v minus the sum of the terms with $a_k < 0$, so that $u - v = 1$. As sums of elements of S_i, $u \in S_i$, and $v \in S_i$ or $v = 0$. If $v = 0$, then $1 \in S_i$, so S_i contains every $t \geq 1$. Otherwise, write $t \geq v^2$ as $t = lv + r$ with $0 \leq r < v \leq l$. Then $t = (l - r)v + ru$ with $l - r \geq 1$, so $t \in S_i$.

So there is an N with $(P^t)_{ii} > 0$ for all states i and all $t \geq N$. By irreducibility, for $i \neq j$ there is a t_{ij} with $(P^{t_{ij}})_{ij} > 0$. Let $m = N + \max_{i \neq j} t_{ij}$. Then $(P^m)_{ii} > 0$, and for $i \neq j$, $(P^m)_{ij} \geq (P^{m - t_{ij}})_{ii}(P^{t_{ij}})_{ij} > 0$. □

Proof of Theorem 8.6. We measure the distance between distributions by $\|\mathbf{d}\|_1 = \sum_i |d_i|$.

Step 1: contraction. Let Q be a stochastic matrix whose entries are all at least $\delta > 0$; since the rows sum to 1, $n\delta \leq 1$. Let $\mathbf{d}$ be a row vector with $\sum_i d_i = 0$. Then $(\mathbf{d}Q)_j = \sum_i d_i Q_{ij} = \sum_i d_i (Q_{ij} - \delta)$, so

$$\|\mathbf{d}Q\|_1 \leq \sum_j \sum_i |d_i|(Q_{ij} - \delta) = (1 - n\delta)\|\mathbf{d}\|_1.$$

The same computation with $\delta = 0$ shows $\|\mathbf{d}P\|_1 \leq \|\mathbf{d}\|_1$ for every stochastic matrix P, and the entries of $\mathbf{d}P$ again sum to 0, since $(\mathbf{d}P)\mathbf{1} = \mathbf{d}(P\mathbf{1}) = \mathbf{d}\mathbf{1} = 0$.

Step 2: irreducible and aperiodic chains. By Lemma 8.7, the entries of $Q = P^m$ are at least some $\delta > 0$. By Proposition 8.5, 1 is an eigenvalue of $P^\top$, so there is a real row vector $\mathbf{v} \neq \mathbf{0}$ with $\mathbf{v}P = \mathbf{v}$, and then $\mathbf{v}Q = \mathbf{v}$. Now $|v_j| = |\sum_i v_i Q_{ij}| \leq \sum_i |v_i| Q_{ij}$, and summing over j gives $\sum_j |v_j| \leq \sum_i |v_i|$. So equality holds for every j, which is only possible if the nonzero numbers $v_i Q_{ij}$, and hence the nonzero entries v_i, all have the same sign. Thus $\boldsymbol{\pi} = \mathbf{v}/\sum_i v_i$ is a stationary distribution. For every initial distribution $\boldsymbol{\pi}_0$, the entries of $\mathbf{d} = \boldsymbol{\pi}_0 - \boldsymbol{\pi}$ sum to 0, and $\boldsymbol{\pi}_0 P^t - \boldsymbol{\pi} = \mathbf{d}P^t$. Writing $t = km + l$ with $0 \leq l < m$, Step 1 gives

$$\|\boldsymbol{\pi}_0 P^t - \boldsymbol{\pi}\|_1 = \|\mathbf{d}Q^k P^l\|_1 \leq (1 - n\delta)^k \|\mathbf{d}\|_1 \leq 2(1 - n\delta)^k \to 0.$$

Every stationary distribution $\boldsymbol{\pi}'$ satisfies $\boldsymbol{\pi}' = \boldsymbol{\pi}' P^t \to \boldsymbol{\pi}$, so $\boldsymbol{\pi}' = \boldsymbol{\pi}$. Moreover, the argument shows that every real row vector $\mathbf{v} \neq \mathbf{0}$ with $\mathbf{v}P = \mathbf{v}$ is a multiple of $\boldsymbol{\pi}$.

Step 3: irreducible chains. Let $P' = \frac{1}{2}(I + P)$, the "lazy" chain that stays where it is with probability $\frac{1}{2}$ and otherwise moves according to P. It is stochastic, and since I and P commute, $P'^t = 2^{-t} \sum_{k=0}^{t} \binom{t}{k} P^k$ has entries $(P'^t)_{ij} \geq 2^{-t}(P^t)_{ij}$ and $(P'^t)_{ii} \geq 2^{-t}$. So P' is irreducible and aperiodic, and by Step 2 it has a unique stationary distribution $\boldsymbol{\pi}$. Since $\boldsymbol{\pi}P' = \frac{1}{2}(\boldsymbol{\pi} + \boldsymbol{\pi}P)$, a distribution is stationary for P' if and only if it is stationary for P, so $\boldsymbol{\pi}$ is the unique stationary distribution of P; in the same way, every real row vector $\mathbf{v} \neq \mathbf{0}$ with $\mathbf{v}P = \mathbf{v}$ is a multiple of $\boldsymbol{\pi}$. Finally, choose i with $\pi_i > 0$. For every state j, there is a t with $(P^t)_{ij} > 0$ (with $t = 0$ if $j = i$), and $\pi_j = (\boldsymbol{\pi}P^t)_j \geq \pi_i (P^t)_{ij} > 0$. □

This is a special case of the Perron–Frobenius theorem on matrices with nonnegative entries. The proof gives the rate $(1-n\delta)^{\lfloor t/m \rfloor}$; more precisely, the speed of convergence is governed by the second largest absolute value of an eigenvalue of P. Both assumptions are needed. For $P = \left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right)$, the chain alternates between the two states: $(\frac{1}{2}, \frac{1}{2})$ is stationary, but $\boldsymbol{\pi}_0 = (1,0)$ gives $\boldsymbol{\pi}_t = (1,0), (0,1), (1,0), \ldots$ (the chain has period 2, and -1 is an eigenvalue). For $P = I$, the chain never moves, and every distribution is stationary. For two states, everything can be computed by hand.

Proposition 8.8 (Two-state chains)**.** *Let* $P = \begin{pmatrix} 1-a & a \\ b & 1-b \end{pmatrix}$ *with* $a+b > 0$. *Then* $\boldsymbol{\pi} = (\frac{b}{a+b}, \frac{a}{a+b})$ *is stationary, and for every initial distribution,*

$$\boldsymbol{\pi}_t - \boldsymbol{\pi} = (1-a-b)^t\,(\boldsymbol{\pi}_0 - \boldsymbol{\pi}).$$

Proof. One checks $\boldsymbol{\pi}P = \boldsymbol{\pi}$ directly. The difference $\mathbf{d} = \boldsymbol{\pi}_0 - \boldsymbol{\pi}$ has entries that sum to 0, so $\mathbf{d} = c(1,-1)$, and $(1,-1)P = (1-a-b,\, a-1+b) = (1-a-b)(1,-1)$: the row vector $(1,-1)$ is a left eigenvector with the eigenvalue $1-a-b$. Hence $\boldsymbol{\pi}_t - \boldsymbol{\pi} = (\boldsymbol{\pi}_0 - \boldsymbol{\pi})P^t = (1-a-b)^t\,\mathbf{d}$. □

For the weather chain, $a = 0.1$ and $b = 0.5$, so the stationary distribution is $(\frac{5}{6}, \frac{1}{6})$, and the distance to it shrinks by the factor 0.4 (the second eigenvalue of P) every day. Starting on a sunny day, $P(X_t = \text{sunny}) = \frac{5}{6} + \frac{1}{6}\,0.4^t$; starting on a rainy day, it is $\frac{5}{6} - \frac{5}{6}\,0.4^t$ (Figure 4).

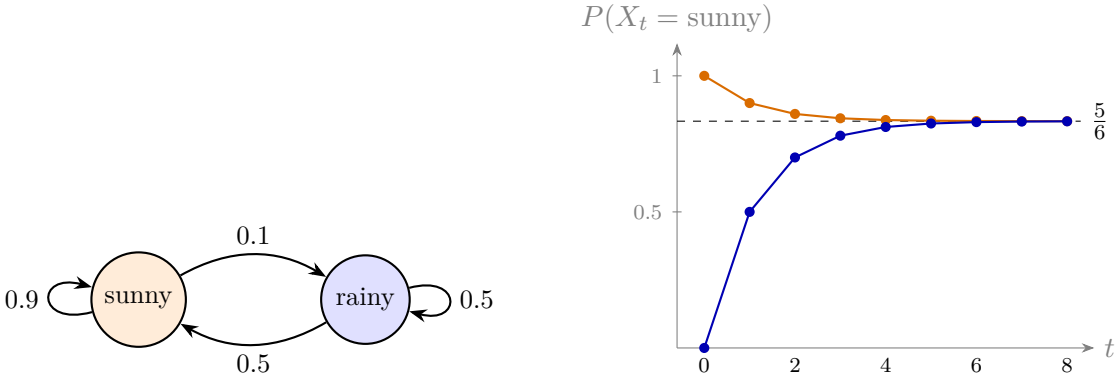


Figure 4: Left: the weather chain of Example 8.2. Right: the probability of a sunny day t days after a sunny day (orange) and after a rainy day (blue); both converge to the stationary probability $\frac{5}{6}$, and the distance shrinks by the factor 0.4 every day.

The ergodic theorem. For an irreducible chain, time averages converge to averages over the stationary distribution. This is a law of large numbers for dependent random variables.

Theorem 8.9 (Ergodic theorem)**.** *Let* $X_0, X_1, \ldots$ *be an irreducible Markov chain with an arbitrary initial distribution and the stationary distribution* $\boldsymbol{\pi}$, *and let* f *be a function on the states. Then* $\frac{1}{T}\sum_{t=0}^{T-1} f(X_t) \to \sum_i \pi_i f(i)$ *in probability as* $T \to \infty$. *In particular, the fraction of time that the chain spends in state* i *converges to* π_i *in probability.*

Proof. Let $\bar{f} = \sum_i \pi_i f(i)$, and let $\mathbf{g}$ be the column vector with the entries $g_i = f(i) - \bar{f}$, so that $\boldsymbol{\pi}\mathbf{g} = 0$. We first solve $\mathbf{g} = (I-P)\mathbf{h}$. By the proof of Theorem 8.6, the null space of $(I-P)^\top$ is spanned by $\boldsymbol{\pi}^\top$, so $\operatorname{rank}(I-P) = \operatorname{rank}\left((I-P)^\top\right) = n-1$ (Lecture 1). Every vector $(I-P)\mathbf{h}$

satisfies $\boldsymbol{\pi}(I-P)\mathbf{h} = (\boldsymbol{\pi}-\boldsymbol{\pi}P)\mathbf{h} = 0$, so the column space of $I-P$ is contained in the null space of the nonzero row vector $\boldsymbol{\pi}$; both have dimension $n-1$, so they are equal (Lecture 1), and $\mathbf{g} = (I-P)\mathbf{h}$ for some $\mathbf{h}$. Let $c = \max_i |h_i|$ and $G = \max_i |g_i|$. Since P^k is stochastic, the entries of $P^k\mathbf{h}$ are at most c in absolute value.

Let $S_T = \sum_{t=0}^{T-1} g_{X_t} = \sum_{t=0}^{T-1} f(X_t) - T\bar{f}$. For $s < t$, Proposition 8.3 gives $P(X_s = i, X_t = j) = P(X_s = i)(P^{t-s})_{ij}$, so $\mathbb{E}[g_{X_s} g_{X_t}] = \sum_i P(X_s = i)\, g_i\, (P^{t-s}\mathbf{g})_i$. The sum over t telescopes:

$$\sum_{t=s+1}^{T-1} P^{t-s}\mathbf{g} = \sum_{k=1}^{T-1-s} \left(P^k\mathbf{h} - P^{k+1}\mathbf{h}\right) = P\mathbf{h} - P^{T-s}\mathbf{h},$$

whose entries are at most $2c$ in absolute value. Hence $\left|\sum_{t>s} \mathbb{E}[g_{X_s} g_{X_t}]\right| \le 2cG$ for every s, and

$$\mathbb{E}[S_T^2] = \sum_{s=0}^{T-1} \mathbb{E}[g_{X_s}^2] + 2\sum_{s=0}^{T-1}\sum_{t=s+1}^{T-1} \mathbb{E}[g_{X_s} g_{X_t}] \le T(G^2 + 4cG).$$

By Markov's inequality for S_T^2 (Proposition 3.6), for every $\varepsilon > 0$,

$$P\Big(\Big|\frac{1}{T}\sum_{t=0}^{T-1} f(X_t) - \bar{f}\Big| > \varepsilon\Big) \le \frac{\mathbb{E}[S_T^2]}{T^2\varepsilon^2} \le \frac{G^2 + 4cG}{T\varepsilon^2} \to 0. \qquad \square$$

As for the strong law of large numbers in the appendix to this lecture, the convergence even holds with probability 1; we only prove the weak form. In the long run, $\frac{5}{6}$ of the days are sunny. Since successive states are correlated, these averages converge more slowly than averages of independent samples.

8.1 Detailed balance and Markov chain Monte Carlo

Finding a stationary distribution means solving a linear system. It is often easier to check a stronger, local condition.

Proposition 8.10 (Detailed balance)**.** *If a distribution $\boldsymbol{\pi}$ satisfies $\pi_i P_{ij} = \pi_j P_{ji}$ for all states i and j, then it is stationary.*

Proof. $(\boldsymbol{\pi}P)_j = \sum_i \pi_i P_{ij} = \sum_i \pi_j P_{ji} = \pi_j \sum_i P_{ji} = \pi_j$. $\square$

Detailed balance says that in the stationary regime, the probability flow from i to j equals the flow from j to i; such chains are called reversible.

Example 8.11 (Random walks on graphs)**.** Let G be a connected undirected graph, and let a walker move from each vertex to a uniformly random neighbor, i.e., $P_{ij} = \frac{1}{\deg(i)}$ for each edge $\{i, j\}$. Then $\pi_i = \frac{\deg(i)}{2|E|}$, where $|E|$ is the number of edges, satisfies detailed balance: $\pi_i P_{ij} = \frac{1}{2|E|} = \pi_j P_{ji}$ for every edge. So in the long run, the walker visits each vertex in proportion to its degree. For the graph with the edges $\{1,2\}$, $\{1,3\}$, $\{2,3\}$, $\{3,4\}$, the degrees are $2, 2, 3, 1$, and $\boldsymbol{\pi} = (\frac{1}{4}, \frac{1}{4}, \frac{3}{8}, \frac{1}{8})$.

Remark 8.12 (PageRank). Google's original PageRank models a "random surfer" who, at each step, follows a uniformly random link on the current page with probability α (typically 0.85), and jumps to a uniformly random page with probability $1-\alpha$. All entries of the resulting transition matrix are positive, so the chain is irreducible and aperiodic, and its unique stationary distribution ranks the pages. It is computed by the power iteration $\boldsymbol{\pi} \leftarrow \boldsymbol{\pi}P$ of the programming exercises of Lecture 1.

Detailed balance is also the key to Markov chain Monte Carlo (MCMC), the most important method for sampling from complicated distributions, such as the posterior distributions of Section 12. These are often only known up to a constant: $p(\theta \mid \mathcal{D}) \propto p(\mathcal{D} \mid \theta)\, p(\theta)$, where the evidence $p(\mathcal{D})$ is an intractable integral. The idea is to construct a Markov chain whose stationary distribution is the target distribution, run it for a long time, and use its states as (dependent) samples.

Theorem 8.13 (Metropolis algorithm)**.** *Let $\boldsymbol{\pi}$ be a distribution with all $\pi_i > 0$, and let Q be a symmetric stochastic matrix ($Q_{ij} = Q_{ji}$), the proposal. From the state i, propose a state j with probability Q_{ij}, and move to j with the acceptance probability $\min\left(1, \frac{\pi_j}{\pi_i}\right)$; otherwise, stay at i. This defines a Markov chain that satisfies detailed balance with respect to $\boldsymbol{\pi}$, so $\boldsymbol{\pi}$ is stationary.*

Proof. For $i \neq j$, $P_{ij} = Q_{ij} \min\left(1, \frac{\pi_j}{\pi_i}\right)$, so $\pi_i P_{ij} = Q_{ij} \min(\pi_i, \pi_j)$, which is symmetric in i and j because Q is. For $i = j$, detailed balance holds trivially. □

The algorithm only uses the ratios $\frac{\pi_j}{\pi_i}$, so the unknown normalizing constant cancels. Moves to more probable states are always accepted, and moves to less probable states sometimes. For densities, the same algorithm works with a symmetric proposal such as $\theta' = \theta + \eta\,\varepsilon$ with $\varepsilon \sim \mathcal{N}(0,1)$, and the ratio of the (unnormalized) densities. If the resulting chain is irreducible and aperiodic, Theorem 8.6 and the ergodic theorem justify using averages along the chain as estimates of expectations under $\boldsymbol{\pi}$.

Example 8.14. Let the states $1, 2, 3, 4$ lie on a cycle, let Q propose each of the two neighbors with probability $\frac{1}{2}$, and let the target be $\boldsymbol{\pi} = (0.1, 0.2, 0.3, 0.4)$. The Metropolis chain has the transition matrix

$$P = \begin{pmatrix} 0 & 0.5 & 0 & 0.5 \\ 0.25 & 0.25 & 0.5 & 0 \\ 0 & 1/3 & 1/6 & 0.5 \\ 0.125 & 0 & 0.375 & 0.5 \end{pmatrix};$$

for example, from state 3, the move to state 2 is proposed with probability $\frac{1}{2}$ and accepted with probability $\frac{0.2}{0.3}$, so $P_{32} = \frac{1}{3}$. One checks that $\boldsymbol{\pi} P = \boldsymbol{\pi}$.

Gibbs sampling. If the state is a vector $\mathbf{x} = (x_1, \ldots, x_d)$, for example the parameters of a Bayesian model or the pixels of an image, it is often easy to sample a single entry from its conditional distribution given all the others, even if the joint distribution π is complicated. The Gibbs sampler updates the entries one at a time: in the update of entry i, it replaces x_i by a sample from $\pi(x_i \mid \mathbf{x}_{-i})$, the conditional distribution of the i-th entry given the other entries $\mathbf{x}_{-i}$, and keeps $\mathbf{x}_{-i}$.

Proposition 8.15 (Gibbs sampling)**.** *Let π be a distribution on a finite set of vectors $\mathbf{x} = (x_1, \ldots, x_d)$. The update of entry i is a Markov chain that satisfies detailed balance with respect to π. Hence every sequence of such updates, for example a sweep through $i = 1, \ldots, d$, leaves π stationary. (The same holds for densities.)*

Proof. The update of entry i moves from $\mathbf{x}$ to $\mathbf{x}'$ only if $\mathbf{x}'_{-i} = \mathbf{x}_{-i}$, and then with the probability $P_{\mathbf{x}\mathbf{x}'} = \pi(x'_i \mid \mathbf{x}_{-i}) = \frac{\pi(\mathbf{x}')}{\pi_{-i}(\mathbf{x}_{-i})}$, where π_{-i} is the marginal distribution of $\mathbf{x}_{-i}$. So $\pi(\mathbf{x}) P_{\mathbf{x}\mathbf{x}'} = \frac{\pi(\mathbf{x})\pi(\mathbf{x}')}{\pi_{-i}(\mathbf{x}_{-i})}$ is symmetric in $\mathbf{x}$ and $\mathbf{x}'$, and π is stationary by Proposition 8.10. If $\pi P_1 = \pi$ and $\pi P_2 = \pi$, then also $\pi P_1 P_2 = \pi P_2 = \pi$. □

Gibbs sampling needs no proposal and no rejections, but it moves only along the coordinate axes, which is slow if the entries are strongly dependent. For a standard bivariate Gaussian with correlation ρ, the conditional distributions are $X_1 \mid X_2 = x_2 \sim \mathcal{N}(\rho x_2, 1-\rho^2)$ and vice versa (Theorem 6.5). A sweep maps X_1 to $\rho^2 X_1$ plus independent Gaussian noise, so successive values of X_1 have the correlation ρ^2. For $\rho = 0.99$, it is 0.98, and it takes about 35 sweeps until the correlation drops below $\frac{1}{2}$ (since $0.98^{35} \approx 0.5$). Gibbs sampling is the method of choice for models with conditionally conjugate priors, such as mixture models and topic models, and for Markov random fields such as the Ising model.

8.2 Markov reward processes and the Bellman equation

Reinforcement learning studies agents that collect rewards while they move through the states of a Markov chain. Before the agent makes any decisions, the basic question is how much reward to expect in the long run.

Definition 8.16 (Markov reward process)**.** A Markov reward process consists of a Markov chain $X_0, X_1, \ldots$ with the transition matrix P on the states $\{1, \ldots, n\}$, a reward vector $\mathbf{r} \in \mathbb{R}^n$ (the reward r_i is collected in every step that the chain spends in state i), and a discount factor $\gamma \in [0, 1)$. The value of the state i is the expected discounted total reward

$$V_i = \mathbb{E}\Big[\sum_{t=0}^{\infty} \gamma^t r_{X_t} \;\Big|\; X_0 = i\Big],$$

and $\mathbf{V} = (V_1, \ldots, V_n)^\top$ is the value vector.

The discount factor makes rewards in the far future count less. One can also read $1-\gamma$ as the probability that the process stops after each step, so that an episode lasts $\frac{1}{1-\gamma}$ steps on average. Since $|r_{X_t}| \le \|\mathbf{r}\|_\infty$, the series converges absolutely, uniformly in the outcomes, so we may take expectations term by term, and $|V_i| \le \frac{\|\mathbf{r}\|_\infty}{1-\gamma}$. We use the maximum norm $\|\mathbf{x}\|_\infty = \max_i |x_i|$ of Lecture 2. For a stochastic matrix P, each entry $(P\mathbf{x})_i = \sum_j P_{ij} x_j$ is a weighted average of the entries of $\mathbf{x}$, so

$$\|P\mathbf{x}\|_\infty \le \|\mathbf{x}\|_\infty \qquad \text{for all } \mathbf{x} \in \mathbb{R}^n.$$

Theorem 8.17 (Bellman equation)**.** *The value vector of a Markov reward process satisfies the Bellman equation*

$$\mathbf{V} = \mathbf{r} + \gamma P \mathbf{V}, \qquad \textit{i.e.,} \qquad V_i = r_i + \gamma \sum_{j=1}^{n} P_{ij} V_j.$$

The matrix $I - \gamma P$ is invertible, with $(I-\gamma P)^{-1} = \sum_{k=0}^{\infty} \gamma^k P^k$, and $\mathbf{V} = (I-\gamma P)^{-1}\mathbf{r}$ is the only solution of the Bellman equation.

Proof. By Proposition 8.3, $\mathbb{E}[r_{X_k} \mid X_0 = i] = \sum_j (P^k)_{ij} r_j = (P^k \mathbf{r})_i$, so $\mathbf{V} = \sum_{k \ge 0} \gamma^k P^k \mathbf{r}$. For every $\mathbf{x}$, $\|\gamma^k P^k \mathbf{x}\|_\infty \le \gamma^k \|\mathbf{x}\|_\infty$, so the series $S = \sum_{k \ge 0} \gamma^k P^k$ converges (apply it to $\mathbf{x} = \mathbf{e}_j$ to get the columns). Since

$$(I - \gamma P) \sum_{k=0}^{K-1} \gamma^k P^k = I - \gamma^K P^K \to I \qquad (K \to \infty),$$

and in the same way with the factors in the other order, $(I-\gamma P)S = S(I-\gamma P) = I$. So $I - \gamma P$ is invertible with the inverse S, and $\mathbf{V} = S\mathbf{r}$ is the only solution of $(I-\gamma P)\mathbf{V} = \mathbf{r}$, which is the Bellman equation. □

The Bellman equation says that the value of a state is the reward now plus the discounted expected value of the next state; this "first-step analysis" is how it is usually derived.

Example 8.18 (The value of sunny days)**.** In the weather chain of Example 8.2, let every sunny day give the reward 1, i.e., $\mathbf{r} = (1, 0)^\top$, and let $\gamma = 0.9$. Then

$$I - \gamma P = \begin{pmatrix} 0.19 & -0.09 \\ -0.45 & 0.55 \end{pmatrix}, \qquad \mathbf{V} = (I - \gamma P)^{-1}\mathbf{r} = \frac{1}{0.064}\begin{pmatrix} 0.55 \\ 0.45 \end{pmatrix} = \begin{pmatrix} 8.59375 \\ 7.03125 \end{pmatrix}.$$

Check: $1 + 0.9(0.9 \cdot 8.59375 + 0.1 \cdot 7.03125) = 8.59375$. With $\frac{1}{1-\gamma} = 10$ days as the effective horizon, the normalized values $(1 - \gamma)\mathbf{V} = (0.859, 0.703)$ are discounted averages of the fraction of sunny days. A sunny start is worth more, but the advantage fades: as $\gamma \to 1$, both entries of $(1 - \gamma)\mathbf{V}$ tend to the long-run fraction $\pi_1 = \frac{5}{6}$ of sunny days (Exercise 25).

For large state spaces, solving the linear system is too expensive, and P may not even be known explicitly. The Bellman equation then suggests an iteration.

Proposition 8.19 (Iterative policy evaluation)**.** *Let $\mathbf{V}_0 \in \mathbb{R}^n$ be arbitrary, and let $\mathbf{V}_{k+1} = \mathbf{r} + \gamma P\mathbf{V}_k$. Then $\|\mathbf{V}_k - \mathbf{V}\|_\infty \le \gamma^k \|\mathbf{V}_0 - \mathbf{V}\|_\infty$.*

Proof. Subtracting the Bellman equation, $\mathbf{V}_{k+1} - \mathbf{V} = \gamma P(\mathbf{V}_k - \mathbf{V})$, so $\|\mathbf{V}_{k+1} - \mathbf{V}\|_\infty \le \gamma \|\mathbf{V}_k - \mathbf{V}\|_\infty$; use induction. □

In other words, the map $B(\mathbf{V}) = \mathbf{r} + \gamma P\mathbf{V}$ is a contraction, $\|B(\mathbf{U}) - B(\mathbf{V})\|_\infty \le \gamma \|\mathbf{U} - \mathbf{V}\|_\infty$, and $\mathbf{V}$ is its fixed point. In Example 8.18, starting from $\mathbf{V}_0 = \mathbf{0}$, the iterates are $(1, 0)$, $(1.81, 0.45)$, $(2.51, 1.02)$, …, and the error bound $0.9^k \cdot 8.6$ drops below 0.01 after 65 steps. The convergence is slow if γ is close to 1, i.e., if the horizon is long.

Markov decision processes. In reinforcement learning, an agent chooses an action a from a finite set A in every state s. It then receives the reward $r(s, a)$, and the next state is s' with the probability $P(s' \mid s, a)$. A (deterministic) policy π chooses the action $\pi(s)$ in the state s. For a fixed policy, the states form a Markov reward process with the transition probabilities $P^\pi_{ss'} = P(s' \mid s, \pi(s))$ and the rewards $r^\pi_s = r(s, \pi(s))$, and its value vector $\mathbf{V}^\pi$ solves the Bellman equation $\mathbf{V}^\pi = \mathbf{r}^\pi + \gamma P^\pi \mathbf{V}^\pi$ (policy evaluation). The agent wants a policy with the largest value. The Bellman optimality operator is

$$(T\mathbf{V})_s = \max_{a \in A} \Big(r(s, a) + \gamma \sum_{s'} P(s' \mid s, a) V_{s'} \Big).$$

Theorem 8.20 (Value iteration)**.** *1. $\|T\mathbf{U} - T\mathbf{V}\|_\infty \le \gamma \|\mathbf{U} - \mathbf{V}\|_\infty$ for all $\mathbf{U}, \mathbf{V} \in \mathbb{R}^n$. Hence T has exactly one fixed point $\mathbf{V}^*$, the solution of the Bellman optimality equation $\mathbf{V}^* = T\mathbf{V}^*$, and the iterates $\mathbf{V}_{k+1} = T\mathbf{V}_k$ of value iteration satisfy $\|\mathbf{V}_k - \mathbf{V}^*\|_\infty \le \gamma^k \|\mathbf{V}_0 - \mathbf{V}^*\|_\infty$.*

2. $\mathbf{V}^\pi \le \mathbf{V}^*$ (entrywise) for every policy π.

3. The greedy policy π^*, which chooses in every state s an action that attains the maximum in $(T\mathbf{V}^*)_s$, has $\mathbf{V}^{\pi^*} = \mathbf{V}^*$. So it is an optimal policy, and $\mathbf{V}^*$ is the optimal value.

Proof. 1. For functions f, g on A, $|\max_a f(a) - \max_a g(a)| \le \max_a |f(a) - g(a)|$: if $\max f$ is attained at a^*, then $\max f - \max g \le f(a^*) - g(a^*)$, and symmetrically. Apply this with $f(a)$ and $g(a)$ the brackets in $(T\mathbf{U})_s$ and $(T\mathbf{V})_s$, whose difference is $\gamma \sum_{s'} P(s' \mid s, a)(U_{s'} - V_{s'})$,

of absolute value at most $\gamma\|\mathbf{U}-\mathbf{V}\|_\infty$. Now let $\mathbf{V}_0$ be arbitrary and $\mathbf{V}_{k+1} = T\mathbf{V}_k$. Then $\|\mathbf{V}_{k+1}-\mathbf{V}_k\|_\infty \le \gamma^k\|\mathbf{V}_1-\mathbf{V}_0\|_\infty$, so for $m > k$, $\|\mathbf{V}_m-\mathbf{V}_k\|_\infty \le (\gamma^k+\cdots+\gamma^{m-1})\|\mathbf{V}_1-\mathbf{V}_0\|_\infty \le \frac{\gamma^k}{1-\gamma}\|\mathbf{V}_1-\mathbf{V}_0\|_\infty$. Each entry of $\mathbf{V}_k$ is a Cauchy sequence of real numbers, so $\mathbf{V}_k$ converges to some $\mathbf{V}^*$, and since T is continuous, $T\mathbf{V}^* = \lim T\mathbf{V}_k = \lim \mathbf{V}_{k+1} = \mathbf{V}^*$. If also $T\mathbf{U} = \mathbf{U}$, then $\|\mathbf{U}-\mathbf{V}^*\|_\infty \le \gamma\|\mathbf{U}-\mathbf{V}^*\|_\infty$, so $\mathbf{U} = \mathbf{V}^*$. The error bound follows as in Proposition 8.19. (This is the Banach fixed-point theorem of the appendix to Lecture 3, for the maximum norm.)

2. T is monotone: if $\mathbf{U} \le \mathbf{V}$, then $T\mathbf{U} \le T\mathbf{V}$, since the probabilities $P(s' \mid s,a)$ are nonnegative. Moreover, $\mathbf{V}^\pi = \mathbf{r}^\pi + \gamma P^\pi \mathbf{V}^\pi \le T\mathbf{V}^\pi$, since the action $\pi(s)$ is one of the candidates in the maximum. By monotonicity and induction, $\mathbf{V}^\pi \le T^k\mathbf{V}^\pi$ for all k, and $T^k\mathbf{V}^\pi \to \mathbf{V}^*$ by part 1.

3. By the choice of π^*, $\mathbf{r}^{\pi^*} + \gamma P^{\pi^*}\mathbf{V}^* = T\mathbf{V}^* = \mathbf{V}^*$. So $\mathbf{V}^*$ solves the Bellman equation of the Markov reward process of π^*, and by the uniqueness in Theorem 8.17, $\mathbf{V}^{\pi^*} = \mathbf{V}^*$. □

The expression in the bracket is the action value $Q^*(s,a) = r(s,a) + \gamma\sum_{s'} P(s' \mid s,a)V^*_{s'}$, the value of taking the action a in the state s and acting optimally afterwards, and $V^*_s = \max_a Q^*(s,a)$. If the transition probabilities are unknown, Q-learning observes single transitions (s,a,r,s') of the agent and updates

$$Q(s,a) \leftarrow Q(s,a) + \eta\big(r + \gamma\max_{a'} Q(s',a') - Q(s,a)\big) :$$

it replaces the expectation over s' in value iteration by a sample, in the same way as SGD replaces the full gradient by a stochastic gradient (Section 7.3). Its convergence, for suitable step sizes η and if every pair (s,a) is tried infinitely often, is proved with the same kind of stochastic approximation argument as for SGD. Deep Q-networks approximate Q by a neural network.

8.3 Stationary time series and autoregressive models

A time series is a sequence of random variables $X_0, X_1, X_2, \ldots$ indexed by time: daily sales, temperatures, server loads, or the training loss of a neural network. Unlike i.i.d. data, neighboring values are dependent, and the simplest description of this dependence is by covariances.

Definition 8.21 (Stationary time series)**.** A sequence $X_0, X_1, \ldots$ of random variables with finite variances is (weakly) stationary if the mean $\mathbb{E}[X_t] = \mu$ and the autocovariances $c(h) = \mathrm{Cov}(X_t, X_{t+h})$ do not depend on t, for all $t, h \ge 0$. Its autocorrelation function is $\rho(h) = \frac{c(h)}{c(0)}$.

So a stationary series has the same mean and variance at all times, and the dependence between two values depends only on their distance in time. (We write $c(h)$ rather than the common $\gamma(h)$, to keep γ for discount factors.) Three basic examples, with uncorrelated noise terms ε_t with mean 0 and variance σ^2 (white noise):

- White noise itself is stationary, with $c(0) = \sigma^2$ and $c(h) = 0$ for $h \ge 1$.
- The moving average $X_t = \varepsilon_t + \theta\varepsilon_{t-1}$, an MA(1) process, is stationary, with $c(0) = (1+\theta^2)\sigma^2$, $c(1) = \theta\sigma^2$ and $c(h) = 0$ for $h \ge 2$: values more than one step apart are uncorrelated. Its autocorrelation $\rho(1) = \frac{\theta}{1+\theta^2}$ is at most $\frac{1}{2}$ in absolute value.
- The random walk $S_t = \varepsilon_1 + \cdots + \varepsilon_t$ (with $S_0 = 0$) has $\mathrm{Var}(S_t) = t\sigma^2$, so it is not stationary. Its differences $S_t - S_{t-1} = \varepsilon_t$ are. Trends of this kind are removed by differencing before a stationary model is fitted; this is the "I" (integrated) in ARIMA models.

The most important model is the autoregressive process, in which each value is a multiple of the previous one plus fresh noise.

Proposition 8.22 (The AR(1) process)**.** *Let $\varepsilon_1, \varepsilon_2, \dots$ be uncorrelated, with mean 0 and variance $\sigma^2 > 0$, let X_0 have a finite variance and be uncorrelated with all ε_t, and let $X_t = \phi X_{t-1} + \varepsilon_t$ for $t \geq 1$.*

1. *$X_t = \phi^t X_0 + \sum_{k=0}^{t-1} \phi^k \varepsilon_{t-k}$. Hence $\mathbb{E}[X_t] = \phi^t \mathbb{E}[X_0]$, $\operatorname{Var}(X_t) = \phi^{2t} \operatorname{Var}(X_0) + \sigma^2 \sum_{k=0}^{t-1} \phi^{2k}$, and $\operatorname{Cov}(X_t, X_{t+h}) = \phi^h \operatorname{Var}(X_t)$.*

2. *Let $|\phi| < 1$. If $\mathbb{E}[X_0] = 0$ and $\operatorname{Var}(X_0) = \frac{\sigma^2}{1-\phi^2}$, then the series is stationary, with*

$$c(h) = \frac{\sigma^2 \phi^h}{1 - \phi^2} \qquad \text{and} \qquad \rho(h) = \phi^h.$$

   For every X_0, $\mathbb{E}[X_t] \to 0$ and $\operatorname{Var}(X_t) \to \frac{\sigma^2}{1-\phi^2}$ as $t \to \infty$.

3. *If $|\phi| \geq 1$, then $\operatorname{Var}(X_t) \geq \operatorname{Var}(X_0) + t\sigma^2 \to \infty$, so the series is not stationary, whatever X_0 is.*

Proof. 1. The formula for X_t follows by induction. The terms $\phi^t X_0, \phi^0 \varepsilon_t, \dots, \phi^{t-1} \varepsilon_1$ are uncorrelated, which gives the variance (Proposition 4.9). In the same way, $X_{t+h} = \phi^h X_t + \sum_{k=0}^{h-1} \phi^k \varepsilon_{t+h-k}$, and the noise terms $\varepsilon_{t+1}, \dots, \varepsilon_{t+h}$ are uncorrelated with X_t, which is a linear combination of $X_0, \varepsilon_1, \dots, \varepsilon_t$. So $\operatorname{Cov}(X_t, X_{t+h}) = \phi^h \operatorname{Var}(X_t)$.

2. By part 1 and the geometric sum, $\operatorname{Var}(X_t) = \phi^{2t} \operatorname{Var}(X_0) + \sigma^2 \frac{1-\phi^{2t}}{1-\phi^2}$, which is $\frac{\sigma^2}{1-\phi^2}$ for all t if $\operatorname{Var}(X_0) = \frac{\sigma^2}{1-\phi^2}$. Then $c(h) = \phi^h \frac{\sigma^2}{1-\phi^2}$, and the mean is $\phi^t \cdot 0 = 0$. For a general X_0, $\phi^t \to 0$ and $\phi^{2t} \to 0$.

3. $\operatorname{Var}(X_t) = \phi^2 \operatorname{Var}(X_{t-1}) + \sigma^2 \geq \operatorname{Var}(X_{t-1}) + \sigma^2$; use induction. □

So for $|\phi| < 1$, the process forgets its starting point and settles into a stationary regime, whose correlations decay geometrically. If the noise is Gaussian and independent, the AR(1) process is a Markov chain with values in $\mathbb{R}$, its stationary distribution is $\mathcal{N}(0, \frac{\sigma^2}{1-\phi^2})$, and the best prediction of X_{t+h} given the past is $\mathbb{E}[X_{t+h} \mid X_t] = \phi^h X_t$ (by the formula for X_{t+h} in the proof), with the error variance $\sigma^2 \frac{1-\phi^{2h}}{1-\phi^2}$. Forecasts revert to the mean, and their uncertainty grows to the stationary variance. For example, for $\phi = 0.8$ and $\sigma^2 = 1$, the stationary variance is $\frac{1}{0.36} \approx 2.78$, and $\rho(3) = 0.512$. The AR(1) process is the time discretization of the Ornstein–Uhlenbeck process of Section 15.

To fit the model to data $x_0, \dots, x_T$, one regresses x_t on x_{t-1}: the least squares estimate, which is the maximum likelihood estimate for Gaussian noise given x_0, is

$$\hat{\phi} = \frac{\sum_{t=1}^{T} x_t x_{t-1}}{\sum_{t=1}^{T} x_{t-1}^2},$$

close to the sample autocorrelation at lag 1.

AR(p) processes and the companion matrix. An AR(p) process $X_t = \phi_1 X_{t-1} + \cdots + \phi_p X_{t-p} + \varepsilon_t$ uses p past values. The vector $\mathbf{Y}_t = (X_t, X_{t-1}, \ldots, X_{t-p+1})^\top$ follows the AR(1)-type recursion

$$\mathbf{Y}_t = A\mathbf{Y}_{t-1} + \varepsilon_t \mathbf{e}_1, \qquad A = \begin{pmatrix} \phi_1 & \phi_2 & \cdots & \phi_{p-1} & \phi_p \\ 1 & 0 & \cdots & 0 & 0 \\ 0 & 1 & \cdots & 0 & 0 \\ \vdots & & \ddots & & \vdots \\ 0 & 0 & \cdots & 1 & 0 \end{pmatrix},$$

with the companion matrix A. As in part 1 of Proposition 8.22, $\mathbf{Y}_t = A^t \mathbf{Y}_0 + \sum_{k=0}^{t-1} A^k \mathbf{e}_1 \varepsilon_{t-k}$, so the process forgets its start and becomes stationary if $A^k \to 0$. This happens if and only if all complex eigenvalues of A have absolute value less than 1; for a diagonalizable $A = QDQ^{-1}$ (with complex entries), this follows from $A^k = QD^kQ^{-1}$ (Lecture 1). The eigenvalues are the roots of $\lambda^p - \phi_1\lambda^{p-1} - \cdots - \phi_p$ (Exercise 26), and this condition is what time series software checks when it fits AR, ARMA and ARIMA models. For example, $X_t = 0.5X_{t-1} + 0.3X_{t-2} + \varepsilon_t$ has the eigenvalues 0.85 and -0.35 and is stationary, while for $X_t = 0.5X_{t-1} + 0.6X_{t-2} + \varepsilon_t$, the eigenvalue 1.06 makes the variance explode. The same condition on the eigenvalues of a matrix that is applied again and again decides whether the gradients of a recurrent neural network vanish or explode (Lecture 3).

Remark 8.23 (Markov models in machine learning). An n-gram language model is a Markov chain on words (or on tuples of $n-1$ words), whose transition probabilities are estimated by counting in a large text, which is maximum likelihood estimation (Section 9). Hidden Markov models, in which only a noisy function of the state is observed, were long the standard model in speech recognition. Reinforcement learning is based on Markov decision processes, in which the transition probabilities depend on the actions of an agent (Section 8.2). Diffusion models for image generation learn to reverse a Markov chain that gradually adds Gaussian noise to an image (Section 15.1).

9 Parameter Estimation

Statistics turns the questions of probability around: instead of computing probabilities from a known distribution, we infer an unknown distribution from data. We assume that the data $x_1, \ldots, x_N$ are i.i.d. samples from a distribution $p(x \mid \theta)$ that belongs to a known family, with an unknown parameter θ (for example, the mean and variance of a Gaussian, or the weights of a model).

Definition 9.1 (Estimator, bias and variance)**.** An estimator is a function $\hat\theta = \hat\theta(X_1, \ldots, X_N)$ of the data; as a function of random data, it is a random variable. Its bias is $\mathbb{E}[\hat\theta] - \theta$, and it is unbiased if the bias is 0 for all θ. Its mean squared error is $\mathrm{MSE}(\hat\theta) = \mathbb{E}[(\hat\theta - \theta)^2]$.

Proposition 9.2 (Bias–variance decomposition)**.** $\mathrm{MSE}(\hat\theta) = \big(\mathbb{E}[\hat\theta] - \theta\big)^2 + \mathrm{Var}(\hat\theta)$.

Proof. Let $m = \mathbb{E}[\hat\theta]$. Then $\mathbb{E}[(\hat\theta - \theta)^2] = \mathbb{E}[(\hat\theta - m)^2] + 2(m-\theta)\mathbb{E}[\hat\theta - m] + (m-\theta)^2$, and the middle term is 0. □

A good estimator has a small bias and a small variance, and sometimes we accept some bias for a much smaller variance. This is the idea behind regularization: ridge regression shrinks the weights towards 0, which introduces a bias but reduces the variance. For a prediction model, a similar decomposition says that the expected squared test error is the sum of the squared bias, the variance and the irreducible noise; very flexible models have a small bias but a large variance (they overfit), and very simple ones the opposite.

Example 9.3 (Shrinkage)**.** Let $X_1, \dots, X_N$ be i.i.d. $\mathcal{N}(\mu, \sigma^2)$, and consider the estimators $\hat{\mu}_c = c\bar{X}_N$ with $0 \le c \le 1$, which shrink the sample mean towards 0. Their bias is $(c-1)\mu$ and their variance $c^2\frac{\sigma^2}{N}$, so

$$\mathrm{MSE}(\hat{\mu}_c) = (1-c)^2\mu^2 + c^2\frac{\sigma^2}{N},$$

which is minimal for $c^* = \frac{\mu^2}{\mu^2+\sigma^2/N}$. For $\mu = 2$, $\sigma^2 = 16$ and $N = 4$, the unbiased sample mean has the MSE 4, while $\hat{\mu}_{1/2}$ has the MSE $1 + 1 = 2$. The best c depends on the unknown μ, so this estimator is not available in practice, but it shows that some bias can pay off. Remarkably, the James–Stein estimator achieves this in a data-driven way: for the means of $d \ge 3$ independent Gaussians, it shrinks the sample means towards 0, and its total MSE is smaller than that of the sample means, whatever the true means are.

By Proposition 7.1, the sample mean is an unbiased estimator of μ, with variance $\frac{\sigma^2}{N}$. The natural estimator of the variance, however, is biased.

Proposition 9.4 (Sample variance)**.** *Let $X_1, \dots, X_N$ be i.i.d. with mean μ and variance σ^2, and $N \ge 2$. Then*

$$\mathbb{E}\Big[\frac{1}{N}\sum_{i=1}^{N}(X_i - \bar{X}_N)^2\Big] = \frac{N-1}{N}\sigma^2, \qquad \textit{so} \qquad s^2 = \frac{1}{N-1}\sum_{i=1}^{N}(X_i - \bar{X}_N)^2$$

is an unbiased estimator of σ^2.

Proof. Expanding $X_i - \mu = (X_i - \bar{X}_N) + (\bar{X}_N - \mu)$ and using $\sum_i (X_i - \bar{X}_N) = 0$ gives

$$\sum_i (X_i - \mu)^2 = \sum_i (X_i - \bar{X}_N)^2 + N(\bar{X}_N - \mu)^2.$$

Taking expectations, $N\sigma^2 = \mathbb{E}\big[\sum_i (X_i - \bar{X}_N)^2\big] + N \cdot \frac{\sigma^2}{N}$. □

Intuitively, the data are on average closer to their own mean $\bar{X}_N$ than to the true mean μ, so the version with $\frac{1}{N}$ underestimates the variance. The factor $\frac{N}{N-1}$ (Bessel's correction) only matters for small N; `np.var` uses $\frac{1}{N}$ by default, and `np.var(x, ddof=1)` uses $\frac{1}{N-1}$.

9.1 The method of moments

The oldest general method of estimation chooses the parameters so that the model reproduces the moments of the data.

Definition 9.5 (Method of moments)**.** Let $p(x \mid \boldsymbol{\theta})$ be a family of distributions with k parameters. The method of moments estimate $\hat{\boldsymbol{\theta}}$ solves the k equations

$$\mathbb{E}_{\boldsymbol{\theta}}[X^j] = \frac{1}{N}\sum_{i=1}^{N} x_i^j, \qquad j = 1, \dots, k,$$

(or equivalent equations for the mean, the variance, and so on).

By the law of large numbers, the sample moments converge to the true moments, so the method of moments gives consistent estimates whenever the moment equations determine $\boldsymbol{\theta}$ in a continuous way.

Example 9.6. 1. Exponential(λ): $\mathbb{E}[X] = \frac{1}{\lambda}$ gives $\hat{\lambda} = \frac{1}{\bar{x}}$.

2. Uniform$(0, \theta)$: $\mathbb{E}[X] = \frac{\theta}{2}$ gives $\hat{\theta} = 2\bar{x}$. This estimator is unbiased, but it can be smaller than the largest observation, which is impossible for the true θ (compare with the MLE in the exercises).

3. The Gamma distribution with shape $k > 0$ and scale $s > 0$ has the density $\frac{x^{k-1}e^{-x/s}}{\Gamma(k)s^k}$ for $x > 0$, the mean ks and the variance ks^2. Matching the sample mean $\bar{x}$ and the sample variance $\hat{\sigma}^2$ gives $\hat{s} = \frac{\hat{\sigma}^2}{\bar{x}}$ and $\hat{k} = \frac{\bar{x}^2}{\hat{\sigma}^2}$; for example, $\bar{x} = 6$ and $\hat{\sigma}^2 = 12$ give $\hat{k} = 3$ and $\hat{s} = 2$. The maximum likelihood estimate of k has no closed form, and the method of moments provides a good starting point for its numerical computation (for example with Newton's method, Lecture 3).

The method of moments is simple, but its estimates usually have a larger variance than maximum likelihood estimates, which we discuss next.

9.2 Maximum likelihood estimation

The most important general method to construct estimators is to choose the parameter under which the observed data are most probable.

Definition 9.7 (Maximum likelihood)**.** The likelihood of a parameter θ for i.i.d. data $x_1, \dots, x_N$ is $L(\theta) = \prod_{i=1}^{N} p(x_i \mid \theta)$, and the log-likelihood is $\ell(\theta) = \log L(\theta) = \sum_{i=1}^{N} \log p(x_i \mid \theta)$. A maximum likelihood estimate (MLE) is a parameter $\hat{\theta}$ that maximizes L, or equivalently ℓ.

Taking logarithms turns the product into a sum, which is easier to differentiate and numerically more stable. Maximizing ℓ is the same as minimizing the negative log-likelihood $-\ell(\theta)$, which is a loss function in the sense of Lecture 3.

Example 9.8 (Bernoulli)**.** Let $x_i \in \{0, 1\}$ with k ones among N observations, and $p(x \mid \theta) = \theta^x(1-\theta)^{1-x}$. Then $\ell(\theta) = k \log\theta + (N-k)\log(1-\theta)$, and $\ell'(\theta) = \frac{k}{\theta} - \frac{N-k}{1-\theta} = 0$ gives $\hat{\theta} = \frac{k}{N}$, the relative frequency. Since $\ell'' < 0$, this is the maximum. Note that after three heads in three flips, $\hat{\theta} = 1$: the MLE claims that tails are impossible. With little data, maximum likelihood overfits; we return to this example in Section 12.

Example 9.9 (Gaussian)**.** For $x_i \sim \mathcal{N}(\mu, \sigma^2)$,

$$\ell(\mu, \sigma^2) = -\frac{N}{2}\log(2\pi\sigma^2) - \frac{1}{2\sigma^2}\sum_{i=1}^{N}(x_i - \mu)^2.$$

Setting the partial derivatives to zero gives $\hat{\mu} = \bar{x}$ and $\hat{\sigma}^2 = \frac{1}{N}\sum_i (x_i - \bar{x})^2$: the MLE of the variance is the biased version of Proposition 9.4. For the data $2.1, 3.4, 1.9, 2.8, 3.3$, we get $\hat{\mu} = 2.7$, $\hat{\sigma}^2 = 0.372$, while $s^2 = 0.465$.

Theorem 9.10 (MLE of a multivariate Gaussian)**.** *For i.i.d. data $\mathbf{x}_1, \dots, \mathbf{x}_N$ from $\mathcal{N}(\boldsymbol{\mu}, \Sigma)$, whose sample covariance matrix S is invertible, the maximum likelihood estimates are the sample mean $\hat{\boldsymbol{\mu}} = \bar{\mathbf{x}}$ and the sample covariance matrix $\hat{\Sigma} = S = \frac{1}{N}\sum_i (\mathbf{x}_i - \bar{\mathbf{x}})(\mathbf{x}_i - \bar{\mathbf{x}})^\top$.*

Proof. Up to a constant, $\ell = -\frac{N}{2}\log\det\Sigma - \frac{1}{2}\sum_i (\mathbf{x}_i - \boldsymbol{\mu})^\top \Sigma^{-1}(\mathbf{x}_i - \boldsymbol{\mu})$. For fixed Σ, this is a concave quadratic function of $\boldsymbol{\mu}$ with gradient $\Sigma^{-1}\sum_i (\mathbf{x}_i - \boldsymbol{\mu})$ (Lecture 3), which vanishes for

$\boldsymbol{\mu} = \bar{\mathbf{x}}$, whatever Σ is. With $\boldsymbol{\mu} = \bar{\mathbf{x}}$, we write ℓ in terms of the precision matrix $\Lambda = \Sigma^{-1}$. Since $\mathbf{a}^\top \Lambda \mathbf{a} = \operatorname{tr}(\Lambda \mathbf{a}\mathbf{a}^\top)$,

$$\ell = \frac{N}{2} \log\det\Lambda - \frac{1}{2}\sum_i \operatorname{tr}\left(\Lambda(\mathbf{x}_i - \bar{\mathbf{x}})(\mathbf{x}_i - \bar{\mathbf{x}})^\top\right) = \frac{N}{2}\big(\log\det\Lambda - \operatorname{tr}(\Lambda S)\big).$$

By the matrix gradients of Lecture 3 ($\nabla \log\det\Lambda = \Lambda^{-\top}$ and $\nabla \operatorname{tr}(\Lambda S) = S^\top$), the gradient with respect to Λ is $\frac{N}{2}(\Lambda^{-1} - S)$, which vanishes for $\Lambda = S^{-1}$, i.e., $\Sigma = S$. (One can show that $\log\det$ is concave on the symmetric positive definite matrices, so this critical point is the maximum.) $\square$

9.3 Consistency and efficiency

How good can an estimator be? Two properties are central: an estimator should converge to the true parameter as the amount of data grows, and its variance should be as small as possible.

Definition 9.11 (Consistency)**.** A sequence of estimators $\hat{\theta}_N$, computed from N data points, is consistent if $\hat{\theta}_N \to \theta$ in probability (Definition 7.8) for every true parameter θ, i.e., if $P(|\hat{\theta}_N - \theta| \geq \varepsilon) \to 0$ as $N \to \infty$ for every $\varepsilon > 0$.

Proposition 9.12. *If the bias and the variance of $\hat{\theta}_N$ tend to 0 as $N \to \infty$, then $\hat{\theta}_N$ is consistent.*

Proof. By Markov's inequality applied to $(\hat{\theta}_N - \theta)^2$ and Proposition 9.2, $P(|\hat{\theta}_N - \theta| \geq \varepsilon) \leq \frac{\operatorname{MSE}(\hat{\theta}_N)}{\varepsilon^2} = \frac{\operatorname{bias}^2 + \operatorname{Var}(\hat{\theta}_N)}{\varepsilon^2} \to 0$. $\square$

For example, the sample mean and both versions of the sample variance are consistent, and so are the method of moments estimators. A lower bound for the variance comes from the curvature of the log-likelihood. For a family $p(x \mid \theta)$ with a single parameter, the score is $s(x;\theta) = \frac{\partial}{\partial\theta}\log p(x \mid \theta)$. Differentiating $\int p(x \mid \theta)\,dx = 1$ under the integral sign gives $\int \frac{\partial p}{\partial\theta}\,dx = \int s\,p\,dx = 0$, i.e., $\mathbb{E}_\theta[s(X;\theta)] = 0$. The Fisher information is

$$I(\theta) = \mathbb{E}_\theta\big[s(X;\theta)^2\big] = \operatorname{Var}_\theta\big(s(X;\theta)\big),$$

and one can show that also $I(\theta) = -\mathbb{E}_\theta\big[\frac{\partial^2}{\partial\theta^2}\log p(X \mid \theta)\big]$: it measures how sharply the log-likelihood is curved around the true parameter, i.e., how much information one observation carries about θ.

Theorem 9.13 (Cramér–Rao bound)**.** *Let $\hat{\theta}$ be an unbiased estimator of θ from N i.i.d. observations, and assume that we may differentiate under the integral sign (Lecture 3; this excludes, e.g., families whose support depends on θ). Then*

$$\operatorname{Var}_\theta(\hat{\theta}) \geq \frac{1}{N\,I(\theta)}.$$

Proof. Let $S = \sum_{i=1}^N s(X_i;\theta)$ be the score of the whole sample. It has mean 0 and, by independence, variance $N\,I(\theta)$. Write $p_N(\mathbf{x} \mid \theta) = \prod_i p(x_i \mid \theta)$, so that $\frac{\partial p_N}{\partial\theta} = p_N S$. Differentiating the unbiasedness condition $\theta = \int \hat{\theta}(\mathbf{x})\,p_N(\mathbf{x} \mid \theta)\,d\mathbf{x}$ gives

$$1 = \int \hat{\theta}(\mathbf{x})\,S(\mathbf{x})\,p_N(\mathbf{x} \mid \theta)\,d\mathbf{x} = \mathbb{E}[\hat{\theta}S] = \operatorname{Cov}(\hat{\theta}, S),$$

since $\mathbb{E}[S] = 0$. By Theorem 4.10 (the Cauchy–Schwarz inequality),

$$1 = \operatorname{Cov}(\hat{\theta}, S)^2 \leq \operatorname{Var}(\hat{\theta})\operatorname{Var}(S) = \operatorname{Var}(\hat{\theta})\,N\,I(\theta).$$

$\square$

An unbiased estimator that attains the bound is called efficient.

Example 9.14. 1. Bernoulli(θ): $s(x;\theta) = \frac{x}{\theta} - \frac{1-x}{1-\theta}$ and $I(\theta) = \frac{1}{\theta(1-\theta)}$. The bound $\frac{\theta(1-\theta)}{N}$ is the variance of the relative frequency $\bar{X}_N$, which is therefore efficient.

2. $\mathcal{N}(\mu, \sigma^2)$ with known σ^2: $s = \frac{x-\mu}{\sigma^2}$ and $I(\mu) = \frac{1}{\sigma^2}$, so the sample mean, with variance $\frac{\sigma^2}{N}$, is efficient.

3. Uniform$(0, \theta)$ does not satisfy the assumptions, since its support depends on θ. Indeed, the unbiased estimator $\frac{N+1}{N} \max_i X_i$ has the variance $\frac{\theta^2}{N(N+2)}$, which decreases like $\frac{1}{N^2}$, faster than any bound of the form $\frac{c}{N}$.

Theorem 9.15 (Asymptotic normality of the MLE)**.** *Under regularity conditions, the maximum likelihood estimator is consistent, and $\sqrt{N}(\hat{\theta}_{MLE} - \theta)$ converges in distribution to $\mathcal{N}(0, \frac{1}{I(\theta)})$.*

We do not prove this theorem. It says that for large N, the MLE is approximately unbiased and attains the Cramér–Rao bound: maximum likelihood is asymptotically efficient, which is the main theoretical justification for it. In practice, the standard error of the MLE is estimated from the curvature of the log-likelihood at its maximum, SE $\approx \left(-\ell''(\hat{\theta})\right)^{-1/2}$ (the observed information). For example, for k successes in N Bernoulli trials, $-\ell''(\hat{\theta}) = \frac{k}{\hat{\theta}^2} + \frac{N-k}{(1-\hat{\theta})^2} = \frac{N}{\hat{\theta}(1-\hat{\theta})}$, which gives the familiar SE $= \sqrt{\hat{\theta}(1-\hat{\theta})/N}$, e.g., 0.046 for 30 successes in 100 trials. With several parameters, the Fisher information becomes a matrix, and the approximate covariance matrix of the MLE is the inverse of the Hessian of $-\ell$ (Lecture 3). The Fisher information matrix also defines the inner product of the natural gradient method mentioned in Lecture 3.

9.4 Sufficiency and the Rao–Blackwell theorem

Often, a few numbers computed from the data contain all the information about the parameter. A statistic $T = T(X_1, \ldots, X_N)$ of discrete data is sufficient for θ if the conditional distribution of the data given T does not depend on θ: once we know T, the remaining details of the data are noise that tells us nothing about θ. We write $\mathbf{x} = (x_1, \ldots, x_N)$.

Theorem 9.16 (Fisher–Neyman factorization)**.** *A statistic T of discrete data is sufficient for θ if and only if the joint pmf of the data can be written as $p(\mathbf{x} \mid \theta) = g\big(T(\mathbf{x}), \theta\big)\, h(\mathbf{x})$ for some functions g and h.*

Proof. If T is sufficient, let $g(t, \theta) = P_\theta(T = t)$, and let $h(\mathbf{x})$ be the conditional probability of $\mathbf{x}$ given $T = T(\mathbf{x})$, which does not depend on θ. Then $p(\mathbf{x} \mid \theta) = P_\theta\big(T = T(\mathbf{x})\big)\, h(\mathbf{x})$ (if $P_\theta(T = T(\mathbf{x})) = 0$, both sides are 0). Conversely, if $p(\mathbf{x} \mid \theta) = g(T(\mathbf{x}), \theta) h(\mathbf{x})$, then for $\mathbf{x}$ with $T(\mathbf{x}) = t$ and $P_\theta(T = t) > 0$,

$$P_\theta(\mathbf{X} = \mathbf{x} \mid T = t) = \frac{p(\mathbf{x} \mid \theta)}{\sum_{\mathbf{x}' : T(\mathbf{x}') = t} p(\mathbf{x}' \mid \theta)} = \frac{g(t, \theta) h(\mathbf{x})}{g(t, \theta) \sum_{\mathbf{x}' : T(\mathbf{x}') = t} h(\mathbf{x}')} = \frac{h(\mathbf{x})}{\sum_{\mathbf{x}' : T(\mathbf{x}') = t} h(\mathbf{x}')},$$

which does not depend on θ. □

For data with densities, conditional distributions given T are more delicate, and we take the factorization $p(\mathbf{x} \mid \theta) = g(T(\mathbf{x}), \theta) h(\mathbf{x})$ of the joint density as the definition of sufficiency. For Bernoulli data, $\prod_i \theta^{x_i}(1-\theta)^{1-x_i} = \theta^k (1-\theta)^{N-k}$ only depends on the number k of ones; for Poisson data, $\prod_i e^{-\lambda} \frac{\lambda^{x_i}}{x_i!} = e^{-N\lambda} \lambda^{\sum_i x_i} \cdot \prod_i \frac{1}{x_i!}$, so $\sum_i x_i$ is sufficient; and for Gaussian data, $\sum_i x_i$ and $\sum_i x_i^2$ together are sufficient for (μ, σ^2). Since the likelihood only depends on the data through $g(T, \theta)$, so does the MLE.

Theorem 9.17 (Rao–Blackwell)**.** *Let T be sufficient for θ, and let $\hat\theta$ be an estimator with $\mathbb{E}[\hat\theta^2] < \infty$. Then $\tilde\theta = \mathbb{E}[\hat\theta \mid T]$ is an estimator (it does not depend on θ), it has the same bias as $\hat\theta$, and $\mathrm{MSE}(\tilde\theta) \le \mathrm{MSE}(\hat\theta)$.*

Proof. By sufficiency, the conditional distribution of the data given T, and hence $\mathbb{E}[\hat\theta \mid T]$, does not depend on θ. By the tower property, $\mathbb{E}[\tilde\theta] = \mathbb{E}[\hat\theta]$, so the bias is the same. By the law of total variance (Example 4.16), $\mathrm{Var}(\hat\theta) = \mathbb{E}\big[\mathrm{Var}(\hat\theta \mid T)\big] + \mathrm{Var}(\tilde\theta) \ge \mathrm{Var}(\tilde\theta)$, and the bias–variance decomposition gives the claim. □

So averaging an estimator over all data sets with the same value of a sufficient statistic never hurts.

Example 9.18 (Estimating the probability of a quiet day)**.** Let $X_1, \dots, X_N$ be the numbers of server failures on N days, i.i.d. Poisson(λ), and suppose that we want to estimate the probability $e^{-\lambda} = P(X_1 = 0)$ of a day without failures. The indicator $\hat\theta = \mathbf{1}_{\{X_1 = 0\}}$ is unbiased, but it only uses the first day. Given $S = X_1 + \cdots + X_N = s$, a short computation with the Poisson pmf shows that X_1 is Binomial(s, $\frac{1}{N}$): each of the s failures happened on day 1 with probability $\frac{1}{N}$, independently. Hence

$$\tilde\theta = \mathbb{E}\big[\mathbf{1}_{\{X_1=0\}} \mid S\big] = \Big(1 - \frac{1}{N}\Big)^S.$$

It is unbiased, since $\mathbb{E}[t^S] = e^{N\lambda(t-1)}$ for $S \sim \mathrm{Poisson}(N\lambda)$ (Example 4.5). For $\lambda = 1$ and $N = 10$, its variance is $\mathbb{E}\big[(0.9)^{2S}\big] - e^{-2} = e^{-1.9} - e^{-2} \approx 0.014$, compared with $e^{-1}(1 - e^{-1}) \approx 0.233$ for $\hat\theta$. The plug-in estimator $e^{-\bar X_N}$, which is the MLE, is slightly biased; one can show that $\tilde\theta$ has the smallest variance among all unbiased estimators of $e^{-\lambda}$ (the Lehmann–Scheffé theorem).

9.5 Loss functions as negative log-likelihoods

Maximum likelihood explains the loss functions of Lectures 2 and 3.

Proposition 9.19 (Least squares as maximum likelihood)**.** *Assume the linear regression model $y_i = \mathbf{w}^\top \mathbf{x}_i + \varepsilon_i$, where the noise terms $\varepsilon_i \sim \mathcal{N}(0, \sigma^2)$ are i.i.d. Then the negative log-likelihood of $\mathbf{w}$ is*

$$-\ell(\mathbf{w}) = \frac{1}{2\sigma^2}\|\mathbf{y} - X\mathbf{w}\|^2 + \frac{N}{2}\log(2\pi\sigma^2),$$

where X has the rows $\mathbf{x}_i^\top$. So the MLE of $\mathbf{w}$ is the least squares solution, and the MLE of the noise variance is the mean squared residual $\frac{1}{N}\|\mathbf{y} - X\hat{\mathbf{w}}\|^2$.

Proof. Given $\mathbf{x}_i$, $y_i \sim \mathcal{N}(\mathbf{w}^\top \mathbf{x}_i, \sigma^2)$, so $-\log p(y_i \mid \mathbf{x}_i, \mathbf{w}) = \frac{(y_i - \mathbf{w}^\top \mathbf{x}_i)^2}{2\sigma^2} + \frac{1}{2}\log(2\pi\sigma^2)$; sum over i. The second statement is proved as in Example 9.9. □

Similarly, if $y_i \in \{0, 1\}$ is Bernoulli with $P(y_i = 1) = \sigma(\mathbf{w}^\top \mathbf{x}_i)$, the negative log-likelihood is $\sum_i \big(-y_i \log p_i - (1 - y_i)\log(1 - p_i)\big)$ with $p_i = \sigma(\mathbf{w}^\top \mathbf{x}_i)$: the cross-entropy loss of logistic regression in Lecture 3. For K classes with $P(y_i = k) = \mathrm{softmax}(\mathbf{z}_i)_k$, it is the softmax cross-entropy loss $-\sum_i \log \mathrm{softmax}(\mathbf{z}_i)_{y_i}$. So choosing a loss function amounts to choosing a noise model. For example, Laplace-distributed noise with density $\frac{1}{2b}e^{-|\varepsilon|/b}$ leads to the absolute loss $\sum_i |y_i - \mathbf{w}^\top \mathbf{x}_i|$, which is less sensitive to outliers than the squared loss.

9.6 Exponential families

The Bernoulli, Poisson, exponential and Gaussian distributions, and the categorical distributions behind softmax, have the same structure, which explains why their maximum likelihood estimates are so simple, and why logistic, softmax and least squares regression have the same gradient.

Definition 9.20 (Exponential family)**.** An exponential family is a family of pmfs or densities of the form

$$p_{\boldsymbol{\eta}}(x) = h(x)\exp\left(\boldsymbol{\eta}^\top T(x) - A(\boldsymbol{\eta})\right), \qquad A(\boldsymbol{\eta}) = \log \sum_x h(x) e^{\boldsymbol{\eta}^\top T(x)} \ \Big(\text{or } \log \int h(x) e^{\boldsymbol{\eta}^\top T(x)}\, dx\Big),$$

where $T(x) \in \mathbb{R}^m$ is the sufficient statistic, $h(x) \geq 0$, the natural parameter $\boldsymbol{\eta} \in \mathbb{R}^m$ ranges over the vectors for which the sum or integral is finite, and the log-partition function A makes the total probability 1.

Example 9.21.

1. Bernoulli(p): $p^x(1-p)^{1-x} = \exp\left(x \log \frac{p}{1-p} + \log(1-p)\right)$, so $T(x) = x$, $\eta = \log \frac{p}{1-p}$ (the log-odds) and $A(\eta) = \log(1+e^\eta)$, since $1 - p = \frac{1}{1+e^\eta}$. The inverse map $p = \frac{1}{1+e^{-\eta}} = \sigma(\eta)$ is the sigmoid function.
2. Poisson(λ): $\frac{\lambda^x}{x!} e^{-\lambda} = \frac{1}{x!}\exp(x \log \lambda - \lambda)$, so $T(x) = x$, $\eta = \log \lambda$, $h(x) = \frac{1}{x!}$ and $A(\eta) = e^\eta$.
3. $\mathcal{N}(\mu, \sigma^2)$: $\frac{1}{\sqrt{2\pi\sigma^2}} e^{-(x-\mu)^2/(2\sigma^2)} = \frac{1}{\sqrt{2\pi}} \exp\left(\frac{\mu}{\sigma^2}x - \frac{1}{2\sigma^2}x^2 - \frac{\mu^2}{2\sigma^2} - \log\sigma\right)$, so $T(x) = (x, x^2)$ and $\boldsymbol{\eta} = \left(\frac{\mu}{\sigma^2}, -\frac{1}{2\sigma^2}\right)$.
4. Categorical on $\{1, \ldots, K\}$ with probabilities softmax($\boldsymbol{\eta}$): $p(k) = \exp\left(\eta_k - \log \sum_j e^{\eta_j}\right)$, so $T(k) = \mathbf{e}_k$ (one-hot) and $A(\boldsymbol{\eta}) = \log \sum_j e^{\eta_j}$, the log-sum-exp function.

Proposition 9.22. *Let $\boldsymbol{\eta}$ lie in the interior of the set of natural parameters, and let $X \sim p_{\boldsymbol{\eta}}$. For every $\mathbf{v} \in \mathbb{R}^m$, the random variable $Y = \mathbf{v}^\top T(X)$ has the cumulant generating function $K_Y(r) = A(\boldsymbol{\eta} + r\mathbf{v}) - A(\boldsymbol{\eta})$ for small $|r|$. Hence*

$$\frac{d}{dr} A(\boldsymbol{\eta} + r\mathbf{v})\Big|_{r=0} = \mathbf{v}^\top \mathbb{E}[T(X)], \qquad \frac{d^2}{dr^2} A(\boldsymbol{\eta} + r\mathbf{v})\Big|_{r=0} = \operatorname{Var}\left(\mathbf{v}^\top T(X)\right) \geq 0,$$

In particular, the partial derivatives of A are $\frac{\partial A}{\partial \eta_j}(\boldsymbol{\eta}) = \mathbb{E}[T_j(X)]$, and A is convex on the interior of the set of natural parameters.

Proof. In the discrete case, $\mathbb{E}[e^{rY}] = \sum_x h(x) e^{(\boldsymbol{\eta}+r\mathbf{v})^\top T(x) - A(\boldsymbol{\eta})} = e^{A(\boldsymbol{\eta}+r\mathbf{v}) - A(\boldsymbol{\eta})}$, and similarly with integrals. This is finite for small $|r|$, since $\boldsymbol{\eta}$ is an interior point. The derivatives follow from Proposition 4.21, with $\mathbf{v} = \mathbf{e}_j$ for the partial derivatives. The set of natural parameters is convex, since $e^{(\theta\boldsymbol{\eta}_1 + (1-\theta)\boldsymbol{\eta}_2)^\top T(x)} \leq \theta e^{\boldsymbol{\eta}_1^\top T(x)} + (1-\theta) e^{\boldsymbol{\eta}_2^\top T(x)}$ by the convexity of exp, and so is its interior. Along every segment in the interior, A has a nonnegative second derivative, so it is convex there (Theorem 11.6 of Lecture 3, in one variable), and a function that is convex on every segment is convex. □

For example, for Poisson(λ), $A'(\eta) = A''(\eta) = e^\eta = \lambda$ is the mean and the variance, and for Bernoulli, $A'(\eta) = \sigma(\eta) = p$ and $A''(\eta) = \sigma(\eta)(1 - \sigma(\eta)) = p(1-p)$.

Theorem 9.23 (Maximum likelihood in exponential families)**.** *For i.i.d. data $x_1, \dots, x_N$ from an exponential family, the log-likelihood $\ell(\boldsymbol{\eta}) = \boldsymbol{\eta}^\top \sum_i T(x_i) - NA(\boldsymbol{\eta}) + \sum_i \log h(x_i)$ is concave on the interior of the set of natural parameters. If an interior parameter $\hat{\boldsymbol{\eta}}$ satisfies*

$$\nabla A(\hat{\boldsymbol{\eta}}) = \mathbb{E}_{\hat{\boldsymbol{\eta}}}[T(X)] = \frac{1}{N}\sum_{i=1}^{N} T(x_i),$$

then it maximizes the likelihood over the interior. Moreover, $\sum_i T(x_i)$ is a sufficient statistic.

Proof. Let $\boldsymbol{\eta}$ be another interior parameter, $\mathbf{v} = \boldsymbol{\eta} - \hat{\boldsymbol{\eta}}$, and $\phi(r) = -\ell(\hat{\boldsymbol{\eta}} + r\mathbf{v})$ for $0 \le r \le 1$. By Proposition 9.22, ϕ is convex, and $\phi'(0) = N\mathbf{v}^\top \mathbb{E}_{\hat{\boldsymbol{\eta}}}[T(X)] - \mathbf{v}^\top \sum_i T(x_i) = 0$. By the first-order characterization of convexity (Theorem 11.3 of Lecture 3, in one variable), $\phi(1) \ge \phi(0) + \phi'(0) = \phi(0)$, i.e., $\ell(\boldsymbol{\eta}) \le \ell(\hat{\boldsymbol{\eta}})$. The likelihood $\prod_i h(x_i) \cdot \exp\left(\boldsymbol{\eta}^\top \sum_i T(x_i) - NA(\boldsymbol{\eta})\right)$ has the form of the factorization theorem (Theorem 9.16). □

So the maximum likelihood estimate matches the expected sufficient statistics with their averages over the data (moment matching). For Poisson data, $\hat{\lambda} = \bar{x}$; for Gaussian data, the mean and the second moment are matched, which gives the estimates of Example 9.9; for categorical data, the predicted probabilities are the observed frequencies.

Generalized linear models. A generalized linear model predicts a response y from features $\mathbf{x}$ by an exponential family with the natural parameter $\eta = \mathbf{w}^\top \mathbf{x}$: the Gaussian with known variance gives linear regression, Bernoulli gives logistic regression, and Poisson gives Poisson regression for counts, with $\mathbb{E}[y \mid \mathbf{x}] = e^{\mathbf{w}^\top \mathbf{x}}$. For data $(\mathbf{x}_i, y_i)$ with $T(y) = y$, the negative log-likelihood is $\sum_i \left(A(\mathbf{w}^\top \mathbf{x}_i) - y_i \mathbf{w}^\top \mathbf{x}_i\right)$ plus a constant, and by the chain rule and Proposition 9.22,

$$\nabla_{\mathbf{w}}\big(-\ell(\mathbf{w})\big) = \sum_{i=1}^{N} \left(A'(\mathbf{w}^\top \mathbf{x}_i) - y_i\right)\mathbf{x}_i = \sum_{i=1}^{N} \left(\mathbb{E}[y \mid \mathbf{x}_i] - y_i\right)\mathbf{x}_i :$$

the gradient is always "prediction minus target" times the input, as we found for linear and logistic regression in Lecture 3. Its Hessian $\sum_i A''(\mathbf{w}^\top \mathbf{x}_i)\mathbf{x}_i\mathbf{x}_i^\top$ is positive semidefinite, so all these losses are convex.

9.7 Confidence intervals

An estimate is more useful with an error bar. By the central limit theorem, $\bar{X}_N$ is approximately $\mathcal{N}(\mu, \frac{\sigma^2}{N})$, so

$$P\Big(|\bar{X}_N - \mu| \le 1.96\frac{\sigma}{\sqrt{N}}\Big) \approx 0.95.$$

Replacing the unknown σ by the sample standard deviation s (which is justified for large N by Slutsky's theorem, Theorem 7.10), the interval $\bar{x} \pm 1.96\frac{s}{\sqrt{N}}$ is an approximate 95% confidence interval for μ. For example, for $N = 100$ data points with $\bar{x} = 5.2$ and $s = 1.5$, it is $[4.91, 5.49]$. Note that μ is fixed and the interval is random: if we repeated the experiment many times, about 95% of the intervals would contain μ. For a classifier with accuracy 0.9 on 1000 independent test examples, the standard error is $\sqrt{0.9 \cdot 0.1/1000} \approx 0.0095$, so the accuracy is known up to about ± 0.019, and a difference of one percentage point between two models on such a test set may well be due to chance.

9.8 Nonparametric estimation

So far, we assumed a parametric family of distributions and estimated its parameters. Nonparametric methods estimate the distribution itself. For i.i.d. data $X_1, \dots, X_N$ with the CDF F, the empirical CDF

$$\hat{F}_N(t) = \frac{1}{N}\#\{i : X_i \le t\} = \frac{1}{N}\sum_{i=1}^{N} \mathbf{1}_{\{X_i \le t\}}$$

is the CDF of the empirical distribution, which puts the probability $\frac{1}{N}$ on each data point. For fixed t, $N\hat{F}_N(t)$ is Binomial$(N, F(t))$, so $\hat{F}_N(t)$ is unbiased with the variance $\frac{F(t)(1-F(t))}{N}$, and Hoeffding's inequality (Theorem 7.5) gives $P(|\hat{F}_N(t) - F(t)| \ge \varepsilon) \le 2e^{-2N\varepsilon^2}$. In fact, the approximation is good for all t at once.

Theorem 9.24 (Uniform convergence of the empirical CDF)**.** *Let F be continuous. For every $\varepsilon \in (0, 1]$,*

$$P\Big(\sup_{t\in\mathbb{R}} |\hat{F}_N(t) - F(t)| > \varepsilon\Big) \le \frac{4}{\varepsilon} e^{-N\varepsilon^2/2}.$$

Proof. Let $m = \lceil 2/\varepsilon \rceil$. Since F is continuous and increases from 0 to 1, the intermediate value theorem gives points $t_1 < \cdots < t_{m-1}$ with $F(t_j) = \frac{j}{m}$; let $t_0 = -\infty$ and $t_m = \infty$, with $\hat{F}_N = F = 0$ at t_0 and $= 1$ at t_m. For $t_{j-1} \le t \le t_j$, both functions are nondecreasing, so

$$\hat{F}_N(t) - F(t) \le \hat{F}_N(t_j) - F(t_{j-1}) = \hat{F}_N(t_j) - F(t_j) + \tfrac{1}{m}, \qquad \hat{F}_N(t) - F(t) \ge \hat{F}_N(t_{j-1}) - F(t_{j-1}) - \tfrac{1}{m}.$$

Hence $\sup_t |\hat{F}_N(t) - F(t)| \le \max_{1\le j\le m-1} |\hat{F}_N(t_j) - F(t_j)| + \frac{\varepsilon}{2}$, and if the supremum exceeds ε, then $|\hat{F}_N(t_j) - F(t_j)| > \frac{\varepsilon}{2}$ for some j. By the union bound and Hoeffding's inequality, this has probability at most $(m-1)\cdot 2e^{-2N(\varepsilon/2)^2} \le \frac{4}{\varepsilon} e^{-N\varepsilon^2/2}$, since $m - 1 < \frac{2}{\varepsilon}$. □

So $\sup_t |\hat{F}_N(t) - F(t)| \to 0$ in probability (the Glivenko–Cantelli theorem; since the bounds are summable over N, the Borel–Cantelli lemma in the appendix to this lecture even gives almost sure convergence). The bound is crude; the Dvoretzky–Kiefer–Wolfowitz inequality, which we do not prove, improves it to $2e^{-2N\varepsilon^2}$ for every F. With it, $\hat{F}_N(t) \pm \sqrt{\log(40)/(2N)}$ is a 95% confidence band for the whole function F; for $N = 1000$, its half-width is 0.043. The largest distance $\sup_t |\hat{F}_N(t) - F_0(t)|$ to a hypothesized CDF F_0 is the statistic of the Kolmogorov–Smirnov test of H_0: $F = F_0$.

Kernel density estimation. A density cannot be estimated by the empirical distribution, which has no density. A histogram counts the data in bins; a smoother estimate replaces each data point by a small bump. With a kernel $K \ge 0$ with $\int K = 1$ and $K(-u) = K(u)$, such as the standard normal density φ, and a bandwidth $h > 0$, the kernel density estimate is

$$\hat{f}_h(x) = \frac{1}{Nh}\sum_{i=1}^{N} K\Big(\frac{x - X_i}{h}\Big),$$

a mixture of N copies of K, scaled by h and centered at the data points.

Proposition 9.25 (Bias and variance of kernel density estimates)**.** *Let the data have a density f with bounded continuous derivatives f', f'' and f''', and let $\kappa_2 = \int u^2 K(u)\,du$, $\kappa_3 = \int |u|^3 K(u)\,du$, $R = \int K(u)^2\,du$ and $R_1 = \int |u| K(u)^2\,du$ be finite. Then for every x,*

$$\Big|\mathbb{E}[\hat{f}_h(x)] - f(x) - \frac{h^2}{2}\kappa_2 f''(x)\Big| \le \frac{h^3}{6}\kappa_3 \sup|f'''|, \qquad \mathrm{Var}\,(\hat{f}_h(x)) \le \frac{f(x)R + hR_1 \sup|f'|}{Nh}.$$

Proof. Each term $Y_i = \frac{1}{h}K(\frac{x-X_i}{h})$ has the mean $\int \frac{1}{h}K(\frac{x-y}{h})f(y)\,dy = \int K(u)f(x-hu)\,du$, with the substitution $y = x-hu$. By Taylor's theorem (Lecture 3), $f(x-hu) = f(x)-huf'(x)+\frac{h^2u^2}{2}f''(x)-\frac{h^3u^3}{6}f'''(\xi)$ for some ξ. Integrating against K, with $\int K = 1$ and $\int uK(u)\,du = 0$ by symmetry, gives the bias. Since $\hat{f}_h(x)$ is the average of the i.i.d. Y_i, $\operatorname{Var}(\hat{f}_h(x)) = \frac{1}{N}\operatorname{Var}(Y_1) \le \frac{1}{N}\mathbb{E}[Y_1^2] = \frac{1}{Nh}\int K(u)^2 f(x-hu)\,du$, and $f(x-hu) \le f(x)+h|u|\sup|f'|$ by the mean value theorem. □

This is a bias–variance tradeoff, controlled by h: a small bandwidth gives a spiky estimate with a large variance, of order $\frac{1}{Nh}$, and a large bandwidth oversmooths, with a bias of order h^2. Minimizing the leading terms $\frac{h^4}{4}\kappa_2^2 f''(x)^2 + \frac{f(x)R}{Nh}$ of the mean squared error over h gives $h \propto N^{-1/5}$ and a mean squared error of order $N^{-4/5}$, slower than the rate $\frac{1}{N}$ of parametric estimates; in practice, h is chosen by cross-validation or by rules of thumb such as $h = 1.06\,\hat{\sigma}N^{-1/5}$ for the Gaussian kernel. In d dimensions, the same computation gives the rate $N^{-4/(4+d)}$, which is very slow for large d: the curse of dimensionality, and a reason why high-dimensional densities are modeled with parametric or neural network models instead.

9.9 Survival analysis and censored data

How long until a customer cancels a subscription, a machine fails, or a patient relapses? Such waiting times are positive random variables T, and their analysis has two peculiarities: they are naturally described by the risk of the event at each moment, and they are often only partially observed, because the study ends before the event happens. Let $T \ge 0$ have a continuous density p, the CDF F, and $F(0) = 0$.

Definition 9.26 (Survival function and hazard rate)**.** The survival function of T is $S(t) = P(T > t) = 1 - F(t)$. For t with $S(t) > 0$, the hazard rate is $h(t) = \frac{p(t)}{S(t)}$, and the cumulative hazard is $H(t) = \int_0^t h(u)\,du$.

The hazard rate is the rate of events among the survivors: $P(t < T \le t+\delta \mid T > t) = \frac{F(t+\delta)-F(t)}{S(t)} \approx h(t)\delta$ for small δ. A constant hazard describes events that strike at random; a decreasing hazard, "infant mortality" (defective units fail early); and an increasing one, wear-out.

Proposition 9.27. *1. $S(t) = e^{-H(t)}$ and $p(t) = h(t)e^{-H(t)}$, so the hazard rate determines the distribution. Moreover, $\mathbb{E}[T] = \int_0^\infty S(t)\,dt$.*

2. The hazard rate is constant, $h(t) = \lambda$, if and only if $T \sim$ Exponential(λ).

Proof. 1. $S' = -p$, so $(\log S)' = -\frac{p}{S} = -h$. Since $S(0) = 1$, integration gives $\log S(t) = -H(t)$, and $p = hS$. The formula for $\mathbb{E}[T]$ is the tail integral of Example 4.7.

2. If $h = \lambda$, then $S(t) = e^{-\lambda t}$, the survival function of Exponential(λ), and conversely $h(t) = \frac{\lambda e^{-\lambda t}}{e^{-\lambda t}} = \lambda$. □

A constant hazard is the memorylessness of the exponential distribution (Section 2). The Weibull distribution with the shape $k > 0$ and the scale $s > 0$ has the hazard $h(t) = \frac{k}{s}\left(\frac{t}{s}\right)^{k-1}$, so $H(t) = (\frac{t}{s})^k$ and $S(t) = e^{-(t/s)^k}$; it has a decreasing hazard for $k < 1$, a constant one for $k = 1$ (the exponential distribution), and an increasing one for $k > 1$.

Censoring. In a study that runs for a limited time, some units have not experienced the event when the study ends, or they leave the study early. For each unit i, we then observe $t_i = \min(T_i, C_i)$, where C_i is the censoring time, and the indicator $\delta_i = 1$ if the event was observed ($T_i \le C_i$) and $\delta_i = 0$ if the observation was censored. For a censored unit, we only know that $T_i > t_i$. Assume that the censoring times are independent of the event times, with a distribution that does not depend on the parameter θ of the event times (non-informative censoring). Then an observed event contributes the density $p_\theta(t_i)$ to the likelihood, a censored observation contributes the probability $S_\theta(t_i)$, and the factors that come from the censoring times do not depend on θ. So the likelihood is

$$L(\theta) = \prod_{i=1}^{N} p_\theta(t_i)^{\delta_i} S_\theta(t_i)^{1-\delta_i} = \prod_{i=1}^{N} h_\theta(t_i)^{\delta_i} e^{-H_\theta(t_i)}.$$

Proposition 9.28 (Exponential survival times with censoring)**.** *For Exponential(λ) event times with non-informative censoring, the maximum likelihood estimate is*

$$\hat{\lambda} = \frac{D}{\sum_{i=1}^{N} t_i},$$

where $D = \sum_i \delta_i$ is the number of observed events: the number of events divided by the total time at risk.

Proof. With $h = \lambda$ and $H(t) = \lambda t$, $\log L(\lambda) = D \log \lambda - \lambda \sum_i t_i$, whose derivative $\frac{D}{\lambda} - \sum_i t_i$ vanishes at $\hat{\lambda}$; the second derivative $-\frac{D}{\lambda^2}$ is negative (for $D > 0$). □

Example 9.29 (Censored waiting times)**.** Six customers are observed until they cancel their subscription or until the study ends. The times (in months) are $2, 3^+, 4, 5^+, 7, 8^+$, where $^+$ marks a censored time: these customers had not canceled yet. With $D = 3$ and the total time 29, $\hat{\lambda} = \frac{3}{29} \approx 0.103$ per month, and the estimated mean lifetime is $\frac{1}{\hat{\lambda}} \approx 9.7$ months. Two naive approaches are biased: dropping the censored customers gives $\frac{3}{2+4+7} \approx 0.23$, and treating the censoring times as cancellations gives $\frac{6}{29} \approx 0.21$. Both ignore that the censored customers stayed at least as long as observed, and both overestimate the risk of canceling by a factor of about 2.

The Kaplan–Meier estimator. Without a parametric model, the survival function is estimated as a product of conditional survival probabilities. Let $\tau_1 < \tau_2 < \cdots$ be the distinct times at which events are observed, let d_j be the number of events at τ_j, and let n_j be the number of units at risk just before τ_j, i.e., with $t_i \ge \tau_j$. Since surviving past t means surviving each event time $\tau_j \le t$, given survival until then, the Kaplan–Meier estimator is

$$\hat{S}(t) = \prod_{j:\tau_j \le t} \Big(1 - \frac{d_j}{n_j}\Big).$$

A censored unit counts as at risk until its censoring time, and then leaves the risk set without causing an event. Without censoring, $n_{j+1} = n_j - d_j$, the product telescopes to $\frac{n_{j+1}}{N}$ for $\tau_j \le t < \tau_{j+1}$, and $\hat{S}(t) = 1 - \hat{F}_N(t)$ is one minus the empirical CDF of Section 9.8. In Example 9.29, $\hat{S}$ drops to $\frac{5}{6}$ at $t = 2$, to $\frac{5}{6} \cdot \frac{3}{4} = \frac{5}{8}$ at $t = 4$ (only 4 customers are at risk then), and to $\frac{5}{8} \cdot \frac{1}{2} = \frac{5}{16}$ at $t = 7$. To compare groups, or to include features $\mathbf{x}$, the Cox proportional hazards model assumes $h(t \mid \mathbf{x}) = h_0(t) e^{\mathbf{w}^\top \mathbf{x}}$, with a baseline hazard h_0 that is common to all units.

10 Hypothesis Testing

Often we want to decide a yes-or-no question from data: is a coin fair? Does a new model have a lower error rate than the old one? Does a new website design increase sales? A hypothesis test makes such decisions while controlling the probability of a false alarm.

Definition 10.1 (Hypothesis test)**.** A test compares a null hypothesis H_0 (usually "no effect") with an alternative hypothesis H_1. It computes a test statistic T from the data and rejects H_0 if T falls into a rejection region. Rejecting a true H_0 is a type I error (a false positive), and keeping a false H_0 is a type II error (a false negative). A test has significance level α if the probability of a type I error is at most α; its power is the probability of rejecting H_0 when H_1 is true.

Definition 10.2 (p-value)**.** If large values of T speak against H_0, the p-value of an observed value t is $p = P_{H_0}(T \geq t)$: the probability, computed under H_0, of a test statistic at least as extreme as the observed one. The test that rejects H_0 if $p \leq \alpha$ has significance level α.

Proposition 10.3. *If T has a continuous, strictly increasing CDF F under H_0, then the p-value $1 - F(T)$ is Uniform$(0,1)$-distributed under H_0. In particular, $P_{H_0}(p \leq \alpha) = \alpha$.*

Proof. $P(F(T) \leq u) = P(T \leq F^{-1}(u)) = F(F^{-1}(u)) = u$, so $F(T)$ is uniform on $(0,1)$, as in Theorem 2.7, and so is $1 - F(T)$. □

Example 10.4 (Testing a coin)**.** A coin shows 60 heads in 100 flips. Is it fair? With H_0: $\theta = \frac{1}{2}$ and the number of heads S as the test statistic, the one-sided p-value is $P_{H_0}(S \geq 60) = 0.028$ (Example 7.4). If we had no reason to expect more heads than tails, a two-sided test is appropriate: it counts deviations in both directions, and its p-value is $P_{H_0}(|S - 50| \geq 10) = 0.057$. At the level $\alpha = 0.05$, the one-sided test rejects H_0, and the two-sided test does not. The test must be chosen before looking at the data; otherwise its significance level is not what it claims.

Example 10.5 (Power)**.** For the coin of Example 10.4, the one-sided test at the level 0.05 rejects H_0: $\theta = \frac{1}{2}$ if $S \geq 59$, since $P_{H_0}(S \geq 59) \approx 0.044 \leq 0.05 < P_{H_0}(S \geq 58) \approx 0.067$. If the coin actually has $\theta = 0.6$, the test rejects H_0 with the probability $P_{\theta=0.6}(S \geq 59) \approx 0.62$: it misses this bias in more than a third of the experiments. With 400 flips, the rejection region is $S \geq 217$, and the power for $\theta = 0.6$ is 0.99. How to plan the sample size for a desired power is explained in Section 11.

The most powerful test. Which test has the largest power at a given level? For two simple hypotheses, which each specify one distribution, the answer is the likelihood ratio test.

Theorem 10.6 (Neyman–Pearson lemma)**.** *Let the data $\mathbf{x}$ have the pmf or density p_0 under H_0 and p_1 under H_1. For $k > 0$, the likelihood ratio test rejects H_0 if $p_1(\mathbf{x}) > kp_0(\mathbf{x})$; let α be its level, $\alpha = P_0\big(p_1(\mathbf{X}) > kp_0(\mathbf{X})\big)$. Then every test with level at most α has at most the power of the likelihood ratio test.*

Proof. Let $\phi^*(\mathbf{x}) = 1$ if $p_1(\mathbf{x}) > kp_0(\mathbf{x})$ and 0 otherwise, and let $\phi(\mathbf{x}) \in \{0,1\}$ indicate the rejection region of another test, with $\mathbb{E}_0[\phi] \leq \alpha$. Then

$$\big(\phi^*(\mathbf{x}) - \phi(\mathbf{x})\big)\big(p_1(\mathbf{x}) - kp_0(\mathbf{x})\big) \geq 0 \qquad \text{for all } \mathbf{x}:$$

if $p_1(\mathbf{x}) > kp_0(\mathbf{x})$, then $\phi^*(\mathbf{x}) = 1 \geq \phi(\mathbf{x})$, and otherwise $\phi^*(\mathbf{x}) = 0 \leq \phi(\mathbf{x})$. Summing or integrating over $\mathbf{x}$ gives $\mathbb{E}_1[\phi^*] - \mathbb{E}_1[\phi] \geq k(\mathbb{E}_0[\phi^*] - \mathbb{E}_0[\phi]) = k(\alpha - \mathbb{E}_0[\phi]) \geq 0$. Here $\mathbb{E}_1[\phi^*]$ and $\mathbb{E}_1[\phi]$ are the powers of the two tests. □

Example 10.7 (The z-test is most powerful)**.** Let $X_1, \dots, X_N$ be i.i.d. $\mathcal{N}(\mu, 1)$, and test H_0: $\mu = 0$ against H_1: $\mu = \mu_1$ with $\mu_1 > 0$. The likelihood ratio is

$$\frac{p_1(\mathbf{x})}{p_0(\mathbf{x})} = \prod_{i=1}^{N} \frac{e^{-(x_i-\mu_1)^2/2}}{e^{-x_i^2/2}} = \exp\Big(N\mu_1\bar{x} - \frac{N\mu_1^2}{2}\Big),$$

an increasing function of $\bar{x}$. So the likelihood ratio tests reject for large $\bar{x}$, and at the level 0.05, the most powerful test rejects if $\bar{x} > \frac{1.645}{\sqrt{N}}$, since $\bar{X}_N \sim \mathcal{N}(0, \frac{1}{N})$ under H_0. This is the one-sided z-test, and it does not depend on μ_1: it is the most powerful test against every $\mu_1 > 0$ at once. For $N = 25$ and $\mu_1 = 0.5$, it rejects if $\bar{x} > 0.329$, and its power is $P(\bar{X}_N > 0.329) = 1 - \Phi(\frac{0.329-0.5}{0.2}) \approx 0.80$.

In classification, the two hypotheses are the two classes, a test is a classifier, its level is the false positive rate, and its power is the true positive rate (Section 11.2). So a classifier that thresholds the likelihood ratio $\frac{p(\mathbf{x}|y=1)}{p(\mathbf{x}|y=0)}$, or equivalently the posterior probability $P(y = 1 \mid \mathbf{x})$, has at each threshold the largest true positive rate among all classifiers whose false positive rate is not larger. Training a classifier to predict $P(y = 1 \mid \mathbf{x})$ well is therefore the right goal for every tradeoff between the two kinds of errors.

Tests for a mean. To test H_0: $\mu = \mu_0$ for i.i.d. data, we use the standardized statistic $Z = \frac{\bar{X}_N - \mu_0}{s/\sqrt{N}}$, which is approximately $\mathcal{N}(0, 1)$ under H_0 by the central limit theorem (the z-test). The two-sided p-value is $2(1 - \Phi(|z|))$. For small samples from a Gaussian distribution, Z has a Student t-distribution with $N - 1$ degrees of freedom (Theorem 10.12 below, with $X = \mathbf{1}$), whose quantiles are larger than 1.96 (for example, 2.78, 2.26 and 2.05 for $N = 5$, 10 and 30); this is the t-test. Tests and confidence intervals are two sides of the same coin: the two-sided test at level 0.05 rejects $\mu = \mu_0$ exactly if μ_0 lies outside the 95% confidence interval $\bar{x} \pm 1.96\frac{s}{\sqrt{N}}$.

Example 10.8 (A paired t-test)**.** A new model B is compared with a model A on 8 benchmark data sets, and the differences of the accuracies (B minus A, in percentage points) are

$$1.2, \quad 0.4, \quad 2.1, \quad -0.3, \quad 1.5, \quad 0.8, \quad 1.1, \quad 0.6.$$

Their mean is $\bar{d} = 0.925$ and their standard deviation is $s = 0.729$, so $t = \frac{\bar{d}}{s/\sqrt{8}} \approx 3.59$. Under H_0: "on average, there is no difference", and if the differences are approximately Gaussian, t has the t-distribution with 7 degrees of freedom, and the two-sided p-value is 0.009. So B is significantly better at the level 0.05, and even at 0.01. Analyzing the differences, rather than the two lists of accuracies, removes the large variation between the data sets (Section 11).

Goodness of fit. To test whether categorical data with the counts $O_1, \dots, O_K$ in N observations follow given probabilities $p_1, \dots, p_K$, Pearson's statistic compares the observed counts with the expected counts $E_k = Np_k$:

$$\chi^2 = \sum_{k=1}^{K} \frac{(O_k - E_k)^2}{E_k}.$$

Under H_0, for large N, it has approximately the χ^2 distribution with $K - 1$ degrees of freedom, the distribution of $Z_1^2 + \cdots + Z_{K-1}^2$ for independent standard normal Z_i; one degree of freedom is lost because the counts add up to N. Large values speak against H_0.

Example 10.9 (Is the die fair?)**.** A die is rolled 120 times, with the counts $18, 25, 16, 21, 13, 27$ for the faces $1, \dots, 6$. The expected counts are 20, so

$$\chi^2 = \frac{4 + 25 + 16 + 1 + 49 + 49}{20} = 7.2.$$

With 5 degrees of freedom, the p-value is $P(\chi^2_5 \geq 7.2) \approx 0.21$: the data are consistent with a fair die, although the counts 13 and 27 may look suspicious at first sight. The same test checks, for example, whether the class frequencies predicted by a classifier match the observed ones, or whether a random number generator is uniform.

Remark 10.10 (What a p-value is not). A p-value is not the probability that H_0 is true. That probability depends on the prior plausibility of H_0, as in the medical test of Example 1.7, and only Bayesian inference can compute it. A small p-value can also come from a tiny effect that is irrelevant in practice, if N is huge (statistical versus practical significance), so effect sizes and confidence intervals should be reported as well. Conversely, a large p-value does not show that there is no effect: the test may just have too little power.

10.1 Inference for linear regression

A fitted regression model reports a coefficient for every feature. Is the effect of a feature real, or could the coefficient be 0? Standard errors, confidence intervals and t-tests answer this, and they are the numbers in the output tables of statistical software. We consider the model $\mathbf{y} = X\mathbf{w} + \boldsymbol{\varepsilon}$ with a fixed $N \times n$ matrix X with linearly independent columns (usually the first column is $\mathbf{1}$, for the intercept), an unknown $\mathbf{w} \in \mathbb{R}^n$, and noise with

$$\mathbb{E}[\boldsymbol{\varepsilon}] = \mathbf{0} \qquad \text{and} \qquad \mathrm{Cov}(\boldsymbol{\varepsilon}) = \sigma^2 I_N :$$

the noise terms are uncorrelated and have the same variance (homoscedasticity). Let $\hat{\mathbf{w}} = (X^\top X)^{-1} X^\top \mathbf{y}$ be the least squares estimate (Lecture 2), $\mathbf{e} = \mathbf{y} - X\hat{\mathbf{w}}$ the residuals, and $\mathrm{RSS} = \|\mathbf{e}\|^2$ the residual sum of squares. The exact distributions of the test statistics are the following.

Definition 10.11 (χ^2- and t-distributions)**.** For independent $Z_1, \dots, Z_k \sim \mathcal{N}(0,1)$, the distribution of $Z_1^2 + \cdots + Z_k^2$ is the χ^2-distribution with k degrees of freedom, χ^2_k. If $Z \sim \mathcal{N}(0,1)$ and $V \sim \chi^2_k$ are independent, the distribution of $\frac{Z}{\sqrt{V/k}}$ is the t-distribution with k degrees of freedom, t_k.

The χ^2_k-distribution has the mean k. The t_k-distribution is symmetric and has heavier tails than $\mathcal{N}(0,1)$, because the denominator is random; as $k \to \infty$, $\frac{V}{k} \to 1$ in probability by the law of large numbers, and t_k tends to $\mathcal{N}(0,1)$ by Slutsky's theorem. Its 97.5% quantiles are $t_{1,0.975} = 12.71$, $t_{2,0.975} = 4.30$, $t_{3,0.975} = 3.18$, $t_{10,0.975} = 2.23$ and $t_{30,0.975} = 2.04$.

Theorem 10.12 (Sampling distribution of least squares)**.** *In the model above:*

1. $\hat{\mathbf{w}}$ *is unbiased,* $\mathbb{E}[\hat{\mathbf{w}}] = \mathbf{w}$*, with the covariance matrix* $\mathrm{Cov}(\hat{\mathbf{w}}) = \sigma^2 (X^\top X)^{-1}$.
2. $\mathbb{E}[\mathrm{RSS}] = (N - n)\sigma^2$*, so* $s^2 = \frac{\mathrm{RSS}}{N-n}$ *is an unbiased estimate of* σ^2.
3. *If* $\boldsymbol{\varepsilon} \sim \mathcal{N}(\mathbf{0}, \sigma^2 I_N)$*, then* $\hat{\mathbf{w}} \sim \mathcal{N}\big(\mathbf{w}, \sigma^2 (X^\top X)^{-1}\big)$*,* $\frac{\mathrm{RSS}}{\sigma^2} \sim \chi^2_{N-n}$*, and* $\hat{\mathbf{w}}$ *and* RSS *are independent. Consequently, with the standard errors* $\mathrm{SE}_j = s\sqrt{\big((X^\top X)^{-1}\big)_{jj}}$*,*

$$\frac{\hat{w}_j - w_j}{\mathrm{SE}_j} \sim t_{N-n} \qquad \text{for } j = 1, \dots, n.$$

Proof. 1. $\hat{\mathbf{w}} = (X^\top X)^{-1}X^\top(X\mathbf{w}+\boldsymbol{\varepsilon}) = \mathbf{w} + (X^\top X)^{-1}X^\top\boldsymbol{\varepsilon}$. So $\mathbb{E}[\hat{\mathbf{w}}] = \mathbf{w}$, and by Proposition 5.2, $\operatorname{Cov}(\hat{\mathbf{w}}) = (X^\top X)^{-1}X^\top(\sigma^2 I)X(X^\top X)^{-1} = \sigma^2(X^\top X)^{-1}$.

2. Let $H = X(X^\top X)^{-1}X^\top$ be the matrix of the orthogonal projection onto $\operatorname{col}(X)$ (Lecture 2). Then $(I-H)X = 0$, so $\mathbf{e} = (I-H)\mathbf{y} = (I-H)\boldsymbol{\varepsilon}$, and since $I-H$ is symmetric with $(I-H)^2 = I-H$, $\mathrm{RSS} = \boldsymbol{\varepsilon}^\top(I-H)\boldsymbol{\varepsilon}$. For every matrix M, $\mathbb{E}[\boldsymbol{\varepsilon}^\top M\boldsymbol{\varepsilon}] = \sum_{i,j} M_{ij}\mathbb{E}[\varepsilon_i\varepsilon_j] = \sigma^2 \operatorname{tr}(M)$, and $\operatorname{tr}(I-H) = N - \operatorname{tr}\left((X^\top X)^{-1}X^\top X\right) = N-n$, since $\operatorname{tr}(AB) = \operatorname{tr}(BA)$.

3. Choose an orthonormal basis $\mathbf{u}_1,\dots,\mathbf{u}_n$ of $\operatorname{col}(X)$ and an orthonormal basis $\mathbf{u}_{n+1},\dots,\mathbf{u}_N$ of $\operatorname{col}(X)^\perp$ (Lecture 2), and let Q be the orthogonal matrix with the columns $\mathbf{u}_1,\dots,\mathbf{u}_N$. By Corollary 6.7, $Q^\top\boldsymbol{\varepsilon} \sim \mathcal{N}(\mathbf{0}, \sigma^2 Q^\top Q) = \mathcal{N}(\mathbf{0},\sigma^2 I)$, so the numbers $Z_i = \frac{1}{\sigma}\mathbf{u}_i^\top\boldsymbol{\varepsilon}$ are independent and $\mathcal{N}(0,1)$ (Theorem 6.4). Now $(I-H)\boldsymbol{\varepsilon} = \sum_{i>n}(\mathbf{u}_i^\top\boldsymbol{\varepsilon})\mathbf{u}_i$, so $\mathrm{RSS} = \sigma^2\sum_{i>n} Z_i^2$, and $\frac{\mathrm{RSS}}{\sigma^2} \sim \chi^2_{N-n}$. On the other hand, $X^\top = X^\top H$ (since $HX = X$), so $\hat{\mathbf{w}} - \mathbf{w} = (X^\top X)^{-1}X^\top H\boldsymbol{\varepsilon}$, and $H\boldsymbol{\varepsilon} = \sigma\sum_{i\le n} Z_i\mathbf{u}_i$ only depends on $Z_1,\dots,Z_n$. Functions of the independent groups $Z_1,\dots,Z_n$ and $Z_{n+1},\dots,Z_N$ are independent, so $\hat{\mathbf{w}}$ and RSS are independent. The matrix $(X^\top X)^{-1}X^\top$ has rank n, so $\hat{\mathbf{w}}$ is Gaussian by Corollary 6.7, with the mean and covariance matrix of part 1. Finally, with $c_j = ((X^\top X)^{-1})_{jj}$, the number $\frac{\hat{w}_j - w_j}{\sigma\sqrt{c_j}}$ is $\mathcal{N}(0,1)$, it is independent of $V = \frac{\mathrm{RSS}}{\sigma^2} \sim \chi^2_{N-n}$, and dividing it by $\sqrt{V/(N-n)} = \frac{s}{\sigma}$ gives $\frac{\hat{w}_j - w_j}{\mathrm{SE}_j}$. □

So $\hat{w}_j \pm t_{N-n,0.975}\,\mathrm{SE}_j$ is an exact 95% confidence interval for w_j under Gaussian noise (and an approximate one in general, for large N, by the central limit theorem), and H_0: $w_j = 0$ ("the feature has no effect") is tested with the t-statistic $\frac{\hat{w}_j}{\mathrm{SE}_j}$. The one-sample t-test of a mean is the special case $X = \mathbf{1}$, with $\hat{w} = \bar{x}$ and $\mathrm{SE} = \frac{s}{\sqrt{N}}$. Two degrees of freedom are lost in a regression with an intercept and a slope, one for each estimated coefficient, as in the factor $\frac{1}{N-1}$ of the sample variance.

Example 10.13 (Standard errors of a regression line)**.** For the five points $x = -2,-1,0,1,2$ with $y = 0.8, 2.3, 2.9, 4.4, 5.6$, the design matrix has the columns $\mathbf{1}$ and $\mathbf{x}$, and $X^\top X = \operatorname{diag}(5,10)$, since $\sum_i x_i = 0$. So $\hat{w}_0 = \bar{y} = 3.2$, $\hat{w}_1 = \frac{\sum_i x_i y_i}{10} = 1.17$, the residuals are $-0.06, 0.27, -0.30, 0.03, 0.06$, $\mathrm{RSS} = 0.171$, and $s^2 = \frac{0.171}{5-2} = 0.057$. The standard error of the slope is $\sqrt{0.057/10} \approx 0.0755$, its t-statistic is 15.5, with a two-sided p-value of 0.0006 (3 degrees of freedom), and the 95% confidence interval is $1.17 \pm 3.18 \cdot 0.0755 = [0.93, 1.41]$.

Least squares is not only unbiased, it is also the most precise among the unbiased estimators that are linear in $\mathbf{y}$.

Proposition 10.14 (Gauss–Markov)**.** *Let $\tilde{\mathbf{w}} = C\mathbf{y}$, with an $n \times N$ matrix C, be unbiased for every $\mathbf{w}$. Then $\operatorname{Cov}(\tilde{\mathbf{w}}) - \operatorname{Cov}(\hat{\mathbf{w}})$ is positive semidefinite; in particular, $\operatorname{Var}(\tilde{w}_j) \ge \operatorname{Var}(\hat{w}_j)$ for every j.*

Proof. $\mathbb{E}[C\mathbf{y}] = CX\mathbf{w} = \mathbf{w}$ for all $\mathbf{w}$ means $CX = I$. Let $D = C - (X^\top X)^{-1}X^\top$, so that $DX = 0$. By Proposition 5.2, $\operatorname{Cov}(\tilde{\mathbf{w}}) = \sigma^2 CC^\top = \sigma^2\big((X^\top X)^{-1} + DD^\top\big)$, since the cross terms $(X^\top X)^{-1}X^\top D^\top = (X^\top X)^{-1}(DX)^\top$ vanish. And $DD^\top$ is positive semidefinite. □

Biased estimators such as ridge regression can still have a smaller mean squared error (Section 9): Gauss–Markov only compares unbiased ones.

Heteroscedasticity. Often the noise is larger for some data points than for others: prices of expensive houses vary more than those of cheap ones, and measurements of large quantities are less precise. If $\operatorname{Cov}(\boldsymbol{\varepsilon}) = \Sigma = \operatorname{diag}(\sigma_1^2, \dots, \sigma_N^2)$ with different variances (heteroscedasticity), the estimate $\hat{\mathbf{w}}$ is still unbiased, but by Proposition 5.2, its covariance matrix is the "sandwich"

$$\operatorname{Cov}(\hat{\mathbf{w}}) = (X^\top X)^{-1} X^\top \Sigma X (X^\top X)^{-1},$$

and the formula $s^2(X^\top X)^{-1}$ is wrong, together with the standard errors, confidence intervals and t-tests based on it. For example, in the model $y_i = w x_i + \varepsilon_i$ without intercept, with the noise standard deviations $\sigma_i = \sigma |x_i|$ that grow with $|x_i|$, the true variance is $\operatorname{Var}(\hat{w}) = \frac{\sigma^2 \sum_i x_i^4}{(\sum_i x_i^2)^2}$. The homoscedastic formula, with the average noise variance $\bar{\sigma}^2 = \frac{\sigma^2}{N} \sum_i x_i^2$, gives $\frac{\bar{\sigma}^2}{\sum_i x_i^2} = \frac{\sigma^2}{N}$. Their ratio is $\frac{N \sum_i x_i^4}{(\sum_i x_i^2)^2} \geq 1$, by the Cauchy–Schwarz inequality for the vectors $(x_i^2)_i$ and $\mathbf{1}$, with equality only if all $|x_i|$ are equal. For $x = 1, 2, \dots, 10$, the ratio is 1.71: the usual standard error is too small by a factor of 1.31, and the test finds effects too often. Heteroscedasticity is detected by plotting the residuals against the fitted values or against a feature, which shows a funnel shape. The remedies are robust (White) standard errors, which replace Σ in the sandwich by the estimate $\operatorname{diag}(e_1^2, \dots, e_N^2)$, and, if the variances are known up to a factor, weighted least squares, which minimizes $\sum_i \frac{(y_i - \mathbf{w}^\top \mathbf{x}_i)^2}{\sigma_i^2}$ and is the maximum likelihood estimate for Gaussian noise with these variances.

10.2 Multiple testing

If we perform many tests, some of them will reject by chance. For m independent tests of true null hypotheses at the level 0.05, the probability of at least one false positive is $1 - 0.95^m$, which is 0.64 for $m = 20$. This happens when we compare many models on the same test set, screen thousands of genes or features, or analyze many metrics of an experiment, and it is a main source of irreproducible results ("p-hacking"). The probability of at least one false positive among all tests is called the family-wise error rate (FWER).

Proposition 10.15 (Bonferroni correction)**.** *If each of m tests rejects its null hypothesis when its p-value is at most $\frac{\alpha}{m}$, then the family-wise error rate is at most α, whatever the dependence between the tests.*

Proof. Let I_0 be the set of true null hypotheses. By the union bound (Proposition 1.3) and Proposition 10.3, $P\big(\bigcup_{i \in I_0} \{p_i \leq \frac{\alpha}{m}\}\big) \leq \sum_{i \in I_0} \frac{\alpha}{m} \leq \alpha$. □

The Bonferroni correction is very conservative when m is large. When we screen many hypotheses, it is often more useful to control the false discovery rate (FDR), the expected fraction of false positives among all rejected hypotheses: if R hypotheses are rejected, V of them true, then $\mathrm{FDR} = \mathbb{E}\big[\frac{V}{\max(R,1)}\big]$. The Benjamini–Hochberg procedure sorts the p-values $p_{(1)} \leq \cdots \leq p_{(m)}$, finds the largest k with $p_{(k)} \leq \frac{k}{m} q$ ($k = 0$ if there is none), and rejects the k hypotheses with the smallest p-values.

Theorem 10.16 (Benjamini–Hochberg)**.** *Suppose that the m p-values are independent, and that the p-value p_i of every true null hypothesis satisfies $P(p_i \leq \alpha) \leq \alpha$ for all $\alpha \in [0,1]$ (as in Proposition 10.3). Then the false discovery rate of the Benjamini–Hochberg procedure is at most $\frac{m_0}{m} q \leq q$, where m_0 is the number of true null hypotheses.*

Proof. Let $N(k)$ be the number of p-values that are at most $\frac{k}{m}q$. Since $p_{(k)} \le \frac{k}{m}q$ if and only if $N(k) \ge k$, the number R of rejections is the largest $k \in \{0, \dots, m\}$ with $N(k) \ge k$. Exactly R p-values are at most $\frac{R}{m}q$: at least R by definition, and if $R+1$ of them were, then $N(R+1) \ge R+1$. So hypothesis i is rejected if and only if $p_i \le \frac{R}{m}q$.

Let i be a true null hypothesis, and let $N^{(i)}(k)$ and $R^{(i)}$ be defined in the same way after replacing p_i by 0; then $R^{(i)}$ is a function of the other p-values, so it is independent of p_i. Let $k \ge 1$, and suppose that $p_i \le \frac{k}{m}q$. Then $N(k') = N^{(i)}(k')$ for all $k' \ge k$. If $R = k$, then $N^{(i)}(k) = N(k) \ge k$ and $N^{(i)}(k') = N(k') < k'$ for $k' > k$, so $R^{(i)} = k$. Conversely, if $R^{(i)} = k$, then $N(k) = N^{(i)}(k) \ge k$ and $N(k') = N^{(i)}(k') < k'$ for $k' > k$, so $R = k$. Hence the events $\{p_i \le \frac{k}{m}q,\, R = k\}$ and $\{p_i \le \frac{k}{m}q,\, R^{(i)} = k\}$ are equal, and

$$\begin{aligned}\mathbb{E}\Big[\frac{\mathbf{1}_{\{i \text{ rejected}\}}}{\max(R,1)}\Big] &= \sum_{k=1}^{m} \frac{1}{k} P\Big(p_i \le \frac{k}{m}q,\, R = k\Big) = \sum_{k=1}^{m} \frac{1}{k} P\Big(p_i \le \frac{k}{m}q\Big) P(R^{(i)} = k) \\ &\le \sum_{k=1}^{m} \frac{q}{m} P(R^{(i)} = k) \le \frac{q}{m}.\end{aligned}$$

Since V is the number of rejected true null hypotheses, adding this over the m_0 true null hypotheses gives FDR $\le \frac{m_0}{m}q$. □

Example 10.17. Ten tests give the p-values

$$0.001,\ 0.008,\ 0.012,\ 0.030,\ 0.041,\ 0.10,\ 0.25,\ 0.40,\ 0.62,\ 0.90.$$

Without correction, five are below 0.05. Bonferroni with $\alpha = 0.05$ uses the threshold 0.005 and rejects only the first. Benjamini–Hochberg with $q = 0.05$ compares $p_{(k)}$ with $0.005k$: $0.001 \le 0.005$, $0.008 \le 0.010$ and $0.012 \le 0.015$, but $p_{(k)} > 0.005k$ for all $k \ge 4$, so it rejects three hypotheses.

10.3 The bootstrap and permutation tests

The formulas for standard errors and p-values so far required either a simple statistic, such as a mean, or a known distribution. Two resampling methods work without such formulas, at the cost of computation.

The bootstrap. To estimate the uncertainty of a statistic $\hat\theta = s(x_1, \dots, x_N)$, such as a median, a correlation, or the AUC of a classifier, the bootstrap replaces the unknown distribution of the data by the empirical distribution (the plug-in principle). It draws B bootstrap samples $x_1^*, \dots, x_N^*$, each consisting of N draws with replacement from the data, and computes $\hat\theta_b^* = s(x_1^*, \dots, x_N^*)$ for $b = 1, \dots, B$. The standard deviation of the $\hat\theta_b^*$ estimates the standard error of $\hat\theta$, and the 2.5% and 97.5% quantiles of the $\hat\theta_b^*$ give an approximate 95% confidence interval (the percentile interval). For the sample mean, everything can be computed exactly: a bootstrap sample consists of N i.i.d. draws from the empirical distribution, whose variance is $\hat\sigma^2 = \frac{1}{N}\sum_i (x_i - \bar x)^2$, so the bootstrap standard error of the mean is $\frac{\hat\sigma}{\sqrt{N}}$, the familiar formula. The bootstrap can fail for very small samples, for statistics such as the maximum, and for dependent data.

Permutation tests. Suppose we observe $x_1, \dots, x_n$ in group A and $y_1, \dots, y_m$ in group B, and want to test H_0: both groups come from the same distribution. Under H_0, the group labels carry no information: given the $n + m$ observed values, each of the $\binom{n+m}{n}$ ways to assign n of them to group A is equally likely. So the p-value of a statistic such as the difference of the group means

is the fraction of all relabelings whose statistic is at least as large as the observed one. For large samples, we use a few thousand random relabelings instead of all of them. The test has exactly the claimed level, and it requires no assumptions about the distribution.

Example 10.18. Let group A have the values $12, 15, 14$ and group B the values $9, 10, 13$. The observed difference of the means is $13\frac{2}{3} - 10\frac{2}{3} = 3$. Of the $\binom{6}{3} = 20$ ways to split the six values into two groups of three, only 2 give a difference of at least 3: the observed split, and the split $\{13, 14, 15\}$ versus $\{9, 10, 12\}$, with the difference $3\frac{2}{3}$. So the one-sided p-value is $\frac{2}{20} = 0.1$: with so few observations, even a clear difference is not significant.

11 Evaluating Models and Experiments

11.1 Cross-validation

To choose between models or hyperparameters, we need to estimate how well a learning algorithm generalizes, and we often cannot afford to set aside a large test set. In k-fold cross-validation, the data are split into k folds of about equal size; for $j = 1, \dots, k$, the model is trained on all folds except the j-th one and evaluated on the j-th fold, which gives an error e_j. The cross-validation estimate is the average $\bar{e} = \frac{1}{k}\sum_j e_j$. Typical choices are $k = 5$ or 10; for $k = N$, it is leave-one-out cross-validation. The statistical properties of $\bar{e}$ are subtle.

- Bias. Each model is trained on only a fraction $\frac{k-1}{k}$ of the data, so $\bar{e}$ slightly overestimates the error of a model trained on all data; this bias is smaller for larger k.
- Variance. The fold errors are not independent, since the training sets overlap. So the naive standard error $\frac{s_e}{\sqrt{k}}$, with the standard deviation s_e of the fold errors, usually underestimates the true uncertainty; one can show that no estimator of the variance of $\bar{e}$ is unbiased for all distributions. For example, the fold errors $0.12, 0.15, 0.10, 0.14, 0.13$ have the mean 0.128, the standard deviation 0.019 and the naive standard error 0.009; the true uncertainty is larger. Repeating cross-validation with different random splits reduces the variance that comes from the split, but not the dependence.
- Comparing two models. Comparing two models on the same folds (or the same test examples) and analyzing the differences is much more precise than comparing their separate error estimates, since $\mathrm{Var}(A - B) = \mathrm{Var}(A) + \mathrm{Var}(B) - 2\,\mathrm{Cov}(A, B)$ and the errors of two models on the same data are positively correlated. This is the idea of paired tests.
- Selection bias. If we evaluate many models or hyperparameter settings and choose the one with the smallest cross-validation error, the selected error estimate is biased downward: by Jensen's inequality, the expected minimum of noisy estimates is smaller than the minimum of their expectations. For example, if 20 models all have the true error 0.10 and their estimates have independent errors with standard deviation 0.01, the smallest estimate is about $0.10 - 1.87 \cdot 0.01 \approx 0.081$ on average (the expected maximum of 20 independent standard normal variables is about 1.87). The remedy is to estimate the error of the selected model on data that played no role in the selection: a separate test set, used once, or nested cross-validation, which repeats the whole selection procedure inside each training fold.

11.2 Classification metrics

The accuracy of a binary classifier hides which errors it makes. A classifier usually computes a score $s(\mathbf{x})$, such as an estimate of $P(y = 1 \mid \mathbf{x})$, and predicts $\hat{y} = 1$ if $s(\mathbf{x}) \geq t$ for a threshold t. On

a test set, the confusion matrix counts the true positives (TP: $\hat{y} = 1$, $y = 1$), false positives (FP: $\hat{y} = 1$, $y = 0$), false negatives (FN: $\hat{y} = 0$, $y = 1$) and true negatives (TN: $\hat{y} = 0$, $y = 0$). From these counts, one computes

$$\text{precision} = \frac{\text{TP}}{\text{TP} + \text{FP}}, \qquad \text{recall} = \text{TPR} = \frac{\text{TP}}{\text{TP} + \text{FN}}, \qquad \text{FPR} = \frac{\text{FP}}{\text{FP} + \text{TN}},$$

and the F_1 score, the harmonic mean $\frac{2}{1/\text{precision}+1/\text{recall}} = \frac{2\text{TP}}{2\text{TP}+\text{FP}+\text{FN}}$ of precision and recall. The recall (true positive rate) is the fraction of the positives that are found, the precision is the fraction of the positive predictions that are correct, and the false positive rate is the fraction of the negatives that raise a false alarm. In the language of hypothesis testing, the FPR is the level of the test "H_0: $y = 0$", and the recall is its power.

When positives are rare, accuracy is misleading: if 1% of the transactions are fraudulent, the classifier that never predicts fraud has an accuracy of 99%, and recall 0. Precision depends on the base rate, as in the medical test of Example 1.7: by Bayes' theorem, precision is $P(y = 1 \mid \hat{y} = 1) = \frac{\text{TPR}\cdot\pi_1}{\text{TPR}\cdot\pi_1+\text{FPR}\cdot(1-\pi_1)}$ for the fraction π_1 of positives, so the same classifier has a much lower precision on data with fewer positives.

ROC curves and the AUC. Lowering the threshold t finds more positives, but also raises more false alarms. The ROC curve (receiver operating characteristic) shows the pairs $(\text{FPR}(t), \text{TPR}(t))$ for all thresholds; it goes from $(0, 0)$ (nothing predicted positive) to $(1, 1)$ (everything predicted positive). A classifier with random scores has the diagonal as its ROC curve, and a perfect one goes through $(0, 1)$. The area under the ROC curve (AUC) summarizes the curve in one number, which has a simple meaning.

Theorem 11.1. *Let a test set have n_+ positives and n_- negatives, with distinct scores. Then the area under the ROC curve is the fraction of the n_+n_- pairs of a positive and a negative example in which the positive example has the higher score.*

Proof. Sort the examples by decreasing score, and lower the threshold past them one at a time. Passing a positive example raises the TPR by $\frac{1}{n_+}$, and passing a negative example raises the FPR by $\frac{1}{n_-}$. So the ROC curve is a staircase, and its area is the sum of the areas of the strips of width $\frac{1}{n_-}$ below its horizontal steps, one for each negative example. The strip of the negative example j has the height $\frac{1}{n_+}$ times the number of positives with a higher score than j. Summing over j gives the number of correctly ordered pairs, divided by n_+n_-. □

So the AUC estimates the probability that a random positive example gets a higher score than a random negative one. It only depends on the order of the scores, not on a threshold, and it does not change if the classes are rebalanced. An AUC of 0.5 is no better than random guessing.

Example 11.2. Ten test examples have the scores $0.95, 0.85, 0.8, 0.7, 0.6, 0.45, 0.4, 0.3, 0.2, 0.1$, and the labels $1, 1, 0, 1, 1, 0, 1, 0, 0, 0$. With the threshold 0.5, TP = 4, FP = 1 (the score 0.8), FN = 1 (the score 0.4) and TN = 4, so the accuracy, precision, recall and F_1 are all 0.8, and FPR = 0.2. Of the 25 positive–negative pairs, 21 are ordered correctly (the positive with the score 0.7 is below one negative, 0.8, and so are the positive 0.6, and the positive 0.4, which is also below 0.45), so the AUC is 0.84.

A classifier is calibrated if, among the examples with the predicted probability q, a fraction q are positive, so that its scores can be read as probabilities. The cross-entropy loss rewards calibration: if the true probability of $y = 1$ is p, the expected loss $-p \log q - (1 - p) \log(1 - q) = H(p, q)$ of

predicting q is smallest for $q = p$, by Gibbs' inequality (Theorem 13.2). Losses with this property are called proper scoring rules; the accuracy is not one, since it only rewards being on the right side of $\frac{1}{2}$.

11.3 Generalization bounds

A model chosen because it fits the training data well will usually do worse on new data. How much worse can be bounded with Hoeffding's inequality and the union bound. Let $\ell(h; \mathbf{x}, y) \in [0, 1]$ be the loss of a model h on the example $(\mathbf{x}, y)$, for example the indicator of a misclassification. The risk $R(h) = \mathbb{E}[\ell(h; \mathbf{X}, Y)]$ is the expected loss on new data, and the empirical risk $\hat{R}(h) = \frac{1}{N}\sum_{i=1}^{N} \ell(h; \mathbf{x}_i, y_i)$ is the average loss on N i.i.d. training examples.

Theorem 11.3 (Generalization bound for finitely many models)**.** *Let $\mathcal{H}$ be a finite set of models, and let $0 < \delta < 1$. With probability at least $1 - \delta$ over the training data,*

$$|R(h) - \hat{R}(h)| \le \varepsilon = \sqrt{\frac{\log(2|\mathcal{H}|/\delta)}{2N}} \qquad \text{for all } h \in \mathcal{H} \text{ simultaneously.}$$

In particular, the empirical risk minimizer $\hat{h}$, the model in $\mathcal{H}$ with the smallest $\hat{R}$, satisfies $R(\hat{h}) \le \min_{h\in\mathcal{H}} R(h) + 2\varepsilon$ with probability at least $1 - \delta$.

Proof. For a fixed h, $\hat{R}(h)$ is a mean of N i.i.d. variables with values in $[0, 1]$ and mean $R(h)$, so by Hoeffding's inequality (Theorem 7.5), $P(|R(h) - \hat{R}(h)| > \varepsilon) \le 2e^{-2N\varepsilon^2} = \frac{\delta}{|\mathcal{H}|}$. By the union bound, the probability that this happens for at least one $h \in \mathcal{H}$ is at most δ. If it happens for none, then for every $h \in \mathcal{H}$, $R(\hat{h}) \le \hat{R}(\hat{h}) + \varepsilon \le \hat{R}(h) + \varepsilon \le R(h) + 2\varepsilon$. □

The price for choosing among many models is the factor $\log|\mathcal{H}|$, which grows slowly. For a single model evaluated on $N = 10\,000$ independent test examples and $\delta = 0.05$, the bound is $\varepsilon = 0.014$; for the best of $|\mathcal{H}| = 10^6$ models, it is 0.030. This is why a model must be evaluated on data that played no role in choosing it (Section 11): on the training data, the factor $\log|\mathcal{H}|$ is not paid, and the error estimate is too optimistic. A model with d parameters stored as 32-bit numbers is one of at most 2^{32d} models, so $\log|\mathcal{H}| \le 32d \log 2$, and the bound is of the order $\sqrt{d/N}$: we need more data points than parameters. Neural networks often generalize much better than this; statistical learning theory replaces $\log|\mathcal{H}|$ by finer measures of complexity, such as the VC dimension, and explaining the generalization of large networks is an active research area.

11.4 A/B testing and experiment design

To find out whether a change causes an improvement, for example a new recommendation model, the most reliable method is a randomized controlled experiment, called an A/B test in industry: each user is randomly assigned to version A (the control) or version B (the treatment), and the outcomes are compared. Randomization makes the two groups comparable in all respects, observed or not, on average. So a significant difference can be attributed to the treatment. Observational comparisons, such as "users who chose the new feature buy more", can be confounded by other differences between the groups.

Confounding and Simpson's paradox. A classic medical study compared two treatments for kidney stones:

	small stones	large stones	all
treatment A	$81/87 = 93\%$	$192/263 = 73\%$	$273/350 = 78\%$
treatment B	$234/270 = 87\%$	$55/80 = 69\%$	$289/350 = 83\%$

Treatment A is more successful for small stones and for large stones, but less successful overall. The reason is that A was mostly given for large stones (263 of 350 patients), which are harder to treat, and B mostly for small ones. By the law of total probability, $P(\text{success} \mid A) = \sum_s P(\text{success} \mid A, s)\, P(s \mid A)$ averages over the stone sizes s with weights $P(s \mid A)$ that differ between the treatments; the stone size is a confounder. Weighting both treatments with the same distribution of stone sizes, that of all 700 patients (51% small), gives $0.51 \cdot 0.93 + 0.49 \cdot 0.73 = 0.83$ for A and $0.51 \cdot 0.87 + 0.49 \cdot 0.69 = 0.78$ for B, the right comparison if the size is the only confounder. Randomization makes $P(s \mid A) = P(s \mid B)$ on average for every confounder s, including the ones we do not know about.

Analyzing an A/B test. Let $\hat{p}_A$ and $\hat{p}_B$ be the conversion rates (for example, the fractions of users who buy something) of n_A and n_B users. Under H_0: $p_A = p_B$, the pooled rate $\hat{p}$ of all users estimates the common rate, and by the central limit theorem,

$$Z = \frac{\hat{p}_B - \hat{p}_A}{\sqrt{\hat{p}(1-\hat{p})\left(\frac{1}{n_A} + \frac{1}{n_B}\right)}}$$

is approximately $\mathcal{N}(0,1)$ (the two-proportion z-test). For example, if 200 of 5000 users convert with version A (4%) and 250 of 5000 with version B (5%), then $\hat{p} = 0.045$, $Z \approx 2.41$, and the two-sided p-value is 0.016. An approximate 95% confidence interval for the improvement $p_B - p_A$ is $0.01 \pm 1.96\sqrt{\frac{\hat{p}_A(1-\hat{p}_A)}{n_A} + \frac{\hat{p}_B(1-\hat{p}_B)}{n_B}} = [0.002, 0.018]$.

Sample size and power. How many users do we need to detect an improvement $\delta = p_B - p_A$? With n users per group, the standard error of $\hat{p}_B - \hat{p}_A$ is about $\sqrt{2\bar{p}(1-\bar{p})/n}$, where $\bar{p}$ is the average rate, and Z is approximately $\mathcal{N}(\delta/\mathrm{SE}, 1)$. The two-sided test at level α rejects if $|Z| > z_{1-\alpha/2}$, so it has power $1-\beta$ approximately when $\frac{\delta}{\mathrm{SE}} = z_{1-\alpha/2} + z_{1-\beta}$, i.e.,

$$n \approx \frac{(z_{1-\alpha/2} + z_{1-\beta})^2 \cdot 2\bar{p}(1-\bar{p})}{\delta^2}$$

per group, where $z_u = \Phi^{-1}(u)$. For $\alpha = 0.05$ and power 0.8, $z_{0.975} + z_{0.8} \approx 1.96 + 0.84$; to detect an increase from 4% to 5%, we need about 6750 users per group. Halving the effect that we want to detect quadruples the sample size.

Design principles and pitfalls.

- Fix the primary metric, the test and the sample size in advance. Peeking at the p-value repeatedly and stopping as soon as it is below 0.05 inflates the false positive rate: in a simulation with two identical versions (conversion rate 5%, up to 10 000 users per group), checking after every 200 users per group declared a significant difference in about one third of the runs. If early stopping is needed, use sequential tests designed for it, such as the likelihood ratio test in the appendix to this lecture.
- Many metrics or user segments mean many tests: correct for multiple testing, or treat secondary results as hypotheses for future experiments.

- Randomize at the right level. If users influence each other (for example in a social network or a marketplace), or one user is counted several times, the observations are not independent, and the standard errors above are too small.
- Reduce variance by design. Paired or blocked designs compare similar units (for example, the same users or the same queries under both versions), and adjusting for measurements taken before the experiment can reduce the variance considerably.
- Watch for novelty effects, which make new versions look better at first, and check that the groups have the planned sizes; a mismatch indicates a bug in the assignment.

The same principles apply to offline comparisons of machine learning models: whether model B is really better than model A is a hypothesis testing question, and many reported improvements are within the noise of the evaluation.

12 Bayesian Inference

Maximum likelihood treats the parameter θ as a fixed unknown number. Bayesian inference treats it as a random variable, which expresses our uncertainty about it. Before seeing data, our beliefs are described by a prior distribution $p(\theta)$; after seeing the data $\mathcal{D}$, Bayes' theorem gives the posterior distribution

$$p(\theta \mid \mathcal{D}) = \frac{p(\mathcal{D} \mid \theta)\, p(\theta)}{p(\mathcal{D})}, \qquad p(\mathcal{D}) = \int p(\mathcal{D} \mid \theta)\, p(\theta)\, d\theta.$$

In words: the posterior is proportional to the likelihood times the prior. The normalizing constant $p(\mathcal{D})$ is called the evidence; it does not depend on θ.

12.1 Coin flips and the Beta distribution

Definition 12.1 (Beta distribution)**.** For $a, b > 0$, the Beta(a, b) distribution on $[0, 1]$ has the density

$$p(\theta) = \frac{\theta^{a-1}(1-\theta)^{b-1}}{B(a,b)}, \qquad \text{where} \quad B(a,b) = \int_0^1 \theta^{a-1}(1-\theta)^{b-1}\, d\theta.$$

For positive integers, $B(a,b) = \frac{(a-1)!\,(b-1)!}{(a+b-1)!}$; for example, Beta$(1,1)$ is the uniform distribution, and Beta$(2,2)$ has the density $6\theta(1-\theta)$. The mean of Beta(a,b) is $\frac{a}{a+b}$, and for $a, b > 1$ its mode (the maximum of the density) is $\frac{a-1}{a+b-2}$.

Theorem 12.2 (Beta–Bernoulli conjugacy)**.** *Let the prior of θ be Beta(a,b), and let the data be N independent Bernoulli(θ) observations with k ones. Then the posterior is Beta$(a+k,\, b+N-k)$.*

Proof. $p(\theta \mid \mathcal{D}) \propto \theta^k(1-\theta)^{N-k} \cdot \theta^{a-1}(1-\theta)^{b-1} = \theta^{a+k-1}(1-\theta)^{b+N-k-1}$. The only density proportional to this function is that of Beta$(a+k, b+N-k)$. □

A prior is called conjugate to a likelihood if the posterior belongs to the same family as the prior; then updating our beliefs amounts to updating a few parameters. Here, the prior acts like a previously seen ones and b previously seen zeros.

Example 12.3. With the prior Beta$(2,2)$, which expresses a mild belief that the coin is fair, and three heads in three flips, the posterior is Beta$(5,2)$. Its mean is $\frac{5}{7} \approx 0.71$ and its mode is $\frac{4}{5} = 0.8$, whereas the MLE was 1 (Example 9.8). After 30 flips with 20 heads, the posterior is Beta$(22,12)$, with mean 0.65 and standard deviation 0.08: with more data, the posterior concentrates, and the prior matters less (Figure 5).

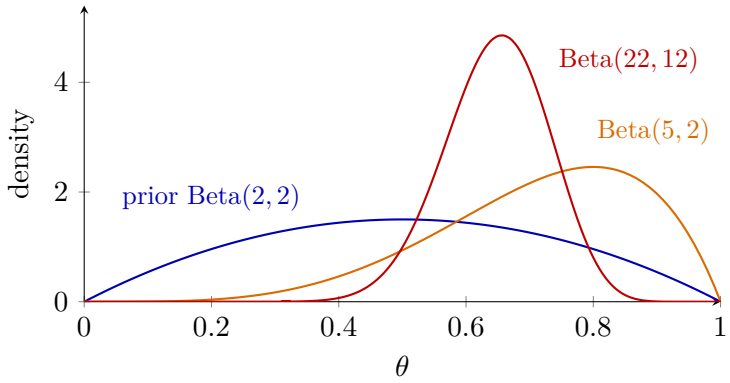


Figure 5: Bayesian updating for the probability θ of heads: the prior Beta$(2,2)$ (blue), the posterior Beta$(5,2)$ after 3 heads in 3 flips (orange), and the posterior Beta$(22,12)$ after 20 heads in 30 flips (red). With more data, the posterior concentrates around the relative frequency of heads.

To predict the next flip, Bayesian inference averages over all values of θ, weighted by the posterior:

$$P(X_{N+1}=1 \mid \mathcal{D}) = \int_0^1 \theta\, p(\theta \mid \mathcal{D})\, d\theta = \mathbb{E}[\theta \mid \mathcal{D}] = \frac{a+k}{a+b+N}.$$

For the uniform prior ($a=b=1$), this is Laplace's rule of succession $\frac{k+1}{N+2}$: after three heads in three flips, the probability of heads is $\frac{4}{5}$, not 1. The same "add pseudo-counts" idea (Laplace smoothing) prevents zero probabilities for unseen words in language models.

Example 12.4 (Credible intervals)**.** A 95% credible interval contains θ with posterior probability 0.95, for example the interval between the 2.5% and the 97.5% quantiles of the posterior. For the posterior Beta$(22,12)$ after 20 heads in 30 flips (Example 12.3), it is $[0.48, 0.80]$ (computed numerically), close to the approximate confidence interval $\frac{2}{3} \pm 1.96\sqrt{\frac{2}{3}\cdot\frac{1}{3}/30} = [0.50, 0.84]$ of Section 9. The interpretations differ: the credible interval makes a probability statement about θ, given the data, while the confidence interval makes a probability statement about the random interval, for a fixed θ. As the amount of data grows, the two intervals agree more and more (the Bernstein–von Mises theorem).

Example 12.5 (Gamma–Poisson conjugacy)**.** Let the numbers of clicks on an ad in N hours be i.i.d. Poisson(λ), and let the prior of λ be the Gamma distribution with the shape a and the rate b, i.e., with the density $\frac{b^a}{\Gamma(a)}\lambda^{a-1}e^{-b\lambda}$ for $\lambda > 0$ and the mean $\frac{a}{b}$ (the Gamma distribution of Example 9.6 with the scale $\frac{1}{b}$). If S is the total number of clicks, the posterior is

$$p(\lambda \mid \mathcal{D}) \propto e^{-N\lambda}\lambda^{S} \cdot \lambda^{a-1}e^{-b\lambda} = \lambda^{a+S-1}e^{-(b+N)\lambda},$$

i.e., Gamma$(a+S, b+N)$: the prior acts like b previous hours with a clicks in total. The posterior mean $\frac{a+S}{b+N}$ lies between the prior mean $\frac{a}{b}$ and the MLE $\frac{S}{N}$. For the prior Gamma$(2,1)$ and the counts $3,5,4,6,2$ in five hours, the posterior is Gamma$(22,6)$, with the mean $\frac{22}{6} \approx 3.67$ (the MLE is 4) and the standard deviation $\frac{\sqrt{22}}{6} \approx 0.78$. The predictive probability of an hour without clicks is $\int e^{-\lambda}\, p(\lambda \mid \mathcal{D})\, d\lambda = \left(\frac{6}{7}\right)^{22} \approx 0.034$, while plugging in the MLE gives $e^{-4} \approx 0.018$: the posterior accounts for the uncertainty about λ.

12.2 Maximum a posteriori estimation and regularization

Often we only want a single parameter value. The maximum a posteriori (MAP) estimate is the mode of the posterior,

$$\hat{\theta}_{\mathrm{MAP}} = \arg\max_{\theta} p(\theta \mid \mathcal{D}) = \arg\min_{\theta} \big(-\log p(\mathcal{D} \mid \theta) - \log p(\theta) \big),$$

since $p(\mathcal{D})$ does not depend on θ. So MAP estimation minimizes the negative log-likelihood plus a penalty $-\log p(\theta)$ coming from the prior: a prior is a regularizer.

Proposition 12.6 (Ridge regression as MAP estimation)**.** *In the linear regression model of Proposition 9.19, with the prior $\mathbf{w} \sim \mathcal{N}(\mathbf{0}, \alpha^{-1} I)$, the MAP estimate minimizes $\|\mathbf{y} - X\mathbf{w}\|^2 + \lambda \|\mathbf{w}\|^2$ with $\lambda = \alpha\sigma^2$, i.e., it is the ridge regression solution $\hat{\mathbf{w}} = (X^\top X + \lambda I)^{-1} X^\top \mathbf{y}$.*

Proof. Up to constants, $-\log p(\mathcal{D} \mid \mathbf{w}) - \log p(\mathbf{w}) = \frac{1}{2\sigma^2}\|\mathbf{y} - X\mathbf{w}\|^2 + \frac{\alpha}{2}\|\mathbf{w}\|^2$; multiply by $2\sigma^2$. The minimum was computed in the exercises of Lecture 3. □

A strong prior belief that the weights are small (large α) means strong regularization. Similarly, the Laplace prior $p(\mathbf{w}) \propto e^{-\|\mathbf{w}\|_1 / b}$ leads to ℓ_1 regularization (the lasso), which favors sparse weight vectors.

12.3 Bayesian linear regression

The MAP estimate discards the uncertainty that the posterior expresses. In linear regression with a Gaussian prior, the full posterior can be computed with the tools of Section 6.

Theorem 12.7 (Posterior of Bayesian linear regression)**.** *In the model of Proposition 9.19 with the prior $\mathbf{w} \sim \mathcal{N}(\mathbf{0}, \alpha^{-1} I)$, the posterior is $\mathcal{N}(\mathbf{m}, S)$ with*

$$S = \big(\alpha I + \sigma^{-2} X^\top X\big)^{-1}, \qquad \mathbf{m} = \sigma^{-2} S X^\top \mathbf{y}.$$

For a new input $\mathbf{x}_*$, the predictive distribution of $y_* = \mathbf{w}^\top \mathbf{x}_* + \varepsilon_*$ is $\mathcal{N}(\mathbf{m}^\top \mathbf{x}_*,\ \mathbf{x}_*^\top S \mathbf{x}_* + \sigma^2)$.

Proof. Up to an additive constant that does not depend on $\mathbf{w}$,

$$\log p(\mathbf{w} \mid \mathcal{D}) = -\frac{1}{2\sigma^2}\|\mathbf{y} - X\mathbf{w}\|^2 - \frac{\alpha}{2}\|\mathbf{w}\|^2 = -\frac{1}{2}\mathbf{w}^\top S^{-1}\mathbf{w} + \sigma^{-2}\mathbf{y}^\top X\mathbf{w} + \text{const}.$$

Completing the square, this is $-\frac{1}{2}(\mathbf{w} - \mathbf{m})^\top S^{-1}(\mathbf{w} - \mathbf{m}) + \text{const}$, since the linear term of the latter is $\mathbf{m}^\top S^{-1}\mathbf{w} = \sigma^{-2}\mathbf{y}^\top X\mathbf{w}$. So the posterior density is proportional to the density of $\mathcal{N}(\mathbf{m}, S)$, hence equal to it. Given the data, $\mathbf{w}^\top \mathbf{x}_*$ is $\mathcal{N}(\mathbf{m}^\top \mathbf{x}_*, \mathbf{x}_*^\top S \mathbf{x}_*)$ by Corollary 6.7, and adding the independent noise $\varepsilon_* \sim \mathcal{N}(0, \sigma^2)$ adds σ^2 to the variance. □

The posterior mean $\mathbf{m} = (X^\top X + \alpha\sigma^2 I)^{-1} X^\top \mathbf{y}$ is the ridge regression solution, which is also the MAP estimate (the mode and the mean of a Gaussian coincide). But the posterior also gives error bars: the predictive variance $\mathbf{x}_*^\top S \mathbf{x}_* + \sigma^2$ consists of the uncertainty about the weights, which shrinks as we collect more data, and the noise, which does not.

Example 12.8. For the data $(1, 2), (2, 3), (3, 5)$, the model $y = wx + \varepsilon$ without intercept, $\alpha = 1$ and $\sigma^2 = 1$: $S = (1 + 1 + 4 + 9)^{-1} = \frac{1}{15}$ and $m = \frac{1}{15}(2 + 6 + 15) = \frac{23}{15} \approx 1.53$. At $x_* = 4$, the predictive distribution is $\mathcal{N}(6.13,\ \frac{16}{15} + 1) \approx \mathcal{N}(6.13, 2.07)$.

12.4 Gaussian processes

Bayesian linear regression puts a prior on the weights of a fixed set of features. A Gaussian process puts a prior directly on the unknown function.

Definition 12.9 (Gaussian process)**.** A Gaussian process with mean function m and covariance function (or kernel) k is a random function f such that for all points $\mathbf{x}_1, \dots, \mathbf{x}_n$, the vector $(f(\mathbf{x}_1), \dots, f(\mathbf{x}_n))$ is Gaussian with the mean vector $(m(\mathbf{x}_i))_i$ and the covariance matrix K with entries $K_{ij} = k(\mathbf{x}_i, \mathbf{x}_j)$. We write $f \sim \mathcal{GP}(m, k)$.

For this to make sense, every such matrix K must be symmetric and positive semidefinite. This holds, for example, for $k(\mathbf{x}, \mathbf{x}') = \boldsymbol{\phi}(\mathbf{x})^\top \boldsymbol{\phi}(\mathbf{x}')$ with any feature map $\boldsymbol{\phi}$, since K is then a Gram matrix (compare with the remark on kernels in Lecture 2).

Example 12.10. 1. (Linear kernel.) If $f(\mathbf{x}) = \mathbf{w}^\top \mathbf{x}$ with $\mathbf{w} \sim \mathcal{N}(\mathbf{0}, \alpha^{-1} I)$, then

$$\operatorname{Cov}(f(\mathbf{x}), f(\mathbf{x}')) = \mathbf{x}^\top \operatorname{Cov}(\mathbf{w}) \mathbf{x}' = \alpha^{-1} \mathbf{x}^\top \mathbf{x}'.$$

So the prior of Bayesian linear regression is the Gaussian process with the kernel $k(\mathbf{x}, \mathbf{x}') = \alpha^{-1} \mathbf{x}^\top \mathbf{x}'$.

2. (Squared exponential kernel.) $k(\mathbf{x}, \mathbf{x}') = \exp\big(-\frac{\|\mathbf{x} - \mathbf{x}'\|^2}{2\ell^2}\big)$ gives smooth random functions, whose values at points closer than the length scale ℓ are strongly correlated. It corresponds to infinitely many features.

3. (Brownian motion.) $k(s, t) = \min(s, t)$ for $s, t \geq 0$ is the covariance function of Brownian motion, a random function that we study in Section 15.

Proposition 12.11 (Gaussian process regression)**.** *Let $f \sim \mathcal{GP}(0, k)$, and let $y_i = f(\mathbf{x}_i) + \varepsilon_i$ for $i = 1, \dots, N$, with i.i.d. noise $\varepsilon_i \sim \mathcal{N}(0, \sigma^2)$ independent of f. Let K be the kernel matrix of the training inputs, and for a new input $\mathbf{x}_*$, let $\mathbf{k}_* = (k(\mathbf{x}_1, \mathbf{x}_*), \dots, k(\mathbf{x}_N, \mathbf{x}_*))^\top$. Then, given the data,*

$$f(\mathbf{x}_*) \sim \mathcal{N}\big(\mathbf{k}_*^\top (K + \sigma^2 I)^{-1} \mathbf{y},\ k(\mathbf{x}_*, \mathbf{x}_*) - \mathbf{k}_*^\top (K + \sigma^2 I)^{-1} \mathbf{k}_*\big).$$

Proof. The vector $(\mathbf{y}, f(\mathbf{x}_*))$ is Gaussian with mean $\mathbf{0}$ and the covariance blocks $\operatorname{Cov}(\mathbf{y}) = K + \sigma^2 I$, $\operatorname{Cov}(\mathbf{y}, f(\mathbf{x}_*)) = \mathbf{k}_*$ and $\operatorname{Var}(f(\mathbf{x}_*)) = k(\mathbf{x}_*, \mathbf{x}_*)$; now apply Theorem 6.5. □

The posterior mean $\sum_i \alpha_i k(\mathbf{x}_i, \mathbf{x}_*)$ with $\boldsymbol{\alpha} = (K + \sigma^2 I)^{-1} \mathbf{y}$ is a combination of kernel functions centered at the data points; it is also the prediction of kernel ridge regression. The posterior variance does not depend on $\mathbf{y}$: it is small near the data, and it returns to the prior variance far away from them (Figure 6). For the linear kernel, the proposition reproduces the predictive distribution of Theorem 12.7.

Example 12.12. With the squared exponential kernel ($\ell = 1$), $\sigma^2 = 0.1$, and the data $(0, 1)$ and $(1, 0)$: $K + \sigma^2 I = \begin{pmatrix} 1.1 & 0.607 \\ 0.607 & 1.1 \end{pmatrix}$, and at $x_* = 0.5$, $\mathbf{k}_* = (0.882, 0.882)^\top$. The posterior mean is 0.517, and the posterior variance is $1 - \mathbf{k}_*^\top (K + \sigma^2 I)^{-1} \mathbf{k}_* \approx 0.087$; a new noisy observation at x_* has the predictive variance $0.087 + 0.1 = 0.187$.

The hyperparameters, such as ℓ and σ^2, are usually chosen by maximizing the marginal likelihood $\log p(\mathbf{y}) = -\frac{1}{2} \mathbf{y}^\top (K + \sigma^2 I)^{-1} \mathbf{y} - \frac{1}{2} \log\det(K + \sigma^2 I) - \frac{N}{2} \log(2\pi)$, since $\mathbf{y} \sim \mathcal{N}(\mathbf{0}, K + \sigma^2 I)$; its gradient can be computed with the matrix calculus of Lecture 3. The cost of solving with the $N \times N$ matrix $K + \sigma^2 I$ grows like N^3, so exact Gaussian processes are limited to moderate data sets. Remarkably, one can show that neural networks with random weights become Gaussian processes in the limit of infinitely wide layers.

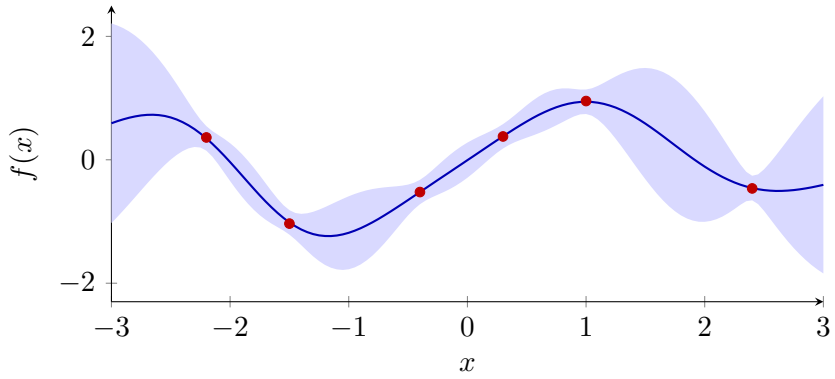


Figure 6: Gaussian process regression with the squared exponential kernel ($\ell = 0.7$, $\sigma^2 = 0.01$) for six data points (red): the posterior mean (blue) and a band of ± 2 posterior standard deviations. The uncertainty is small near the data and grows back to the prior standard deviation 1 between and beyond them.

Remark 12.13 (Beyond conjugate models). For most models, such as neural networks, the posterior has no closed form, and the evidence $p(\mathcal{D})$ is an intractable integral. Approximate methods include Markov chain Monte Carlo sampling (Section 8) and variational inference, which approximates the posterior by a simpler distribution by minimizing a Kullback–Leibler divergence (Section 13).

13 Information Theory

Information theory measures uncertainty and the difference between distributions. Its quantities appear in machine learning as loss functions (cross-entropy), as regularizers (the KL term of variational autoencoders), and in the analysis of learning algorithms.

Definition 13.1 (Entropy, cross-entropy and KL divergence)**.** Let $p = (p_1, \dots, p_K)$ and $q = (q_1, \dots, q_K)$ be probability distributions on $\{1, \dots, K\}$. The entropy of p, the cross-entropy of q relative to p, and the Kullback–Leibler (KL) divergence of q from p are

$$H(p) = -\sum_k p_k \log p_k, \qquad H(p, q) = -\sum_k p_k \log q_k, \qquad \mathrm{KL}(p \,\|\, q) = \sum_k p_k \log \frac{p_k}{q_k},$$

with the conventions $0 \log 0 = 0$ and $\mathrm{KL}(p \,\|\, q) = \infty$ if $q_k = 0 < p_k$ for some k.

The number $-\log p_k$ measures the surprise of seeing the outcome k: rare outcomes are surprising. The entropy is the expected surprise. With $\log_2$ instead of log, entropy is measured in bits, and it is the minimal average number of bits per symbol needed to encode outcomes of p (Shannon's source coding theorem). A fair coin has entropy 1 bit, a coin with $P(\text{heads}) = 0.9$ only 0.47 bits, and a deterministic outcome 0 bits. Clearly $H(p) \geq 0$, and

$$H(p, q) = H(p) + \mathrm{KL}(p \,\|\, q) :$$

the cross-entropy is the average number of bits if we encode outcomes of p with a code designed for q, and the KL divergence is the number of extra bits we pay for using the wrong distribution.

Theorem 13.2 (Gibbs' inequality)**.** $\mathrm{KL}(p \,\|\, q) \geq 0$*, with equality if and only if* $p = q$*.*

Proof. We may assume $q_k > 0$ whenever $p_k > 0$. Let T be the set of k with $p_k > 0$, and let R be the random variable that takes the value $\frac{q_k}{p_k}$ with probability p_k, for $k \in T$. By Jensen's inequality for the concave function log (Theorem 3.7),

$$-\operatorname{KL}(p \,\|\, q) = \sum_{k \in T} p_k \log \frac{q_k}{p_k} = \mathbb{E}[\log R] \le \log \mathbb{E}[R] = \log \sum_{k \in T} q_k \le \log 1 = 0.$$

If equality holds, then $\sum_{k \in T} q_k = 1$, and equality holds in Jensen's inequality, which for the strictly concave log happens only if R is constant; so $q_k = c\, p_k$ for $k \in T$ and $q_k = 0$ otherwise, and $c = 1$ since both sum to 1. □

Two consequences: the cross-entropy $H(p, q)$ is minimal for $q = p$, and for the uniform distribution u on K outcomes, $0 \le \operatorname{KL}(p \,\|\, u) = \log K - H(p)$, so $H(p) \le \log K$: the uniform distribution has maximal entropy (compare with the exercise on maximum entropy in Lecture 3). The KL divergence is not a distance in the sense of Lecture 2, since it is not symmetric. For $p = (\frac{1}{2}, \frac{1}{2})$ and $q = (0.9, 0.1)$, $\operatorname{KL}(p \,\|\, q) \approx 0.511$ but $\operatorname{KL}(q \,\|\, p) \approx 0.368$.

Theorem 13.3 (Maximum likelihood minimizes the KL divergence)**.** *Let $x_1, \dots, x_N \in \{1, \dots, K\}$ be data with the empirical distribution $\hat{p}$, i.e., $\hat{p}_k$ is the fraction of data points equal to k, and let q_θ be a model distribution. Then*

$$\frac{1}{N} \sum_{i=1}^{N} \log q_\theta(x_i) = -H(\hat{p}, q_\theta) = -H(\hat{p}) - \operatorname{KL}(\hat{p} \,\|\, q_\theta).$$

So maximizing the likelihood is the same as minimizing the cross-entropy $H(\hat{p}, q_\theta)$, and the same as minimizing $\operatorname{KL}(\hat{p} \,\|\, q_\theta)$.

Proof. Grouping the data points by their values, $\frac{1}{N} \sum_i \log q_\theta(x_i) = \sum_k \hat{p}_k \log q_\theta(k) = -H(\hat{p}, q_\theta)$. The term $H(\hat{p})$ does not depend on θ. □

This is why the loss of classifiers is called the cross-entropy loss: for each training example, it is the cross-entropy between the one-hot distribution of the true label and the predicted distribution, and training fits the model distribution to the data in the sense of the KL divergence.

Continuous distributions. For densities, the sums become integrals: the differential entropy is $h(p) = -\int p(x) \log p(x)\, dx$, and $\operatorname{KL}(p \,\|\, q) = \int p(x) \log \frac{p(x)}{q(x)}\, dx \ge 0$. The differential entropy of $\mathcal{N}(\mu, \sigma^2)$ is $\frac{1}{2} \log(2\pi e \sigma^2)$, and among all densities with variance σ^2, the Gaussian has the largest entropy.

Example 13.4 (Maximum entropy)**.** Among the distributions on $\mathbb{R}$ with the variance 1, the uniform distribution on $[-\sqrt{3}, \sqrt{3}]$ has the differential entropy $\log(2\sqrt{3}) \approx 1.24$, the Laplace distribution with the density $\frac{1}{\sqrt{2}} e^{-\sqrt{2}|x|}$ has $1 + \frac{1}{2} \log 2 \approx 1.35$, and $\mathcal{N}(0, 1)$ has the largest value, $\frac{1}{2} \log(2\pi e) \approx 1.42$ nats. In fact, for every density p with mean 0 and variance 1, $\operatorname{KL}(p \,\|\, \mathcal{N}(0, 1)) = -h(p) + \mathbb{E}_p[\frac{1}{2} \log(2\pi) + \frac{X^2}{2}] = \frac{1}{2} \log(2\pi e) - h(p)$, which is nonnegative by Gibbs' inequality. This proves the maximum entropy property of the Gaussian, which is one justification for Gaussian noise models: they assume nothing beyond the variance.

Proposition 13.5 (KL divergence between Gaussians)**.** *For $p = \mathcal{N}(\mu_1, \sigma_1^2)$ and $q = \mathcal{N}(\mu_2, \sigma_2^2)$,*

$$\operatorname{KL}(p \,\|\, q) = \log \frac{\sigma_2}{\sigma_1} + \frac{\sigma_1^2 + (\mu_1 - \mu_2)^2}{2\sigma_2^2} - \frac{1}{2}.$$

Proof. $\log \frac{p(x)}{q(x)} = \log \frac{\sigma_2}{\sigma_1} - \frac{(x-\mu_1)^2}{2\sigma_1^2} + \frac{(x-\mu_2)^2}{2\sigma_2^2}$. Take the expectation for $X \sim p$, using $\mathbb{E}[(X-\mu_1)^2] = \sigma_1^2$ and $\mathbb{E}[(X-\mu_2)^2] = \sigma_1^2 + (\mu_1 - \mu_2)^2$. □

In a variational autoencoder, each data point is encoded as a Gaussian $\mathcal{N}(\mu, \sigma^2)$ (in each latent dimension), and the loss contains the term $\mathrm{KL}\left(\mathcal{N}(\mu, \sigma^2) \,\|\, \mathcal{N}(0,1)\right) = \frac{1}{2}(\mu^2 + \sigma^2 - 1 - \log \sigma^2)$, which keeps the codes close to a standard Gaussian. For example, for $\mu = 1$ and $\sigma = 2$, it is $\frac{1}{2}(4 - \log 4) \approx 1.31$.

Mutual information. The mutual information of two random variables is the KL divergence between their joint distribution and the product of their marginals,

$$I(X;Y) = \mathrm{KL}\left(p(x,y) \,\|\, p_X(x) p_Y(y)\right) = \sum_{x,y} p(x,y) \log \frac{p(x,y)}{p_X(x) p_Y(y)}.$$

By Gibbs' inequality, $I(X;Y) \geq 0$, with equality if and only if X and Y are independent. Unlike the correlation, it detects every kind of dependence. For Example 4.3, $I(X;Y) \approx 0.086$ nats.

14 Latent Variable Models and the EM Algorithm

Many data sets are explained by quantities that we do not observe: the cluster to which a customer belongs, the topic of a document, or the object in an image. A latent variable model describes the joint distribution $p(\mathbf{x}, z \mid \theta)$ of an observation $\mathbf{x}$ and a hidden variable z. Only $\mathbf{x}$ is observed, so the likelihood is the marginal

$$p(\mathbf{x} \mid \theta) = \sum_{z} p(\mathbf{x}, z \mid \theta) \qquad \text{or} \qquad \int p(\mathbf{x}, z \mid \theta)\, dz,$$

and the logarithm of a sum is hard to maximize directly. This section derives a general method for this problem from the KL divergence of Section 13.

14.1 Gaussian mixture models

Definition 14.1 (Gaussian mixture model)**.** A Gaussian mixture model with K components draws a component $z \in \{1, \dots, K\}$ with probabilities $\pi_1, \dots, \pi_K$, and then $\mathbf{x} \sim \mathcal{N}(\boldsymbol{\mu}_z, \Sigma_z)$. So the density of $\mathbf{x}$ is

$$p(\mathbf{x} \mid \theta) = \sum_{k=1}^{K} \pi_k \, \varphi(\mathbf{x}; \boldsymbol{\mu}_k, \Sigma_k),$$

where $\varphi(\cdot\,; \boldsymbol{\mu}, \Sigma)$ is the density of $\mathcal{N}(\boldsymbol{\mu}, \Sigma)$, and θ consists of the weights π_k, the means $\boldsymbol{\mu}_k$ and the covariance matrices Σ_k.

Given an observation, Bayes' theorem gives the posterior probabilities of the components,

$$r_k(\mathbf{x}) = P(z = k \mid \mathbf{x}, \theta) = \frac{\pi_k \, \varphi(\mathbf{x}; \boldsymbol{\mu}_k, \Sigma_k)}{\sum_{j} \pi_j \, \varphi(\mathbf{x}; \boldsymbol{\mu}_j, \Sigma_j)},$$

called the responsibilities: a soft assignment of $\mathbf{x}$ to the clusters. If we knew the components z_i of the data points, maximum likelihood would be easy: $\hat{\pi}_k$ would be the fraction of points in cluster k, and $\hat{\boldsymbol{\mu}}_k$ and $\hat{\Sigma}_k$ the mean and covariance matrix of these points (Theorem 9.10). The EM algorithm alternates between guessing the components with the current parameters and fitting the parameters to the guess.

14.2 The evidence lower bound

Theorem 14.2 (Evidence lower bound)**.** *For every distribution q on the values of z (with $q(z) > 0$ only where $p(\mathbf{x}, z \mid \theta) > 0$),*

$$\log p(\mathbf{x} \mid \theta) = \underbrace{\sum_z q(z) \log \frac{p(\mathbf{x}, z \mid \theta)}{q(z)}}_{\mathrm{ELBO}(q,\theta)} + \mathrm{KL}\left(q \,\|\, p(\cdot \mid \mathbf{x}, \theta)\right).$$

Hence $\log p(\mathbf{x} \mid \theta) \geq \mathrm{ELBO}(q, \theta)$, with equality if and only if q is the posterior $p(z \mid \mathbf{x}, \theta)$.

Proof. Since $p(\mathbf{x}, z \mid \theta) = p(z \mid \mathbf{x}, \theta)\, p(\mathbf{x} \mid \theta)$,

$$\begin{aligned} \sum_z q(z) \log \frac{p(\mathbf{x}, z \mid \theta)}{q(z)} &= \sum_z q(z) \log p(\mathbf{x} \mid \theta) + \sum_z q(z) \log \frac{p(z \mid \mathbf{x}, \theta)}{q(z)} \\ &= \log p(\mathbf{x} \mid \theta) - \mathrm{KL}\left(q \,\|\, p(\cdot \mid \mathbf{x}, \theta)\right). \end{aligned}$$

The rest follows from Gibbs' inequality (Theorem 13.2). □

The ELBO is the evidence lower bound; the marginal likelihood $p(\mathbf{x} \mid \theta)$ is also called the evidence. For data $\mathbf{x}_1, \ldots, \mathbf{x}_N$ with independent latent variables, we choose one distribution q_i for each data point, and then $\ell(\theta) = \sum_i \log p(\mathbf{x}_i \mid \theta) \geq \sum_i \mathrm{ELBO}(q_i, \theta)$.

Definition 14.3 (EM algorithm)**.** Starting from parameters θ_0, the expectation-maximization (EM) algorithm repeats two steps:

1. (E-step) $q_i = p(\cdot \mid \mathbf{x}_i, \theta_t)$, the posterior of the latent variable of each data point;
2. (M-step) θ_{t+1} maximizes $\sum_i \mathrm{ELBO}(q_i, \theta)$. Since the terms $-q_i(z) \log q_i(z)$ do not depend on θ, this is the same as maximizing the expected complete-data log-likelihood

$$\sum_{i=1}^{N} \sum_z q_i(z) \log p(\mathbf{x}_i, z \mid \theta).$$

Theorem 14.4. *The EM algorithm never decreases the log-likelihood: $\ell(\theta_{t+1}) \geq \ell(\theta_t)$.*

Proof. By Theorem 14.2, the choice of q_i in the E-step, and the M-step,

$$\ell(\theta_{t+1}) \geq \sum_i \mathrm{ELBO}(q_i, \theta_{t+1}) \geq \sum_i \mathrm{ELBO}(q_i, \theta_t) = \ell(\theta_t).$$ □

So EM is coordinate ascent on the ELBO: the E-step maximizes it over the q_i, which makes the bound tight, and the M-step maximizes it over θ. Like gradient ascent, it finds a local maximum (or a saddle point) of the likelihood, not necessarily the global one, so it is run from several starting points.

Theorem 14.5 (EM for Gaussian mixtures)**.** *For a Gaussian mixture model, let $r_{ik} = r_k(\mathbf{x}_i)$ be the responsibilities computed with θ_t in the E-step, and $N_k = \sum_i r_{ik}$. The M-step gives*

$$\pi_k = \frac{N_k}{N}, \qquad \boldsymbol{\mu}_k = \frac{1}{N_k} \sum_{i=1}^{N} r_{ik} \mathbf{x}_i, \qquad \Sigma_k = \frac{1}{N_k} \sum_{i=1}^{N} r_{ik} (\mathbf{x}_i - \boldsymbol{\mu}_k)(\mathbf{x}_i - \boldsymbol{\mu}_k)^\top.$$

Proof. The M-step maximizes

$$\sum_{i=1}^{N}\sum_{k=1}^{K} r_{ik}\big(\log\pi_k + \log\varphi(\mathbf{x}_i;\boldsymbol{\mu}_k,\Sigma_k)\big) = \sum_k N_k \log\pi_k + \sum_k\sum_i r_{ik}\log\varphi(\mathbf{x}_i;\boldsymbol{\mu}_k,\Sigma_k).$$

The weights only appear in the first sum, which is $-N\,H(\hat{\mathbf{p}},\boldsymbol{\pi})$ for the distribution $\hat{p}_k = \frac{N_k}{N}$ (note $\sum_k N_k = N$); by Gibbs' inequality, it is largest for $\boldsymbol{\pi} = \hat{\mathbf{p}}$. The parameters of component k only appear in $\sum_i r_{ik}\log\varphi(\mathbf{x}_i;\boldsymbol{\mu}_k,\Sigma_k)$, the log-likelihood of a Gaussian in which the data point $\mathbf{x}_i$ has the weight r_{ik}. The computation of Theorem 9.10, with these weights, gives the weighted mean and covariance matrix. □

So EM for Gaussian mixtures alternates between computing the responsibilities and refitting each Gaussian to all data points, weighted by their responsibilities. Each step costs $O(NK)$ density evaluations.

Example 14.6. Fit a mixture of two Gaussians with variance 1 and equal weights to the four numbers $1, 2, 4, 5$, starting with the means $\mu_1 = 0$ and $\mu_2 = 3$; only the means are fitted. The responsibilities of the first component are $r_{i1} = \frac{e^{-(x_i-\mu_1)^2/2}}{e^{-(x_i-\mu_1)^2/2}+e^{-(x_i-\mu_2)^2/2}}$: for $x = 1$, $\frac{1}{1+e^{-1.5}} = 0.818$; for $x = 2$, $\frac{1}{1+e^{1.5}} = 0.182$; for $x = 4$ and $x = 5$, 0.0006 and 0.00003. The M-step gives the new means $\mu_1 = \frac{0.818\cdot 1+0.182\cdot 2+\cdots}{0.818+0.182+\cdots} \approx 1.18$ and $\mu_2 \approx 3.61$, and the log-likelihood increases from -9.54 to -7.47. Further steps converge to $\mu_1 \approx 1.56$ and $\mu_2 \approx 4.44$ (not exactly 1.5 and 4.5, since the two Gaussians overlap), with the log-likelihood -6.84.

14.3 k-means clustering

The k-means algorithm is the "hard" version of EM. It looks for K centers $\boldsymbol{\mu}_1,\ldots,\boldsymbol{\mu}_K$ and an assignment $c(i) \in \{1,\ldots,K\}$ of the data points to the centers that minimize

$$J = \sum_{i=1}^{N} \|\mathbf{x}_i - \boldsymbol{\mu}_{c(i)}\|^2.$$

Lloyd's algorithm alternates two steps: assign each point to its nearest center, and move each center to the mean of its points.

Proposition 14.7. *Neither step of Lloyd's algorithm increases J. Hence the values of J after the update steps never increase, and they are eventually constant.*

Proof. For fixed centers, the assignment step minimizes each term of J separately. For a fixed assignment, the terms of cluster k are $\sum_{i:c(i)=k}\|\mathbf{x}_i - \boldsymbol{\mu}_k\|^2 = \sum_{i:c(i)=k}\|\mathbf{x}_i - \bar{\mathbf{x}}_k\|^2 + n_k\|\bar{\mathbf{x}}_k - \boldsymbol{\mu}_k\|^2$, where $\bar{\mathbf{x}}_k$ is the mean of the n_k points of the cluster (expand the squares; the mixed terms add up to 0). This is smallest for $\boldsymbol{\mu}_k = \bar{\mathbf{x}}_k$. After an update step, J only depends on the assignment, which can take finitely many values, so a nonincreasing sequence of these values is eventually constant. □

k-means is EM for a Gaussian mixture with equal weights and covariance matrices $\sigma^2 I$, in the limit $\sigma \to 0$: the responsibilities are proportional to $e^{-\|\mathbf{x}_i-\boldsymbol{\mu}_k\|^2/(2\sigma^2)}$, and as $\sigma \to 0$, the ratio of the responsibilities of two components k and j is $e^{(\|\mathbf{x}_i-\boldsymbol{\mu}_j\|^2-\|\mathbf{x}_i-\boldsymbol{\mu}_k\|^2)/(2\sigma^2)}$, which tends to ∞ if $\boldsymbol{\mu}_k$ is closer to $\mathbf{x}_i$. So the responsibility of the nearest center tends to 1, and the M-step becomes the update of the centers. Like EM, k-means only finds a local minimum and depends on the initial centers; the method k-means++ chooses them at random, far apart from each other. Since it measures distances with the Euclidean norm, k-means finds round clusters of similar size; for other shapes, Gaussian mixtures with general covariance matrices, or spectral clustering (Lecture 2), work better.

14.4 Probabilistic PCA

Principal component analysis (Lecture 2) can also be derived from a latent variable model, with a continuous latent variable: each data point is a linear function of k hidden factors plus noise.

Definition 14.8 (Probabilistic PCA)**.** Let $k < d$. In probabilistic PCA, the latent vector is $\mathbf{z} \sim \mathcal{N}(\mathbf{0}, I_k)$, and the observation is $\mathbf{x} = W\mathbf{z} + \boldsymbol{\mu} + \boldsymbol{\varepsilon}$ with $\boldsymbol{\varepsilon} \sim \mathcal{N}(\mathbf{0}, \sigma^2 I_d)$ independent of $\mathbf{z}$. The parameters are $W \in \mathbb{R}^{d\times k}$, $\boldsymbol{\mu} \in \mathbb{R}^d$ and $\sigma^2 > 0$.

The vector $(\mathbf{z}, \boldsymbol{\varepsilon})$ is $\mathcal{N}(\mathbf{0}, \operatorname{diag}(I_k, \sigma^2 I_d))$, and $(\mathbf{z}, \mathbf{x})$ is an affine function of it, so it is Gaussian (Theorem 6.2), with

$$\mathbf{x} \sim \mathcal{N}(\boldsymbol{\mu}, C), \qquad C = WW^\top + \sigma^2 I_d, \qquad \operatorname{Cov}(\mathbf{z}, \mathbf{x}) = \mathbb{E}[\mathbf{z}(W\mathbf{z} + \boldsymbol{\varepsilon})^\top] = W^\top.$$

So the model describes data whose covariance matrix is a rank-k matrix plus isotropic noise. By Theorem 6.5, the posterior of $\mathbf{z}$ is Gaussian with the mean $W^\top C^{-1}(\mathbf{x} - \boldsymbol{\mu})$ and the covariance $I_k - W^\top C^{-1} W$. With $M = W^\top W + \sigma^2 I_k$, the identity $W^\top(WW^\top + \sigma^2 I_d) = MW^\top$ gives $W^\top C^{-1} = M^{-1} W^\top$, so

$$\mathbb{E}[\mathbf{z} \mid \mathbf{x}] = M^{-1} W^\top (\mathbf{x} - \boldsymbol{\mu}), \qquad \operatorname{Cov}(\mathbf{z} \mid \mathbf{x}) = I_k - M^{-1} W^\top W = \sigma^2 M^{-1},$$

which only needs the inverse of a $k \times k$ matrix. As $\sigma^2 \to 0$, the posterior mean tends to $(W^\top W)^{-1} W^\top (\mathbf{x} - \boldsymbol{\mu})$, the coordinates of the orthogonal projection of $\mathbf{x} - \boldsymbol{\mu}$ onto the column space of W (Lecture 2).

Theorem 14.9 (Maximum likelihood for probabilistic PCA)**.** *Let $\mathbf{x}_1, \dots, \mathbf{x}_N \in \mathbb{R}^d$ have the mean $\bar{\mathbf{x}}$ and the sample covariance matrix $S = \frac{1}{N}\sum_i (\mathbf{x}_i - \bar{\mathbf{x}})(\mathbf{x}_i - \bar{\mathbf{x}})^\top$, with eigenvalues $\lambda_1 \geq \cdots \geq \lambda_d$ and orthonormal eigenvectors $\mathbf{u}_1, \dots, \mathbf{u}_d$, and assume $\lambda_{k+1} > 0$. Then the likelihood is maximized by*

$$\hat{\boldsymbol{\mu}} = \bar{\mathbf{x}}, \qquad \hat{\sigma}^2 = \frac{1}{d-k} \sum_{j=k+1}^{d} \lambda_j, \qquad \hat{W} = U_k (\Lambda_k - \hat{\sigma}^2 I_k)^{1/2},$$

where U_k has the columns $\mathbf{u}_1, \dots, \mathbf{u}_k$ and $\Lambda_k = \operatorname{diag}(\lambda_1, \dots, \lambda_k)$, and also by $\hat{W}R$ for every orthogonal $k \times k$ matrix R.

Proof. As in the proof of Theorem 9.10, $\boldsymbol{\mu} = \bar{\mathbf{x}}$ is optimal for every C, and then $\ell = \frac{N}{2}\big(-\log\det C - \operatorname{tr}(C^{-1}S)\big)$ plus a constant. By the spectral theorem, $WW^\top = \sum_{j=1}^{k} s_j \mathbf{v}_j \mathbf{v}_j^\top$ with $s_1 \geq \cdots \geq s_k \geq 0$ and orthonormal vectors $\mathbf{v}_1, \dots, \mathbf{v}_d$ (its rank is at most k). So C has the eigenvalues $c_j = \sigma^2 + s_j$ for $j \leq k$ and σ^2 for $j > k$, $\log\det C = \sum_{j\leq k} \log c_j + (d-k)\log\sigma^2$, and

$$C^{-1} = \frac{1}{\sigma^2} I_d - \sum_{j=1}^{k} a_j \mathbf{v}_j \mathbf{v}_j^\top, \qquad a_j = \frac{1}{\sigma^2} - \frac{1}{c_j}, \qquad a_1 \geq \cdots \geq a_k \geq 0.$$

With $q_j = \mathbf{v}_j^\top S \mathbf{v}_j$ and $a_{k+1} = 0$, we have $\sum_{j\leq k} a_j q_j = \sum_{j\leq k} (a_j - a_{j+1})(q_1 + \cdots + q_j) \leq \sum_{j\leq k} (a_j - a_{j+1})(\lambda_1 + \cdots + \lambda_j) = \sum_{j\leq k} a_j \lambda_j$, by Ky Fan's inequality (Lecture 2) and since $a_j - a_{j+1} \geq 0$. Hence

$$\operatorname{tr}(C^{-1}S) = \frac{\operatorname{tr} S}{\sigma^2} - \sum_{j\leq k} a_j q_j \geq \sum_{j\leq k} \frac{\lambda_j}{c_j} + \sum_{j>k} \frac{\lambda_j}{\sigma^2},$$

and with $\bar{\sigma}^2 = \frac{1}{d-k}\sum_{j>k} \lambda_j$,

$$-\log\det C - \operatorname{tr}(C^{-1}S) \leq -\sum_{j\leq k} \Big(\log c_j + \frac{\lambda_j}{c_j}\Big) - (d-k)\Big(\log\sigma^2 + \frac{\bar{\sigma}^2}{\sigma^2}\Big).$$

For $\lambda > 0$, the function $c \mapsto \log c + \frac{\lambda}{c}$ on $(0, \infty)$ has the derivative $\frac{c-\lambda}{c^2}$, so its minimum is $\log \lambda + 1$, at $c = \lambda$. So the right side is at most $-\sum_{j \le k}(\log \lambda_j + 1) - (d-k)(\log \bar{\sigma}^2 + 1)$, for all parameters. For $\sigma^2 = \bar{\sigma}^2$ and $W = \hat{W}$ (note $\lambda_j \ge \lambda_k \ge \bar{\sigma}^2$ for $j \le k$), we get $WW^\top = \sum_{j \le k}(\lambda_j - \bar{\sigma}^2)\mathbf{u}_j\mathbf{u}_j^\top$, so $\mathbf{v}_j = \mathbf{u}_j$ and $c_j = \lambda_j$, and all inequalities are equalities. Finally, $(\hat{W}R)(\hat{W}R)^\top = \hat{W}\hat{W}^\top$. □

So the maximum likelihood estimate spans the principal subspace $\operatorname{span}\{\mathbf{u}_1, \dots, \mathbf{u}_k\}$ of PCA, reproduces the variances $\lambda_1, \dots, \lambda_k$ in these directions exactly, and replaces the variances in the remaining $d - k$ directions by their average $\hat{\sigma}^2$, the noise level. For example, for $S = \operatorname{diag}(4, 2, 1)$ and $k = 1$, $\hat{\sigma}^2 = 1.5$ and $\hat{W} = (\sqrt{2.5}, 0, 0)^\top$, so the fitted covariance matrix is $\hat{C} = \operatorname{diag}(4, 1.5, 1.5)$. Unlike PCA, probabilistic PCA has a likelihood, which can be compared across different k or used in mixtures of such models, and it can be fitted with EM when some entries of the data are missing. Its decoder $p(\mathbf{x} \mid \mathbf{z}) = \mathcal{N}(W\mathbf{z} + \boldsymbol{\mu}, \sigma^2 I)$ is linear; a variational autoencoder replaces it by a neural network.

14.5 Variational inference and variational autoencoders

In more complex models, such as deep generative models, the posterior $p(z \mid \mathbf{x}, \theta)$ cannot be computed, and the E-step is impossible. Variational inference maximizes the ELBO over a family of tractable distributions q instead, for example Gaussians $q(z) = \mathcal{N}(\boldsymbol{\mu}, \operatorname{diag}(\boldsymbol{\sigma}^2))$; by Theorem 14.2, this finds the member of the family that is closest to the posterior in the sense of $\mathrm{KL}(q \,\|\, p(\cdot \mid \mathbf{x}, \theta))$. Writing $p(\mathbf{x}, z \mid \theta) = p(\mathbf{x} \mid z, \theta)p(z)$ gives the form

$$\mathrm{ELBO}(q, \theta) = \mathbb{E}_{z \sim q}\big[\log p(\mathbf{x} \mid z, \theta)\big] - \mathrm{KL}\left(q \,\|\, p(z)\right) :$$

a reconstruction term, which rewards latent codes that explain $\mathbf{x}$, and a KL term, which keeps q close to the prior $p(z)$. A variational autoencoder trains an encoder network that outputs the parameters of $q(z \mid \mathbf{x})$ for each data point and a decoder network that defines $p(\mathbf{x} \mid z, \theta)$, both by stochastic gradient ascent on the ELBO. For a Gaussian q and the prior $\mathcal{N}(\mathbf{0}, I)$, the KL term has the closed form of Proposition 13.5, and the gradient of the reconstruction term is estimated with the reparameterization trick, $z = \boldsymbol{\mu} + \boldsymbol{\sigma} \odot \boldsymbol{\varepsilon}$ with $\boldsymbol{\varepsilon} \sim \mathcal{N}(\mathbf{0}, I)$ (Section 6).

15 Stochastic Differential Equations

Markov chains (Section 8) move in discrete steps. Many random processes change continuously in time: the position of a particle in a fluid, the price of a stock, the parameters of a network during training with stochastic gradients, or the gradually noised images of a diffusion model. Stochastic differential equations (SDEs) describe such processes by adding random noise to ordinary differential equations (Section 15 of Lecture 3). The source of the noise is Brownian motion.

Definition 15.1 (Brownian motion)**.** A Brownian motion (or Wiener process) is a family of random variables $B(t)$, $t \ge 0$, with $B(0) = 0$, such that

1. for $0 \le t_0 < t_1 < \cdots < t_k$, the increments $B(t_1) - B(t_0), \dots, B(t_k) - B(t_{k-1})$ are independent;
2. $B(t) - B(s) \sim \mathcal{N}(0, t - s)$ for $s < t$;
3. the paths $t \mapsto B(t)$ are continuous.

That such a process exists is a theorem of Norbert Wiener, which we do not prove. On the grid $t_k = kh$, Brownian motion is simulated exactly by the random walk $B(t_{k+1}) = B(t_k) + \sqrt{h}\, Z_k$ with i.i.d. $Z_k \sim \mathcal{N}(0, 1)$ (Figure 7). Conversely, by the central limit theorem, every random walk with many small independent steps looks like a Brownian motion from far away.

Proposition 15.2. $\mathbb{E}[B(t)] = 0$, $\mathrm{Var}(B(t)) = t$, *and* $\mathrm{Cov}(B(s), B(t)) = \min(s, t)$.

Proof. The first two statements are property (2) with $s = 0$. For $s \le t$, $\mathrm{Cov}(B(s), B(t)) = \mathrm{Cov}\left(B(s), B(s)+(B(t)-B(s))\right) = \mathrm{Var}(B(s))+0 = s$, since the increment $B(t)-B(s)$ is independent of $B(s) = B(s) - B(0)$. □

So Brownian motion is a Gaussian process with mean 0 and covariance function $\min(s, t)$ (Section 12.4). Its typical displacement in a time h is $\sqrt{h}$, which is much larger than h for small h: the difference quotient $\frac{B(t+h)-B(t)}{h}$ has the variance $\frac{1}{h} \to \infty$. In fact, the paths of Brownian motion are continuous but nowhere differentiable, and ordinary calculus does not apply to them.

Definition 15.3 (Stochastic differential equation)**.** An SDE $dX = \mu(X)\,dt + \sigma(X)\,dB$, with drift μ and diffusion coefficient σ, describes a process with $X(t) = X(0) + \int_0^t \mu(X(s))\,ds + \int_0^t \sigma(X(s))\,dB(s)$, where the last integral is a stochastic (Itô) integral. Its Euler–Maruyama approximation with step h is

$$X_{k+1} = X_k + \mu(X_k)\,h + \sigma(X_k)\sqrt{h}\,Z_k, \qquad Z_k \sim \mathcal{N}(0,1) \text{ i.i.d.}$$

We do not define the stochastic integral; for our purposes, the Euler–Maruyama scheme, which converges to the solution as $h \to 0$, can serve as the definition. It is Euler's method for the ODE $x' = \mu(x)$ (Section 15.2 of Lecture 3), plus Gaussian noise whose standard deviation is proportional to $\sqrt{h}$, and it is a Markov chain with continuous states. For $\sigma = 0$, the SDE is an ODE.

Example 15.4 (The Ornstein–Uhlenbeck process)**.** The SDE $dX = -\theta X\,dt + \sigma\,dB$ with $\theta > 0$ pulls X towards 0 while the noise pushes it around. Its Euler–Maruyama approximation is $X_{k+1} = (1 - \theta h)X_k + \sigma\sqrt{h}\,Z_k$. Its mean satisfies $\mathbb{E}[X_{k+1}] = (1 - \theta h)\mathbb{E}[X_k]$, so it follows Euler's method for the ODE $x' = -\theta x$, with the solution $e^{-\theta t}X(0)$. If $X_k \sim \mathcal{N}(0, v)$ is independent of Z_k, then $X_{k+1} \sim \mathcal{N}(0, (1 - \theta h)^2 v + \sigma^2 h)$ by Corollary 6.7. So for $0 < \theta h < 2$, the chain has the stationary distribution $\mathcal{N}(0, v_h)$ with

$$v_h = \frac{\sigma^2 h}{1 - (1 - \theta h)^2} = \frac{\sigma^2}{\theta(2 - \theta h)} \to \frac{\sigma^2}{2\theta} \qquad \text{as } h \to 0.$$

For $\theta = 1$, $\sigma = \sqrt{2}$, the stationary distribution of the SDE is $\mathcal{N}(0, 1)$ (Figure 7); for $\theta = \sigma = 1$ and $h = 0.01$, $v_h \approx 0.5025$ instead of 0.5.

15.1 Langevin dynamics and diffusion models

The Ornstein–Uhlenbeck process is an example of a general principle: a gradient flow with the right amount of noise samples from a probability distribution.

Theorem 15.5 (Langevin dynamics)**.** *Let $U : \mathbb{R}^d \to \mathbb{R}$ be smooth with $\int e^{-U(\mathbf{x})}\,d\mathbf{x} < \infty$. Then the SDE $d\mathbf{X} = -\nabla U(\mathbf{X})\,dt + \sqrt{2}\,d\mathbf{B}$, where $\mathbf{B}$ is a vector of d independent Brownian motions, has the stationary density $p(\mathbf{x}) \propto e^{-U(\mathbf{x})}$, and under mild conditions, the distribution of $\mathbf{X}(t)$ converges to it.*

We do not prove the convergence, but we can see why e^{-U} is stationary. Without noise, the density p_t of $\mathbf{X}(t)$ under the flow of the vector field $-\nabla U$ satisfies the continuity equation $\frac{\partial p_t}{\partial t} = -\operatorname{div}(-p_t \nabla U)$ (Proposition 16.3 of Lecture 3). The noise adds a diffusion term, and one can show that p_t satisfies the Fokker–Planck equation

$$\frac{\partial p_t}{\partial t} = \operatorname{div}(p_t \nabla U) + \Delta p_t = \operatorname{div}(p_t \nabla U + \nabla p_t),$$

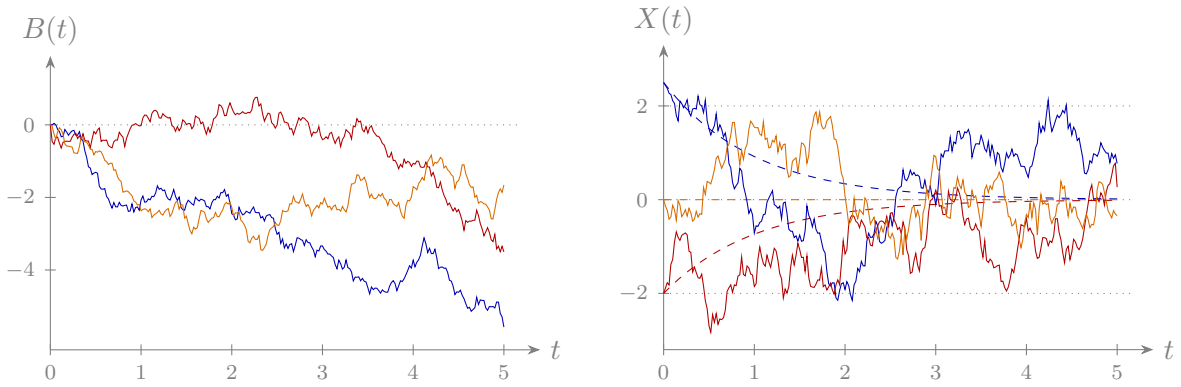


Figure 7: Left: three paths of Brownian motion, simulated with $h = 0.02$. Right: three paths of the Ornstein–Uhlenbeck process $dX = -X\,dt + \sqrt{2}\,dB$, started at 2.5, -2 and 0, together with the solutions $X(0)e^{-t}$ of the ODE without noise (dashed). The process forgets its starting point and fluctuates around 0 with the stationary distribution $\mathcal{N}(0,1)$; about 95% of the time, it stays between -2 and 2.

with the Laplacian $\Delta p_t = \operatorname{div} \nabla p_t$ (Lecture 3). For $p \propto e^{-U}$, we have $\nabla p = -p \nabla U$, so the right side vanishes, and p does not change in time. For $U = 0$, the Fokker–Planck equation is the heat equation $\frac{\partial p_t}{\partial t} = \Delta p_t$, which describes how the density of $\sqrt{2}\,\mathbf{B}(t)$ spreads out. For $U(x) = \frac{x^2}{2s^2}$, the SDE is the Ornstein–Uhlenbeck process with $\theta = \frac{1}{s^2}$ and $\sigma = \sqrt{2}$, whose stationary variance $\frac{\sigma^2}{2\theta} = s^2$ agrees with the density $e^{-U} \propto e^{-x^2/(2s^2)}$ of $\mathcal{N}(0, s^2)$. The Euler–Maruyama scheme gives the unadjusted Langevin algorithm

$$\mathbf{x}_{k+1} = \mathbf{x}_k - h \nabla U(\mathbf{x}_k) + \sqrt{2h}\,\mathbf{z}_k, \qquad \mathbf{z}_k \sim \mathcal{N}(\mathbf{0}, I),$$

which is gradient descent on U plus Gaussian noise. To sample from a posterior $p(\boldsymbol{\theta} \mid \mathcal{D}) \propto e^{-U(\boldsymbol{\theta})}$, we take $U(\boldsymbol{\theta}) = -\log p(\mathcal{D} \mid \boldsymbol{\theta}) - \log p(\boldsymbol{\theta})$, the training loss plus the regularizer. Without the noise, the iteration converges to the MAP estimate; with the noise, it explores the posterior. The discretization introduces a small bias: for the target $\mathcal{N}(2, 0.25)$ and $h = 0.01$, the stationary variance of the algorithm is 0.2551 instead of 0.25, by Example 15.4. The Metropolis-adjusted Langevin algorithm removes this bias by accepting or rejecting each step as in Theorem 8.13, and stochastic gradient Langevin dynamics uses mini-batch gradients to make the method feasible for large models.

Remark 15.6 (SDEs in machine learning).

1. Diffusion models for image generation use an SDE of Ornstein–Uhlenbeck type that gradually turns images into Gaussian noise. The time-reversed process is again an SDE, whose drift involves the score $\nabla \log p_t(\mathbf{x})$ of the noised data distribution at time t. A neural network learns this score, with a variant of the score matching loss of Lecture 3, and new images are generated by simulating the reverse SDE, starting from pure noise.

2. The noise of mini-batch gradients makes SGD with a small step size behave approximately like the Euler–Maruyama scheme for an SDE $d\boldsymbol{\theta} = -\nabla L(\boldsymbol{\theta})\,dt + \sqrt{\eta\,\Sigma(\boldsymbol{\theta})}\,d\mathbf{B}$, where Σ is the covariance matrix of the mini-batch gradients. This heuristic is used to study which minima SGD prefers.

16 Key Takeaways

- A probability assigns numbers in $[0,1]$ to events, additively for disjoint events. Conditional probability $P(A \mid B) = \frac{P(A \cap B)}{P(B)}$ and Bayes' theorem $P(B \mid A) = \frac{P(A|B)P(B)}{P(A)}$ update beliefs with evidence; ignoring the prior $P(B)$ is the base rate fallacy.

- Random variables are described by a pmf or a density. The sum rule $p(x) = \sum_y p(x,y)$ and the product rule $p(x,y) = p(y \mid x)p(x)$ are the basic rules of probabilistic modeling; independence means that the joint distribution factorizes, and the density of a sum of independent random variables is the convolution of their densities.

- The expectation is linear; $\mathrm{Var}(X) = \mathbb{E}[X^2] - \mathbb{E}[X]^2$; Markov's, Chebyshev's, Chernoff's and Jensen's inequalities relate them. For $X \geq 0$, $\mathbb{E}[X] = \sum_{k \geq 0} P(X > k)$ for integer values and $\mathbb{E}[X] = \int_0^\infty P(X > t)\, dt$ for densities. For densities, $p_Y(y) = \frac{p_X(x)}{|g'(x)|}$ under $y = g(x)$, and $F^{-1}(U)$, with the quantile function $F^{-1}(u) = \min\{x : F(x) \geq u\}$, samples from the CDF F.

- The moment generating function $M_X(t) = \mathbb{E}[e^{tX}]$ has the derivatives $M_X^{(k)}(0) = \mathbb{E}[X^k]$, and $M_{X+Y} = M_X M_Y$ for independent X and Y; for $\mathcal{N}(\mu, \sigma^2)$, it is $e^{\mu t + \sigma^2 t^2/2}$. Optimizing the Chernoff bound $e^{-ta} M_X(t)$ over t gives exponential tail bounds.

- Covariance is an inner product on centered random variables, and correlation is the cosine of their angle, so $|\,\mathrm{Corr}\,| \leq 1$. Uncorrelated does not imply independent. $\mathbb{E}[Y \mid X]$ is the best mean-squared-error predictor of Y, the orthogonal projection of Y onto functions of X, and $\mathrm{Var}(Y) = \mathbb{E}[\mathrm{Var}(Y \mid X)] + \mathrm{Var}(\mathbb{E}[Y \mid X])$ (law of total variance).

- A unit $z = \sum_{j=1}^n W_j x_j$ with independent weights of mean 0 and variance v has $\mathrm{Var}(z) = nv\mathbb{E}[x_j^2]$, and a ReLU halves the second moment of a symmetric z. So signals keep their size through deep ReLU networks with $v = \frac{2}{n_{\text{in}}}$ (He initialization), and vanish or explode geometrically otherwise; Xavier initialization $v = \frac{2}{n_{\text{in}} + n_{\text{out}}}$ balances the forward and the backward pass.

- $X \perp\!\!\!\perp Y \mid Z$ means $p(x, y \mid z) = p(x \mid z)p(y \mid z)$; it neither implies nor follows from independence (common causes, explaining away). Directed graphical models factor $p(\mathbf{x}) = \prod_i p(x_i \mid x_{\mathrm{pa}(i)})$; in chains and forks, the middle variable makes the ends conditionally independent, in a collider, it makes independent causes dependent. Naive Bayes assumes conditionally independent features, $p(y \mid \mathbf{x}) \propto p(y) \prod_j p(x_j \mid y)$, and has linear log-odds like logistic regression.

- In a Poisson process with rate λ, the waiting times are i.i.d. Exponential(λ), the n-th arrival time has the density $\frac{\lambda^n t^{n-1}}{(n-1)!} e^{-\lambda t}$, and the number of arrivals in the time t is Poisson(λt).

- The covariance matrix $\Sigma = \mathbb{E}[(\mathbf{X} - \boldsymbol{\mu})(\mathbf{X} - \boldsymbol{\mu})^\top]$ is symmetric positive semidefinite, $\mathrm{Cov}(A\mathbf{X} + \mathbf{b}) = A\Sigma A^\top$, and $\mathrm{Var}(\mathbf{u}^\top \mathbf{X}) = \mathbf{u}^\top \Sigma \mathbf{u}$ is maximal for the top eigenvector (PCA). Whitening $\Sigma^{-1/2}(\mathbf{X} - \boldsymbol{\mu})$ gives identity covariance.

- $\mathcal{N}(\boldsymbol{\mu}, \Sigma)$ is the distribution of $\boldsymbol{\mu} + A\mathbf{Z}$ with $AA^\top = \Sigma$ and $\mathbf{Z} \sim \mathcal{N}(\mathbf{0}, I)$; its level sets are ellipsoids along the eigenvectors of Σ. Affine maps, marginals and conditionals of Gaussians are Gaussian, with $\mathbf{X}_a \mid \mathbf{x}_b \sim \mathcal{N}(\boldsymbol{\mu}_a + \Sigma_{ab}\Sigma_{bb}^{-1}(\mathbf{x}_b - \boldsymbol{\mu}_b), \Sigma_{aa} - \Sigma_{ab}\Sigma_{bb}^{-1}\Sigma_{ba})$; for jointly Gaussian vectors, uncorrelated means independent.

- The sample mean of N i.i.d. variables has standard error $\frac{\sigma}{\sqrt{N}}$; it converges to μ in probability (law of large numbers), its fluctuations are approximately Gaussian (central limit theorem), and

for bounded variables Hoeffding's inequality bounds large deviations exponentially. Slutsky's theorem allows us to replace σ by s in confidence intervals, and the delta method gives the approximate standard error $|g'(\theta)|\,\mathrm{SE}$ of a smooth function $g(\hat{\theta})$ of an estimator.

- Averages of samples underlie Monte Carlo methods, test set evaluation, and stochastic gradients. Rejection sampling draws exact samples using an envelope $p \le Mq$ (acceptance rate $\frac{1}{M}$); importance sampling reweights samples from q by the likelihood ratio $\frac{p}{q}$, which can reduce the variance dramatically for rare events.

- SGD with unbiased stochastic gradients converges for strongly convex functions with step sizes $\eta_k = \frac{1}{\mu(k+1)}$, at the rate $\mathbb{E}\|\mathbf{x}_k - \mathbf{x}^*\|^2 \le \frac{G^2}{\mu^2 k}$; with a constant step size, it converges to a noise ball whose size is proportional to the step size.

- A Markov chain with transition matrix P evolves by $\boldsymbol{\pi}_{t+1} = \boldsymbol{\pi}_t P$. Stationary distributions are left eigenvectors of P with eigenvalue 1; an irreducible, aperiodic chain converges to its unique stationary distribution, at a speed governed by the second largest eigenvalue in absolute value. Detailed balance $\pi_i P_{ij} = \pi_j P_{ji}$ implies stationarity; the Metropolis algorithm uses it to sample from distributions that are only known up to a constant (MCMC).

- Gibbs sampling updates one coordinate at a time from its conditional distribution; each update satisfies detailed balance, so the target distribution is stationary. It is slow when the coordinates are strongly correlated.

- The value $\mathbf{V}$ of a Markov reward process, the expected discounted sum of rewards, solves the Bellman equation $\mathbf{V} = \mathbf{r} + \gamma P\mathbf{V}$, so $\mathbf{V} = (I - \gamma P)^{-1}\mathbf{r} = \sum_k \gamma^k P^k \mathbf{r}$. Since $\|P\mathbf{x}\|_\infty \le \|\mathbf{x}\|_\infty$, the Bellman maps are γ-contractions in the maximum norm: policy evaluation and value iteration converge like γ^k, the Bellman optimality equation has exactly one solution $\mathbf{V}^*$, and the greedy policy with respect to $\mathbf{V}^*$ is optimal. Q-learning is a stochastic version of value iteration.

- A time series is (weakly) stationary if its mean and its autocovariances $c(h) = \mathrm{Cov}(X_t, X_{t+h})$ do not depend on t. The AR(1) process $X_t = \phi X_{t-1} + \varepsilon_t$ is stationary for $|\phi| < 1$, with the variance $\frac{\sigma^2}{1-\phi^2}$ and the autocorrelations ϕ^h; for $|\phi| \ge 1$, its variance grows without bound, like that of a random walk, which differencing makes stationary. An AR(p) process is stationary if the eigenvalues of its companion matrix lie inside the unit circle.

- Estimators have a bias and a variance, and MSE = bias^2 + variance, so some bias (shrinkage) can pay off; if both tend to 0, the estimator is consistent. The sample variance needs the factor $\frac{1}{N-1}$ to be unbiased. No unbiased estimator has a variance below the Cramér–Rao bound $\frac{1}{N\,I(\theta)}$, where the Fisher information $I(\theta)$ is the variance of the score $\frac{\partial}{\partial\theta}\log p(x \mid \theta)$; the MLE attains it asymptotically, and its standard error is estimated from the curvature of the log-likelihood. A sufficient statistic carries all the information about θ, and conditioning an estimator on it never increases the MSE (Rao–Blackwell).

- Maximum likelihood maximizes $\sum_i \log p(x_i \mid \theta)$. Least squares is maximum likelihood with Gaussian noise, and the cross-entropy loss is maximum likelihood for categorical labels; the MLE of a Gaussian is the sample mean and the (biased) sample covariance. The method of moments matches sample moments instead; it is simpler but usually less efficient. Confidence intervals $\bar{x} \pm 1.96\frac{s}{\sqrt{N}}$ quantify uncertainty.

- Exponential families $h(x)e^{\boldsymbol{\eta}^\top T(x) - A(\boldsymbol{\eta})}$ include the Bernoulli, Poisson, Gaussian and categorical distributions. The log-partition function A is convex, with $\nabla A = \mathbb{E}[T]$, so the log-likelihood is

concave and the MLE matches $\mathbb{E}[T]$ with the average of the $T(x_i)$. Generalized linear models have the gradient "prediction minus target" times the input.

- The empirical CDF converges to F uniformly in t, with exponentially small failure probabilities, which gives confidence bands and the Kolmogorov–Smirnov test. Kernel density estimates $\frac{1}{Nh}\sum_i K(\frac{x-X_i}{h})$ have a bias of order h^2 and a variance of order $\frac{1}{Nh}$; the best bandwidth is of order $N^{-1/5}$, and the rate $N^{-4/(4+d)}$ in d dimensions shows the curse of dimensionality.

- A waiting time T is described by its survival function $S(t) = P(T > t) = e^{-H(t)}$ and its hazard rate $h = \frac{p}{S}$, the rate of events among the survivors; a constant hazard means an exponential distribution. Censored observations, of which we only know $T > t_i$, contribute $S(t_i)$ to the likelihood; for exponential times, $\hat{\lambda}$ is the number of events divided by the total time at risk. Ignoring censoring biases the estimates. The Kaplan–Meier estimator $\prod_{\tau_j \le t}(1 - \frac{d_j}{n_j})$ generalizes the empirical survival function to censored data.

- A test rejects H_0 if the p-value, the probability under H_0 of data at least as extreme as observed, is at most α; under H_0, p-values are uniform, so the false positive rate is α, and the power is the probability of detecting a real effect. A p-value is not the probability that H_0 is true. Paired t-tests compare two models on the same data sets, and Pearson's χ^2 test compares observed counts with expected counts. Many tests need corrections: Bonferroni ($\frac{\alpha}{m}$) controls the family-wise error rate, Benjamini–Hochberg the false discovery rate. The bootstrap (resampling the data with replacement) estimates standard errors of any statistic, and permutation tests (relabeling the groups) give exact p-values without distributional assumptions.

- Neyman–Pearson: for two simple hypotheses, the likelihood ratio test is the most powerful test at its level; for a Gaussian mean, it is the one-sided z-test.

- In linear regression with uncorrelated noise of constant variance σ^2, least squares is unbiased with $\mathrm{Cov}(\hat{\mathbf{w}}) = \sigma^2(X^\top X)^{-1}$, and it is the best linear unbiased estimator (Gauss–Markov); $s^2 = \frac{\mathrm{RSS}}{N-n}$ estimates σ^2 without bias. For Gaussian noise, $\frac{\hat{w}_j - w_j}{\mathrm{SE}_j}$ has the t_{N-n}-distribution, which gives exact confidence intervals and t-tests for the coefficients. With heteroscedastic noise, $\hat{\mathbf{w}}$ is still unbiased, but its covariance is the sandwich $(X^\top X)^{-1}X^\top \Sigma X (X^\top X)^{-1}$, and the usual standard errors are wrong; robust standard errors or weighted least squares fix this.

- Cross-validation estimates the generalization error, but its naive standard error is too small, and the best of many cross-validated models is optimistically biased; compare models with paired differences and evaluate the final choice on untouched data. A/B tests randomize to measure causal effects, since a confounder can reverse a comparison in every group (Simpson's paradox); plan the sample size $n \approx (z_{1-\alpha/2} + z_{1-\beta})^2\, 2\bar{p}(1-\bar{p})/\delta^2$ per group in advance, and do not stop at the first significant peek.

- Precision, recall and the false positive rate describe the errors of a classifier at a threshold; the ROC curve shows all thresholds, and the AUC is the fraction of correctly ordered positive–negative pairs. The cross-entropy loss is a proper scoring rule. For $|\mathcal{H}|$ models and N i.i.d. examples, all empirical risks are within $\sqrt{\log(2|\mathcal{H}|/\delta)/(2N)}$ of the risks with probability $1 - \delta$ (Hoeffding and the union bound).

- Bayesian inference updates a prior to the posterior $p(\theta \mid \mathcal{D}) \propto p(\mathcal{D} \mid \theta)p(\theta)$. Conjugate priors (Beta for Bernoulli, Gamma for Poisson, Gaussian for linear regression) give closed-form posteriors, and credible intervals are quantiles of the posterior. MAP estimation adds the regularizer

$-\log p(\theta)$: a Gaussian prior gives ridge regression, a Laplace prior the lasso. The Bayesian posterior also gives predictive uncertainty. A Gaussian process is a prior on functions given by a kernel; its posterior follows from Gaussian conditioning, with the mean $\mathbf{k}_*^\top (K + \sigma^2 I)^{-1}\mathbf{y}$ and a variance that is small near the data.

- Entropy measures uncertainty, cross-entropy $H(p, q) = H(p) + \mathrm{KL}(p \,\|\, q)$, and $\mathrm{KL}(p \,\|\, q) \geq 0$ with equality only for $p = q$ (Gibbs). Maximum likelihood minimizes the cross-entropy and the KL divergence from the empirical distribution to the model. Among all densities with a given variance, the Gaussian has the largest entropy.

- In latent variable models, $\log p(\mathbf{x} \mid \theta) = \mathrm{ELBO}(q, \theta) + \mathrm{KL}(q \,\|\, p(z \mid \mathbf{x}, \theta))$. EM alternates between setting q to the posterior (E-step) and maximizing the ELBO over θ (M-step), and never decreases the likelihood; for Gaussian mixtures, it alternates responsibilities and weighted means and covariances. k-means is its hard-assignment limit. Probabilistic PCA, $\mathbf{x} = W\mathbf{z} + \boldsymbol{\mu} + \boldsymbol{\varepsilon}$, has the maximum likelihood solution $\hat{W} = U_k(\Lambda_k - \hat{\sigma}^2 I)^{1/2}$, which spans the principal subspace, with $\hat{\sigma}^2$ the average of the remaining eigenvalues. Variational autoencoders maximize the ELBO over an encoder and a decoder.

- Brownian motion has independent Gaussian increments $B(t) - B(s) \sim \mathcal{N}(0, t - s)$ and continuous, nowhere differentiable paths. An SDE $dX = \mu(X)\,dt + \sigma(X)\,dB$ is simulated by Euler–Maruyama, $X_{k+1} = X_k + \mu(X_k)h + \sigma(X_k)\sqrt{h}\,Z_k$: noise of size $\sqrt{h}$, not h. Langevin dynamics $d\mathbf{X} = -\nabla U\,dt + \sqrt{2}\,d\mathbf{B}$ samples from $p \propto e^{-U}$, the stationary solution of the Fokker–Planck equation; its discretization is gradient descent plus noise, and diffusion models generate data by reversing a noising SDE.

17 Exercises

1. (The birthday problem.) Assume that birthdays are independent and uniformly distributed over 365 days. Show that the probability that among n people at least two share a birthday is $1 - \frac{365 \cdot 364 \cdots (365 - n + 1)}{365^n}$, and that it exceeds $\frac{1}{2}$ for $n = 23$. How many people are needed for 99%?

2. (a) Prove the inclusion–exclusion formula $P(A \cup B \cup C) = P(A) + P(B) + P(C) - P(A \cap B) - P(A \cap C) - P(B \cap C) + P(A \cap B \cap C)$.

   (b) Two fair coins are flipped. Let A = "the first coin shows heads", B = "the second coin shows heads", and C = "exactly one coin shows heads". Show that any two of these events are independent, but the three events are not independent.

3. A spam filter knows that 40% of all emails are spam, that the word "free" appears in 30% of the spam emails and in 2% of the other emails. What is the probability that an email containing "free" is spam?

4. Let X be Geometric(p). Show that $\mathbb{E}[X] = \frac{1}{p}$ and that $P(X > s + t \mid X > s) = P(X > t)$ for all positive integers s and t. How many times do you expect to roll a fair die until you see a six?

5. Let U and V be independent and Uniform$(0, 1)$. Find the densities of (a) U^2, (b) $-\log U$, (c) $\max(U, V)$ (Hint: compute the CDF first.)

6. Let X be Poisson(λ). Compute $\mathbb{E}[X(X - 1)]$, and use it to show that $\mathrm{Var}(X) = \lambda$.

7. (a) A nonnegative random variable has mean 2. Bound $P(X \geq 10)$ with Markov's inequality, and compare with the exact value for $X \sim$ Exponential($\frac{1}{2}$).

(b) Use Jensen's inequality to show that $\mathbb{E}[\frac{1}{X}] \geq \frac{1}{\mathbb{E}[X]}$ for every positive random variable X, and that $\mathbb{E}[e^X] \geq e^{\mathbb{E}[X]}$.

8. Let (X, Y) have the joint pmf $p(0,0) = 0.1$, $p(0,1) = 0.2$, $p(0,2) = 0.1$, $p(1,0) = 0.2$, $p(1,1) = 0.1$, $p(1,2) = 0.3$. Compute the marginal distributions, $\mathbb{E}[Y \mid X = 0]$ and $\mathbb{E}[Y \mid X = 1]$, and check the tower property. Compute $\mathrm{Cov}(X, Y)$. Are X and Y independent?

9. Let $X \sim \mathcal{N}(0,1)$. Show that X and X^2 are uncorrelated, but not independent. Show that $\mathrm{Corr}(X, aX + b) = \mathrm{sign}(a)$ for $a \neq 0$.

10. (Initialization.)

    (a) Let $z \sim \mathcal{N}(0, s^2)$. Show that $\mathbb{E}[\mathrm{ReLU}(z)] = \frac{s}{\sqrt{2\pi}}$, $\mathbb{E}[\mathrm{ReLU}(z)^2] = \frac{s^2}{2}$ and $\mathrm{Var}(\mathrm{ReLU}(z)) = s^2(\frac{1}{2} - \frac{1}{2\pi})$.

    (b) (The backward pass.) Let W be an $m \times n$ matrix with independent entries of mean 0 and variance v, and let $\boldsymbol{\delta} \in \mathbb{R}^m$ be a random vector independent of W, with $\mathbb{E}[\delta_i^2] = q$ for all i. Show that every entry of $W^\top \boldsymbol{\delta}$, the gradient that backpropagation passes to the inputs of a linear layer $\mathbf{y} = W\mathbf{x}$, has mean 0 and variance mvq. Which condition on v keeps the size of the gradients constant?

    (c) Find the standard deviations of the He and Xavier normal initializations for a layer with 256 inputs and 128 outputs, and the bound b of the uniform He initialization on $[-b, b]$.

11. Let X and Y be independent and Exponential(λ). Show that $X + Y$ has the density $\lambda^2 z e^{-\lambda z}$ for $z > 0$.

12. (The Poisson process.)

    (a) Show that the minimum of independent $X \sim$ Exponential(λ) and $Y \sim$ Exponential(μ) is Exponential($\lambda + \mu$), and that $P(X < Y) = \frac{\lambda}{\lambda+\mu}$.

    (b) Requests arrive at a server as a Poisson process with the rate 2 per second. Find the probability of no request in one second, the probability of at least 3 requests in two seconds, and the expected waiting time until the 10th request.

13. (Conditional independence.)

    (a) Show that $X \perp\!\!\!\perp Y \mid Z$ if and only if $p(x, y, z) = g(x, z)h(y, z)$ for some functions g and h.

    (b) In a chain $X \to Z \to Y$ of binary variables, let $P(Z = 1 \mid X = 1) = 0.9$, $P(Z = 1 \mid X = 0) = 0.2$, $P(Y = 1 \mid Z = 1) = 0.7$ and $P(Y = 1 \mid Z = 0) = 0.1$. Compute $P(Y = 1 \mid X = 1)$, $P(Y = 1 \mid X = 0)$ and $P(Y = 1 \mid X = 1, Z = 1)$. Are X and Y independent? Are they independent given Z?

    (c) A naive Bayes classifier has seen the word "meeting" in 0 of 50 spam emails. What is the maximum likelihood estimate of $P(\text{meeting} \mid \text{spam})$, and what happens to an email that contains "meeting" and many typical spam words? What does Laplace smoothing, adding 1 to each of the counts of "contains the word" and "does not contain the word", estimate instead?

14. (Moment generating functions.) Let $X \sim$ Exponential(λ). Use the cumulant generating function $K_X(t) = \log \frac{\lambda}{\lambda - t}$ to find $\mathbb{E}[X]$ and $\mathrm{Var}(X)$. Find the moment generating function of the sum of n independent Exponential(λ) variables, and its mean and variance. Show that $\mathbb{E}[X^k] = \frac{k!}{\lambda^k}$ by expanding $M_X(t) = \frac{1}{1 - t/\lambda}$ as a geometric series.

15. Let $\mathbf{X}$ have the covariance matrix $\Sigma = \begin{pmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{pmatrix}$. Compute $\mathrm{Var}(X_1 + X_2 + X_3)$ and $\mathrm{Var}(X_1 - X_3)$. In which direction $\mathbf{u}$ (with $\|\mathbf{u}\| = 1$) is $\mathrm{Var}(\mathbf{u}^\top \mathbf{X})$ largest, and what is it? Find a matrix A such that $A\mathbf{X}$ has covariance matrix I.

16. Let $(X, Y) \sim \mathcal{N}\Big(\mathbf{0}, \begin{pmatrix} 4 & 2 \\ 2 & 2 \end{pmatrix}\Big)$. Find the conditional distributions of Y given $X = x$ and of X given $Y = y$, the distribution of $X - Y$, and $P(X > Y)$. Describe the level curves of the density.

17. Let $\mathbf{X} \sim \mathcal{N}(\boldsymbol{\mu}, \Sigma)$ in $\mathbb{R}^d$. Show that the squared Mahalanobis distance $(\mathbf{X} - \boldsymbol{\mu})^\top \Sigma^{-1} (\mathbf{X} - \boldsymbol{\mu})$ has the same distribution as $\|\mathbf{Z}\|^2$ with $\mathbf{Z} \sim \mathcal{N}(\mathbf{0}, I)$, and that its expectation is d. What does this say about where the probability mass of a high-dimensional Gaussian lies?

18. How many times do we have to flip a fair coin so that the fraction of heads is within 0.01 of $\frac{1}{2}$ with probability at least 95%? Answer with Chebyshev's inequality, with Hoeffding's inequality, and with the central limit theorem, and compare.

19. (Delta method.) Let $X_1, \dots, X_N$ be i.i.d. Exponential(λ). Show that $\sqrt{N}(\frac{1}{\bar{X}_N} - \lambda) \to \mathcal{N}(0, \lambda^2)$ in distribution, and compare with the Fisher information of the exponential distribution (Exercise 29).

20. Let $p(x) = \sqrt{2/\pi}\, e^{-x^2/2}$ for $x > 0$ be the density of $|Z|$ for $Z \sim \mathcal{N}(0,1)$, and let $q(x) = e^{-x}$ be the Exponential(1) density.

    (a) Find the smallest M with $p \le Mq$, and the acceptance rate of rejection sampling with this proposal.

    (b) Write down the importance sampling estimator of $\mathbb{E}_p[X^2]$ with samples from q, and explain why its variance is finite.

21. (SGD with a constant step size.) For $f(\theta) = \frac{1}{2N}\sum_i (\theta - y_i)^2$, SGD with mini-batches of size 1 and a constant step size $\eta \in (0,1)$ is $\theta_{k+1} = \theta_k - \eta(\theta_k - y_{I_k})$. Show that $e_k = \theta_k - \bar{y}$ satisfies $e_{k+1} = (1-\eta)e_k + \eta(y_{I_k} - \bar{y})$, and that $\mathbb{E}[e_k^2] \to \frac{\eta s^2}{2-\eta}$, where $s^2 = \frac{1}{N}\sum_i (y_i - \bar{y})^2$. Compare with Theorem 7.16.

22. Consider the Markov chain with the transition matrix

$$P = \begin{pmatrix} 0.7 & 0.2 & 0.1 \\ 0.3 & 0.5 & 0.2 \\ 0.2 & 0.3 & 0.5 \end{pmatrix}.$$

    Draw its transition graph, compute $P(X_2 = 3 \mid X_0 = 1)$, and find the stationary distribution. Is the chain irreducible and aperiodic? In the long run, what fraction of the time does it spend in state 1?

23. (Random walks on graphs.)

    (a) Let the walker of Example 8.11 move on the cycle with the four vertices $1, 2, 3, 4$ (and the edges $\{1,2\}, \{2,3\}, \{3,4\}, \{4,1\}$). Find the stationary distribution and the eigenvalues of P. Show that the distribution of X_t does not converge if $X_0 = 1$.

(b) The lazy walk stays at the current vertex with probability $\frac{1}{2}$, i.e., it has the transition matrix $\frac{1}{2}(I+P)$. Show that it has the same stationary distribution, find its eigenvalues, and explain why it converges.

24. (Gibbs sampling.) Let (X_1, X_2) be uniformly distributed on the four points $(0,0)$, $(0,1)$, $(1,1)$ and $(1,2)$. Find the conditional distributions of X_1 given X_2 and of X_2 given X_1, write down the 4×4 transition matrix of one sweep of the Gibbs sampler (first update X_1, then X_2), and check that the uniform distribution is stationary. Is the chain irreducible?

25. (Markov reward processes and decision processes.)

(a) For the weather chain of Example 8.18, compute $\mathbf{V}$ as a function of γ, and show that $(1-\gamma)\mathbf{V} \to (\frac{5}{6}, \frac{5}{6})^\top$ as $\gamma \to 1$. For a general irreducible, aperiodic chain with the stationary distribution $\boldsymbol{\pi}$, explain with Theorem 8.6 why $(1-\gamma)V_i \to \boldsymbol{\pi}\mathbf{r}$ for every state i: the long-run average reward does not depend on the start.

(b) Show that the Bellman optimality operator satisfies $T(\mathbf{V}+c\mathbf{1}) = T\mathbf{V}+\gamma c\mathbf{1}$ for every $c \in \mathbb{R}$, and that value iteration started at $\mathbf{V}_0 = \mathbf{0}$ produces nondecreasing iterates if all rewards are nonnegative.

(c) (Policy iteration.) Starting from a policy π_0, compute $\mathbf{V}^{\pi_k}$ by solving the Bellman equation, and let π_{k+1} be greedy with respect to $\mathbf{V}^{\pi_k}$, i.e., let it maximize $r(s,a)+\gamma \sum_{s'} P(s' \mid s,a) V_{s'}^{\pi_k}$ over a in every state s. Show that $\mathbf{V}^{\pi_{k+1}} \geq \mathbf{V}^{\pi_k}$, and that if $\mathbf{V}^{\pi_{k+1}} = \mathbf{V}^{\pi_k}$, then π_k is optimal. Conclude that policy iteration finds an optimal policy after finitely many steps.

26. (Time series.)

(a) Compute the autocovariances of the MA(2) process $X_t = \varepsilon_t + \theta_1 \varepsilon_{t-1} + \theta_2 \varepsilon_{t-2}$, with white noise ε_t of variance σ^2.

(b) Show that the companion matrix A of an AR(p) process has the characteristic polynomial $\pm(\lambda^p - \phi_1 \lambda^{p-1} - \cdots - \phi_p)$. (Hint: show that $(\lambda^{p-1}, \ldots, \lambda, 1)^\top$ is an eigenvector of A if λ is a root of this polynomial, or expand the determinant along the last column.)

(c) Show that the AR(2) process $X_t = 1.2X_{t-1} - 0.5X_{t-2} + \varepsilon_t$ satisfies the stationarity condition. Its eigenvalues are complex; what does this suggest about the shape of its autocorrelation function?

(d) For a stationary AR(1) process, show that $\operatorname{Cov}(X_t, X_{t-1}) = \phi \operatorname{Var}(X_{t-1})$, so that $\phi = \rho(1)$ (the Yule–Walker equation), and compare with the least squares estimate $\hat{\phi}$.

27. Let $X_1, \ldots, X_N$ be i.i.d. Uniform$(0, \theta)$.

(a) Show that the MLE of θ is $M = \max_i X_i$, and that $\mathbb{E}[M] = \frac{N}{N+1}\theta$. (Hint: $P(M \leq t) = (t/\theta)^N$.)

(b) Show that $\frac{N+1}{N} M$ and $2\bar{X}_N$ are unbiased, and compare their variances, $\frac{\theta^2}{N(N+2)}$ and $\frac{\theta^2}{3N}$. Which estimator is better?

28. Find the maximum likelihood estimates for i.i.d. data from (a) Poisson(λ), (b) Exponential(λ), (c) Geometric(p).

29. (Fisher information.)

(a) Show that the Fisher information of Poisson(λ) is $I(\lambda) = \frac{1}{\lambda}$, and that the MLE $\bar{X}_N$ is efficient.

(b) Show that $I(\lambda) = \frac{1}{\lambda^2}$ for Exponential(λ). The sum S of $N \geq 3$ i.i.d. Exponential(λ) variables has the Gamma density $\frac{\lambda^N s^{N-1} e^{-\lambda s}}{(N-1)!}$ for $s > 0$. Use it to show that the MLE $\frac{1}{\bar{X}_N}$ has the expectation $\frac{N\lambda}{N-1}$, that $\frac{N-1}{N\bar{X}_N}$ is unbiased, and that its variance $\frac{\lambda^2}{N-2}$ is larger than the Cramér–Rao bound. Is the MLE asymptotically efficient?

(c) Counts of $N = 50$ independent Poisson observations have the sample mean 3. Estimate the standard error of $\hat{\lambda}$ from the observed information $-\ell''(\hat{\lambda})$, and give an approximate 95% confidence interval for λ.

30. (Rao–Blackwell.) Let $X_1, \ldots, X_N$ be i.i.d. Uniform$(0, \theta)$. Use the factorization criterion to show that $M = \max_i X_i$ is sufficient for θ. Show that $\mathbb{E}[2X_1 \mid M] = \frac{N+1}{N} M$: Rao–Blackwellizing the unbiased estimator $2X_1$ gives the unbiased estimator of Exercise 27. (Hint: given $M = m$, $X_1 = m$ with probability $\frac{1}{N}$, and otherwise X_1 is uniform on $[0, m]$.)

31. Assume the regression model $y_i = \mathbf{w}^\top \mathbf{x}_i + \varepsilon_i$ with i.i.d. Laplace noise, with density $\frac{1}{2b} e^{-|\varepsilon|/b}$. Show that the MLE of $\mathbf{w}$ minimizes $\sum_i |y_i - \mathbf{w}^\top \mathbf{x}_i|$. In the special case $\mathbf{x}_i = 1$ (estimating a single number w), show that every median of the y_i is a minimizer.

32. Find the method of moments estimators of (a) a and b for Uniform(a, b), using the mean and the variance; (b) the shape k and the scale s of a Gamma distribution, for data with the sample mean 4 and the sample variance 8.

33. (Exponential families.) Write the Geometric(p) distribution, $P(X = x) = p(1-p)^{x-1}$ for $x = 1, 2, \ldots$, as an exponential family, and find T, η and $A(\eta)$. Use Proposition 9.22 to find the mean and the variance, and Theorem 9.23 to find the MLE of p. Show also that the Exponential(λ) and the Gamma distributions with a known shape are exponential families.

34. (Nonparametric estimation.)

(a) Show that a kernel density estimate $\hat{f}_h$ is a probability density.

(b) For the uniform kernel $K = \frac{1}{2}\mathbf{1}_{[-1,1]}$, show that $\hat{f}_h(x) = \frac{1}{2Nh} \#\{i : |x - X_i| \leq h\}$, a histogram with a moving bin centered at x.

(c) For $\mathcal{N}(0,1)$ data and the Gaussian kernel, show that $\mathbb{E}[\hat{f}_h(x)]$ is the density of $\mathcal{N}(0, 1+h^2)$ at x. (Hint: it is the density of $X + hU$ with independent $X, U \sim \mathcal{N}(0,1)$.) Compare $\mathbb{E}[\hat{f}_h(0)] - \varphi(0)$ with the bias $\frac{h^2}{2}\kappa_2 f''(0)$ of Proposition 9.25 for small h.

35. (Survival analysis.)

(a) For the Weibull distribution with the shape k and the scale s, find the density, the median $s(\log 2)^{1/k}$, and the mean $s\,\Gamma(1 + \frac{1}{k})$, using $\mathbb{E}[T] = \int_0^\infty S(t)\,dt$ and the Gamma function $\Gamma(a) = \int_0^\infty u^{a-1} e^{-u}\,du$.

(b) (Discrete time.) A customer who is still subscribed at the beginning of month j cancels during that month with probability h_j. Show that the probability to be still subscribed after t months is $\prod_{j=1}^{t}(1 - h_j)$, and compare with the Kaplan–Meier estimator.

(c) For Weibull event times with a known shape k and non-informative censoring, show that the maximum likelihood estimate of $\theta = s^{-k}$ is $\hat{\theta} = \frac{D}{\sum_i t_i^k}$, where D is the number of observed events.

(d) Show that without censoring, the Kaplan–Meier estimator equals $1 - \hat{F}_N$.

36. A die is rolled 120 times and shows 30 sixes. Test at the level 0.05 whether the die favors sixes: state H_0 and H_1, and compute the exact one-sided p-value $P(S \geq 30)$ for $S \sim \text{Binomial}(120, \frac{1}{6})$ with a computer, and its normal approximation.

37. (Neyman–Pearson.) Let $X_1, \dots, X_N$ be i.i.d. Exponential(λ). Find the most powerful test at the level α of H_0: $\lambda = 1$ against H_1: $\lambda = 2$, and show that it rejects if $\sum_i X_i$ is small. For $N = 1$ and $\alpha = 0.05$, find the rejection region and the power.

38. (Inference for linear regression.)

    (a) For the simple regression $y_i = w_0 + w_1 x_i + \varepsilon_i$ with homoscedastic noise, show that $\operatorname{Var}(\hat{w}_1) = \frac{\sigma^2}{\sum_i (x_i - \bar{x})^2}$. How should the inputs x_i be chosen in an experiment to estimate the slope precisely?

    (b) For heteroscedastic noise with $\operatorname{Cov}(\boldsymbol{\varepsilon}) = \operatorname{diag}(\sigma_1^2, \dots, \sigma_N^2)$, show that $\mathbb{E}[\text{RSS}] = \sum_i \sigma_i^2 (1 - H_{ii})$, where $H = X(X^\top X)^{-1} X^\top$.

    (c) In the model $y_i = w x_i + \varepsilon_i$ with $\sigma_i = \sigma |x_i|$ and all $x_i \neq 0$, show that weighted least squares gives $\tilde{w} = \frac{1}{N} \sum_i \frac{y_i}{x_i}$, with the variance $\frac{\sigma^2}{N}$, and that this is at most the variance $\frac{\sigma^2 \sum_i x_i^4}{(\sum_i x_i^2)^2}$ of ordinary least squares. Which theorem explains this, applied to the rescaled data $\frac{y_i}{x_i}$?

39. Out of 200 users, 70, 80 and 50 prefer the layouts A, B and C. Use Pearson's χ^2 test to test at the level 0.05 whether the three layouts are equally popular. (The χ^2_2 distribution has the tail $P(\chi^2_2 \geq c) = e^{-c/2}$.)

40. (a) Show that for m independent tests of true null hypotheses at the level $\frac{\alpha}{m}$, the probability of at least one rejection is $1 - (1 - \frac{\alpha}{m})^m \leq \alpha$.

    (b) In a permutation test with two groups of three observations each, what is the smallest possible one-sided p-value? What does this mean for very small experiments?

    (c) Show that the bootstrap standard error of the sample mean (for $B \to \infty$) is $\frac{\hat{\sigma}}{\sqrt{N}}$ with $\hat{\sigma}^2 = \frac{1}{N} \sum_i (x_i - \bar{x})^2$.

41. An online shop wants to test whether a new design increases the click-through rate from 10% to 11%. How many users per group are needed for a two-sided test at the level 0.05 with power 0.8? After the experiment, the two groups of 15 000 users have 1500 and 1650 clicks. Compute the z-statistic, the p-value, and a 95% confidence interval for the difference.

42. (Classification metrics.) A test for a disease with a prevalence of 2% has a true positive rate of 90% and a false positive rate of 5%. Compute its precision. Which threshold-independent number does not change when the prevalence changes, and why? For the scores $0.9, 0.7, 0.6, 0.4, 0.2$ of five examples with the labels $1, 0, 1, 0, 0$, draw the ROC curve and compute the AUC in two ways.

43. (Generalization.) How many i.i.d. test examples are needed to estimate the error rate of one classifier within ± 0.01 with probability at least 0.95, using Hoeffding's inequality? How many are needed to estimate the error rates of 1000 classifiers simultaneously with the same accuracy?

44. (Simpson's paradox.) Two treatments are compared in groups $s = 1, \dots, m$, with the success rates $a_s = P(\text{success} \mid A, s)$ and $b_s = P(\text{success} \mid B, s)$, and $a_s > b_s$ for every s. Show that if $P(s \mid A) = P(s \mid B)$ for all s, as in a randomized experiment, then $P(\text{success} \mid A) > P(\text{success} \mid B)$. Then construct an example with two groups in which $a_s > b_s$ for both groups, but $P(\text{success} \mid A) < P(\text{success} \mid B)$.

45. With the uniform prior Beta$(1,1)$, a coin shows 7 heads in 10 flips. Find the posterior distribution, its mean, the MAP estimate, the MLE, and the predictive probability of heads in the next flip.

46. (Gaussian–Gaussian conjugacy.) Let $x_1, \dots, x_N$ be i.i.d. $\mathcal{N}(\mu, \sigma^2)$ with known σ^2, and let the prior be $\mu \sim \mathcal{N}(\mu_0, \tau^2)$. Show that the posterior is Gaussian with precision $\frac{1}{\tau^2} + \frac{N}{\sigma^2}$ and mean

$$\frac{\mu_0/\tau^2 + N\bar{x}/\sigma^2}{1/\tau^2 + N/\sigma^2},$$

a weighted average of the prior mean and the sample mean. What happens as $N \to \infty$?

47. (Gaussian processes and Bayesian linear regression.)

    (a) Show that $(X^\top X + \lambda I)X^\top = X^\top(XX^\top + \lambda I)$ for every matrix X and $\lambda > 0$, and conclude that $(X^\top X + \lambda I)^{-1}X^\top = X^\top(XX^\top + \lambda I)^{-1}$.

    (b) Use (a) to show that for the linear kernel $k(\mathbf{x}, \mathbf{x}') = \alpha^{-1}\mathbf{x}^\top\mathbf{x}'$, the posterior mean of Proposition 12.11 equals $\mathbf{m}^\top\mathbf{x}_*$ from Theorem 12.7. Which of the two formulas is cheaper to evaluate when $N \gg d$, and which when $d \gg N$?

    (c) Let $f \sim \mathcal{GP}(0, k)$ with $k(s,t) = \min(s,t)$, and observe $f(1) = 1$ without noise. Find the posterior mean and variance of $f(t)$ for $0 \le t \le 1$ and for $t > 1$, and sketch them.

48. (a) Compute the entropy of a fair die in bits.

    (b) Compute $\mathrm{KL}(\text{Bernoulli}(p) \,\|\, \text{Bernoulli}(q))$, and show that

$$\mathrm{KL}\left(\mathcal{N}(\mu_1, \sigma^2) \,\|\, \mathcal{N}(\mu_2, \sigma^2)\right) = \frac{(\mu_1 - \mu_2)^2}{2\sigma^2}.$$

    (c) Show that $I(X;Y) = H(X) + H(Y) - H(X,Y)$ for discrete random variables, where $H(X,Y)$ is the entropy of the joint distribution. Compute $I(X;Y)$ for Exercise 8.

49. (EM for a mixture of two coins.) In each of 5 experiments, one of two coins with the unknown head probabilities θ_A and θ_B is chosen with probability $\frac{1}{2}$ and flipped 10 times. The numbers of heads are $5, 9, 8, 4, 7$, and we do not know which coin was used. Starting from $\theta_A = 0.6$ and $\theta_B = 0.5$, carry out one step of EM: compute the posterior probabilities that coin A was used in each experiment, and the new estimates. Show that the log-likelihood increased.

50. (k-means.) Show that every assignment step and every update step of Lloyd's algorithm can only decrease J, and find an example with $N = 4$ points in $\mathbb{R}$ and $K = 2$ for which Lloyd's algorithm, started from a bad pair of centers, stops at an assignment that is not optimal.

51. (Probabilistic PCA.)

    (a) Show that the maximal log-likelihood of probabilistic PCA is $-\frac{N}{2}\big(d\log(2\pi) + \sum_{j \le k}\log\lambda_j + (d-k)\log\hat{\sigma}^2 + d\big)$.

    (b) For data with $S = \mathrm{diag}(4, 2, 1)$, compute it (per data point) for $k = 0$, 1 and 2. Why does it always increase with k, and why is it nevertheless not a good idea to choose k by the likelihood alone?

    (c) Show that the posterior covariance $\sigma^2 M^{-1}$ of $\mathbf{z}$ is $\sigma^2 I_k$ minus a positive semidefinite matrix, so that observing $\mathbf{x}$ reduces the uncertainty about $\mathbf{z}$ in every direction.

52. (Brownian motion and the Ornstein–Uhlenbeck process.)

(a) Compute $\mathrm{Var}(B(2) - B(0.5))$, $\mathrm{Cov}(B(1), B(3))$ and $\mathrm{Corr}(B(1), B(3))$.

(b) Let $c > 0$. Show that $W(t) = \frac{1}{\sqrt{c}} B(ct)$ has the same mean and covariance functions as $B(t)$. (In fact, W is again a Brownian motion: zooming into a Brownian path by the factor c in time and $\sqrt{c}$ in space gives a Brownian path.)

(c) Let $X_{k+1} = (1 - \theta h)X_k + \sigma\sqrt{h}\, Z_k$ with $X_0 = x_0$ and i.i.d. $Z_k \sim \mathcal{N}(0, 1)$. Show that $\mathbb{E}[X_k] = (1 - \theta h)^k x_0$ and $\mathrm{Var}(X_k) = v_h(1 - (1 - \theta h)^{2k})$, with v_h from Example 15.4. What happens for $\theta h > 2$?

18 Programming Exercises

As in the previous lectures, implement the methods yourself with NumPy, and use library routines (such as `rng.normal` or `scipy.stats`) only to check your results. Start each file with

```
import numpy as np
import matplotlib.pyplot as plt
rng = np.random.default_rng(0)
```

P1. Monte Carlo estimation. Estimate π from N uniform random points in $[-1, 1]^2$, for $N = 10^2, 10^3, \ldots, 10^7$. Plot the absolute error against N on a log-log scale, together with the standard error $4\sqrt{\frac{\pi}{4}(1 - \frac{\pi}{4})/N}$. What is the slope? Then estimate the volume of the unit ball in $\mathbb{R}^d$ for $d = 2, 5, 10, 20$ in the same way. What goes wrong in high dimensions?

P2. Sampling. Generate 10^5 samples from each of the following distributions, using only `rng.random()` (uniform samples), and compare histograms with the densities.

- Exponential(λ) by inverse transform sampling (Theorem 2.7).
- A discrete distribution with the values $x_1 < \cdots < x_K$ and the probabilities $p_1, \ldots, p_K$, by its quantile function $F^{-1}(u) = \min\{x : F(x) \geq u\}$ (Theorem 2.7), using the cumulative sums and `np.searchsorted`. Compare the frequencies with the probabilities.
- $\mathcal{N}(0, 1)$ with the Box–Muller method: for independent $U_1, U_2 \sim \mathrm{Uniform}(0, 1)$, the numbers $\sqrt{-2\log U_1}\cos(2\pi U_2)$ and $\sqrt{-2\log U_1}\sin(2\pi U_2)$ are independent standard normal.
- $\mathcal{N}(\boldsymbol{\mu}, \Sigma)$ in $\mathbb{R}^2$ with $\Sigma = \begin{pmatrix} 3 & 1 \\ 1 & 3 \end{pmatrix}$, via $\boldsymbol{\mu} + L\mathbf{z}$ with the Cholesky factor L. Plot the samples together with the ellipses of Figure 2, and check the sample mean and covariance matrix. What fraction of the samples lies inside the ellipse with $r = 1$, and with $r = 2$? (Compare with $1 - e^{-r^2/2}$.)

P3. The law of large numbers and the central limit theorem. For $X_i \sim$ Exponential(1), simulate 10^4 sample means $\bar{X}_N$ for $N = 1, 2, 5, 30$, and plot histograms of the standardized means $\sqrt{N}(\bar{X}_N - 1)$ together with the standard normal density. Repeat with the Cauchy distribution (the ratio of two independent standard normal variables), which has no mean: plot $\bar{X}_N$ against N for one long sequence. What happens?

P4. Gaussian classifiers. Generate two classes in $\mathbb{R}^2$ from Gaussians with means $(0, 0)$ and $(2, 1)$ and the same covariance matrix $\Sigma = \begin{pmatrix} 1 & 0.5 \\ 0.5 & 1 \end{pmatrix}$, with 200 points each.

- Estimate the means and the shared covariance matrix by maximum likelihood (Theorem 9.10).
- Show (on paper) that with equal class priors, Bayes' theorem gives $P(y = 1 \mid \mathbf{x}) = \sigma(\mathbf{w}^\top \mathbf{x} + b)$ with $\mathbf{w} = \Sigma^{-1}(\boldsymbol{\mu}_1 - \boldsymbol{\mu}_0)$ and $b = -\frac{1}{2}(\boldsymbol{\mu}_1^\top \Sigma^{-1} \boldsymbol{\mu}_1 - \boldsymbol{\mu}_0^\top \Sigma^{-1} \boldsymbol{\mu}_0)$: a logistic regression model.
- Plot the data and the decision boundary, and compare with logistic regression trained by gradient descent (Lecture 3).

P5. Maximum likelihood and bias. Draw 10^5 data sets of size $N = 5$ from $\mathcal{N}(2, 9)$. Compute the average of the MLE $\hat{\sigma}^2$ and of s^2 over the data sets, and compare with $\frac{N-1}{N}\sigma^2 = 7.2$ and $\sigma^2 = 9$. Then estimate the mean squared errors of both estimators. Which one has the smaller MSE?

P6. Rao–Blackwell and the delta method.

- For 10^5 data sets of $N = 10$ Poisson(1) counts, compute the three estimators $\mathbf{1}_{\{X_1 = 0\}}$, $(1 - \frac{1}{N})^S$ and $e^{-\bar{X}_N}$ of e^{-1} (Example 9.18), and estimate their biases, variances and mean squared errors.
- For $N = 30$ Bernoulli(p) observations with $p = 0.1$, estimate the coverage probabilities of the interval $\hat{p} \pm 1.96\sqrt{\hat{p}(1-\hat{p})/N}$ and of the log-odds interval of Example 7.12 by simulation (skip the data sets with $\hat{p} \in \{0, 1\}$, and report how often this happens). Which interval keeps its promise of 95% better?

P7. The bootstrap. For $N = 50$ samples from Exponential(1), estimate the standard error of the sample median by resampling the data 2000 times with replacement and computing the median of each resample. Compare with the standard error of the median over many independent data sets. Use the same method to compute a 95% interval for the mean (the 2.5% and 97.5% quantiles of the bootstrapped means), and compare with $\bar{x} \pm 1.96\frac{s}{\sqrt{N}}$.

P8. Bayesian updating and Thompson sampling.

- Simulate 100 flips of a coin with $\theta = 0.3$, and plot the Beta posterior (starting from Beta$(1, 1)$) after $0, 1, 5, 20$ and 100 flips, as in Figure 5.
- A slot machine has three arms with unknown success probabilities 0.3, 0.5 and 0.6. Thompson sampling keeps a Beta posterior for each arm, and in each round draws one sample from each posterior and plays the arm with the largest sample. Simulate 1000 rounds, and plot how often each arm is played. Compare the total reward with always playing a random arm and with always playing the best arm.

P9. Bayesian linear regression. Generate 20 points $y = \sin(3x) + \varepsilon$ with x uniform in $[-1, 1]$ and $\varepsilon \sim \mathcal{N}(0, 0.1^2)$, and use the features $1, x, \ldots, x^7$. Compute the posterior of Theorem 12.7 with $\alpha = 1$ and $\sigma = 0.1$. Plot the data, the posterior mean, the band of ± 2 predictive standard deviations, and a few functions with weights sampled from the posterior. Where is the uncertainty largest? Check that the posterior mean equals the ridge solution with $\lambda = \alpha\sigma^2$.

P10. Cross-entropy and KL divergence.

- Write functions for $H(p)$, $H(p, q)$ and $\mathrm{KL}(p \,\|\, q)$, and check $H(p, q) = H(p) + \mathrm{KL}(p \,\|\, q)$ and $\mathrm{KL} \geq 0$ for random distributions (e.g., `rng.dirichlet(np.ones(5))`).

- Let p be the mixture $\frac{1}{2}\mathcal{N}(-2, 0.5^2) + \frac{1}{2}\mathcal{N}(2, 0.5^2)$. On a grid, find the Gaussian $q = \mathcal{N}(\mu, s^2)$ that minimizes $\mathrm{KL}(p \,\|\, q)$, and the one that minimizes $\mathrm{KL}(q \,\|\, p)$ (try several starting points). Plot both. Why does the first cover both modes, while the second picks one of them?

P11. Gaussian process regression (Section 12.4), with the squared exponential kernel $k(x, x') = \exp(-\frac{(x-x')^2}{2\ell^2})$.

- Sample and plot a few functions from the prior $\mathcal{GP}(0, k)$ on a grid of 200 points in $[-3, 3]$, for $\ell = 0.1, 0.3, 1$. (Add $10^{-8}I$ to the kernel matrix before the Cholesky factorization.) Do the same for the Brownian motion kernel $\min(s, t)$ (Section 15) on $[0, 3]$.
- For 8 noisy observations $y_i = \sin(3x_i) + \varepsilon_i$ ($\varepsilon_i \sim \mathcal{N}(0, 0.1^2)$), compute the posterior mean and variance of Proposition 12.11 on the grid, and plot them with a band of ± 2 standard deviations, as in Figure 6. Use `np.linalg.solve` (or a Cholesky factorization) rather than the inverse. Sample a few functions from the posterior.
- Compute the log marginal likelihood $\log p(\mathbf{y})$ for ℓ on a grid from 0.05 to 3, and plot it. Which length scale does it select? Compare the posterior with the one for $\ell = 0.1$ and $\ell = 1$.

P12. Markov chains and MCMC.

- (A bigram language model.) Take a text of a few thousand words, split it into words, and estimate the transition probabilities $P(\text{next word} = j \mid \text{current word} = i)$ by counting. Generate random sentences from the chain. Compute the stationary distribution by power iteration, and compare it with the word frequencies in the text.
- (Metropolis sampling.) For the probability θ of heads, use the prior density $p(\theta) \propto e^{-10(\theta-0.5)^2}$ on $[0, 1]$, which is not a Beta distribution, and the data "7 heads in 10 flips". Sample from the posterior $p(\theta \mid \mathcal{D}) \propto \theta^7(1-\theta)^3 e^{-10(\theta-0.5)^2}$ with the Metropolis algorithm, using the proposals $\theta' = \theta + \eta\varepsilon$ with $\varepsilon \sim \mathcal{N}(0, 1)$ (and target 0 outside $[0, 1]$). Compare the histogram of 50 000 samples with the posterior density, normalized numerically on a grid. Plot the chain for $\eta = 0.01$, 0.1 and 1, and report the acceptance rates. Which step size works best, and why?

P13. Rejection and importance sampling.

- Sample from the Beta$(2, 5)$ distribution by rejection sampling with the uniform proposal. Find the best M numerically, compare the acceptance rate with $\frac{1}{M}$, and compare a histogram with the density.
- Estimate $P(Z > 4)$ and $P(Z > 6)$ for $Z \sim \mathcal{N}(0, 1)$ with plain Monte Carlo and with importance sampling from $\mathcal{N}(a, 1)$ for $a = 4$ (or 6), using $N = 10^4$ samples. Repeat 100 times to estimate the relative standard errors, and compare with the exact values $1-\Phi(4)$ and $1 - \Phi(6)$. What happens with the proposal $\mathcal{N}(0, 1)$ shifted to $a = 8$? Compute the effective sample size $\frac{(\sum_i w_i)^2}{\sum_i w_i^2}$ in each case.

P14. Hypothesis tests by simulation.

- Simulate 10 000 data sets of size $N = 30$ from $\mathcal{N}(0, 1)$, compute the two-sided z-test p-value for H_0: $\mu = 0$ in each, and plot the histogram of the p-values. Repeat with data from $\mathcal{N}(0.5, 1)$, and estimate the power of the test at the level 0.05.

- Generate 1000 tests, of which 50 have a real effect (data from $\mathcal{N}(0.8, 1)$) and 950 do not (data from $\mathcal{N}(0, 1)$), each with $N = 20$. Count the true and false discoveries without correction, with Bonferroni, and with Benjamini–Hochberg at the level 0.05.
- Implement a permutation test for the difference of the means of two groups, and apply it to two groups of 15 samples from Exponential(1) and Exponential(0.6). Compare its p-value with the z-test.
- (Peeking.) Simulate A/B tests with two identical versions (conversion rate 5%) and up to 10 000 users per group, computing the two-proportion z-test after every 200 users per group. What fraction of the experiments ever reaches $p < 0.05$? Compare with a single test at the end.

P15. Cross-validation. Use the polynomial regression data of the programming exercises of Lecture 2 (20 points $y = \sin(3x) + \varepsilon$).

- Implement k-fold cross-validation, and choose the polynomial degree from 1 to 12 with 5-fold cross-validation. Plot the cross-validation error and the training error against the degree.
- Repeat the cross-validation with 50 different random splits. How much does the selected degree, and the cross-validation error of a fixed degree, vary? Compare the spread with the naive standard error $\frac{s_e}{\sqrt{k}}$.
- Generate a large independent test set from the same distribution. Is the cross-validation error of the selected degree optimistic? Implement nested cross-validation and compare.

P16. Stochastic differential equations.

- Simulate 10 000 Brownian paths on $[0, 1]$ with the step $h = 0.001$. Check $\mathrm{Var}(B(t)) = t$ and $\mathrm{Cov}(B(0.3), B(0.8)) = 0.3$ empirically, and plot a few paths. Zoom into one path: plot it on $[0.5, 0.5 + \delta]$ for $\delta = 0.1, 0.01, 0.001$ (simulating the path with a step much smaller than δ). Why does it never look smooth?
- Simulate the Ornstein–Uhlenbeck process $dX = -\theta X\,dt + \sigma\,dB$ with Euler–Maruyama for $\theta = 1$, $\sigma = \sqrt{2}$, starting 10 000 paths at $x_0 = 3$. Plot the mean and the variance of $X(t)$ against t, together with $x_0 e^{-t}$ and $1 - e^{-2t}$. Compare the histogram at $t = 5$ with $\mathcal{N}(0, 1)$, and check v_h from Example 15.4 for $h = 0.5$.
- (Langevin sampling.) Use the unadjusted Langevin algorithm to sample from the mixture $p = \frac{1}{2}\mathcal{N}(-2, 0.5^2) + \frac{1}{2}\mathcal{N}(2, 0.5^2)$, i.e., $U = -\log p$ (compute ∇U by hand). Compare a histogram of one long chain with p, and count how often the chain moves between the two modes. Then sample the posterior of the logistic regression model of Lecture 3 on a small two-dimensional data set, with the prior $\mathcal{N}(\mathbf{0}, I)$, and plot the samples of $\mathbf{w}$ together with the MAP estimate.

P17. ROC curves. Generate 500 examples of each class with the scores $s \sim \mathcal{N}(0, 1)$ for the negatives and $s \sim \mathcal{N}(1, 1)$ for the positives.

- Compute the TPR and the FPR for all thresholds, plot the ROC curve, and compute the AUC as the area under the curve and as the fraction of correctly ordered pairs (Theorem 11.1). Compare with the exact value $\Phi(1/\sqrt{2}) \approx 0.76$ for these distributions.
- Plot the precision and the recall as functions of the threshold. Repeat with only 25 positives and 500 negatives: which curves change, and which do not?

P18. Gaussian mixtures and k-means. Generate 300 points in $\mathbb{R}^2$ from a mixture of three Gaussians with different means and covariance matrices.

- Implement Lloyd's algorithm for k-means with $K = 3$. Plot J after each step, and check that it never increases (Proposition 14.7). Run it from 10 random initializations; do they all give the same J?
- Implement EM for Gaussian mixtures (Theorem 14.5), initialized with the k-means result. Compute the responsibilities with logarithms and the log-sum-exp trick to avoid underflow. Plot the log-likelihood after each step, and check that it never decreases (Theorem 14.4).
- Plot the data colored by their most likely component, and the ellipses of the fitted covariance matrices. Where do EM and k-means disagree, and why?

P19. Naive Bayes. Generate 2000 examples with a label $y \in \{0, 1\}$ ($P(y = 1) = 0.3$) and 20 binary features that are conditionally independent given y, with $P(x_j = 1 \mid y)$ drawn uniformly from $[0.1, 0.9]$ for each j and y.

- Fit naive Bayes by counting, with Laplace smoothing, and logistic regression by gradient descent (Lecture 3), on 1000 training examples. Compare their test accuracies, and plot the predicted probabilities of one model against those of the other.
- Now copy the first feature 10 times, so that the features are no longer conditionally independent, and repeat. Which model's probabilities become overconfident, and why?

P20. Nonparametric estimation and probabilistic PCA.

- Draw $N = 100$ and $N = 1000$ samples from Exponential(1), plot the empirical CDF with the band $\hat{F}_N \pm \sqrt{\log(40)/(2N)}$ and the true CDF, and check in 1000 repetitions how often the true CDF leaves the band.
- Draw 500 samples from the mixture $0.5\,\mathcal{N}(-2, 1) + 0.5\,\mathcal{N}(2, 0.5^2)$, and plot kernel density estimates with the Gaussian kernel and $h = 0.05$, 0.3 and 1, together with the true density. Choose h by maximizing the leave-one-out log-likelihood $\sum_i \log \hat{f}_{h,-i}(x_i)$, where $\hat{f}_{h,-i}$ omits the i-th point, and compare with the rule of thumb $h = 1.06\,\hat{\sigma} N^{-1/5}$.
- Generate $N = 1000$ points $\mathbf{x} = W\mathbf{z} + \boldsymbol{\varepsilon}$ in $\mathbb{R}^{10}$ with a random 10×2 matrix W, $\mathbf{z} \sim \mathcal{N}(\mathbf{0}, I_2)$ and $\boldsymbol{\varepsilon} \sim \mathcal{N}(\mathbf{0}, 0.25I)$. Fit probabilistic PCA with Theorem 14.9, compare $\hat{\sigma}^2$ with 0.25, and compare the orthogonal projections onto the column spaces of $\hat{W}$ and W.

P21. Initialization of deep networks (Example 4.13).

- Pass a batch of 1000 standard normal inputs in $\mathbb{R}^{512}$ through 50 ReLU layers of width 512 with weights drawn from $\mathcal{N}(0, v)$, for $v = \frac{2}{512}$ (He), $\frac{1}{512}$ and $\frac{4}{512}$. Plot the mean of the squared activations against the layer on a logarithmic scale, and compare with the prediction $q_l = q_0(\frac{nv}{2})^l$.
- Repeat with tanh activations and $v = \frac{1}{512}$. Then backpropagate the gradient of the loss $\sum_i y_i$ of the last layer, and plot the mean squared gradient with respect to the activations of each layer. Which initialization keeps both the forward signal and the gradients at a constant size?

P22. Policy evaluation and value iteration (Section 8.2). A robot moves on a 4×4 grid with the actions up, down, left and right; it moves in the chosen direction with probability 0.8 and

in one of the two perpendicular directions with probability 0.1 each, and stays in place if it would leave the grid. Entering the corner $(4, 4)$ gives the reward 1 and restarts the robot at $(1, 1)$, and every other step gives the reward -0.04. Let $\gamma = 0.95$.

- Compute the value of the uniformly random policy by solving the Bellman equation with `np.linalg.solve`, and by iterative policy evaluation. Plot the error $\|\mathbf{V}_k - \mathbf{V}\|_\infty$ against k on a logarithmic scale, together with $\gamma^k \|\mathbf{V}_0 - \mathbf{V}\|_\infty$.
- Find the optimal values with value iteration, and draw the greedy policy as arrows on the grid. Compare with policy iteration (Exercise 25): how many iterations does each method need?
- Implement Q-learning with ε-greedy exploration (a random action with probability $\varepsilon = 0.1$, otherwise the action with the largest Q), using simulated transitions only, and compare the learned greedy policy and $\max_a Q(s, a)$ with the results of value iteration.

P23. Time series (Section 8.3).

- Simulate AR(1) processes with $\phi = 0.5$, 0.9, 0.99 and 1 (a random walk) for $T = 1000$ steps with standard normal noise, starting at $X_0 = 0$, and plot them. Compute the sample autocorrelation $\hat{\rho}(h) = \frac{\sum_t (x_t - \bar{x})(x_{t+h} - \bar{x})}{\sum_t (x_t - \bar{x})^2}$ for $h = 0, \dots, 30$, and compare with ϕ^h. What goes wrong for the random walk, and what happens after differencing?
- Estimate ϕ by least squares for 100 simulated series with $\phi = 0.9$ and $T = 50$, and for $T = 500$. Is $\hat{\phi}$ biased? Compare its standard deviation with $\sqrt{(1 - \phi^2)/T}$.
- Simulate the AR(2) processes of Section 8.3 and of Exercise 26, compute the eigenvalues of their companion matrices with `np.linalg.eigvals`, and relate them to the behavior of the simulated paths and of their autocorrelations.

P24. Survival analysis (Section 9.9). Generate $N = 200$ lifetimes from the Weibull distribution with the shape $k = 1.5$ and the scale $s = 10$ (by inverse transform sampling), and independent censoring times uniform on $[0, 15]$.

- Implement the Kaplan–Meier estimator, and plot it together with the true survival function $e^{-(t/s)^k}$ and with the empirical survival function of the observed times t_i that ignores the censoring. Which one is biased, and in which direction?
- Fit an exponential model with Proposition 9.28, and a Weibull model by maximizing the censored log-likelihood numerically over k and s (for example, with gradient ascent on $\log k$ and $\log s$). Plot both fitted survival functions together with the Kaplan–Meier estimate. Which model fits, and how do the fitted hazard rates differ?

P25. Inference for linear regression (Section 10.1).

- Generate $N = 20$ points $y = 1 + 2x + \varepsilon$ with x uniform on $[0, 10]$ and $\varepsilon \sim \mathcal{N}(0, 4)$. Compute $\hat{\mathbf{w}}$, s^2, the standard errors, the t-statistics and the 95% confidence intervals, and check them against `statsmodels` if you have it installed. In 10 000 repetitions, estimate the coverage probability of the interval for the slope.
- Repeat with heteroscedastic noise $\varepsilon \sim \mathcal{N}(0, (0.3x)^2)$. Compare the coverage of the usual interval with that of the interval based on the robust standard errors, and plot the residuals against x. How do the results change with $N = 200$?